1. Distributional Properties of Categorical Processes

1.1. General Notation

Let (X_t) be a *categorical* time series process, where each X_t takes values in

$$S := \{s_0, \dots, s_m\}, \qquad m \in \mathbb{N}.$$

(X_t) may be *nominal*, in which case the order of magnitude of the elements in S is arbitrary and solely used for notational consistency, or *ordinal*, in which case the order of the elements in S reflects the inherent order of the elements.

1.2. Stationarity

Furthermore, (X_t) is assumed to be stationary. A distinction can be made between two forms of stationarity:

Definition 1.1 (Weak Stationarity)**.** A process (X_t) is said to be *weakly stationary* if:

$$\begin{aligned} \mathrm{E}(X_t) &= \mu_X && \forall t, \\ \mathrm{E}(X_t^2) &< \infty && \forall t, \\ \mathrm{Cov}(X_t, X_{t+h}) &= \gamma(h) && \forall t, h. \end{aligned}$$

Definition 1.2 (Strong/ Strict Stationarity)**.** Strong/ Strict stationarity requires that all finite-dimensional distributions are invariant under time shifts, i.e.,

$$F_{X_t,\dots,X_{t+j}} = F_{X_{t+h},\dots,X_{t+h+j}} \quad \forall\, t, j, h. \tag{1.1}$$

1.3. Marginal Distributions

Regardless of whether we consider a simple categorical process with few categories or attempt to describe a complex, temporally evolving process using moments and measures of serial dependence, the analysis of categorical processes is always built on the examination of the marginal probabilities. However, because ordinal and nominal data encode different types of information, their marginals are typically represented in different ways:

- **Ordinal data:** Most analytical tools for ordinal processes take into account the natural ordering of the elements in S. For this reason, most methods for ordinal data rely on the marginal cumulative distribution function (CDF), which accumulates probabilities in accordance with the natural ordering of the categories:

$$\boldsymbol{f} = (f_0, \ldots, f_{m-1})^\top, \qquad f_i := P(X_t \leq s_i).$$

  For ordinal data, it is sufficient to consider the cumulative probabilities of the categories $0, \ldots, m-1$ since the cumulative probability of the last category m is always equal to 1.

  Although one could equivalently estimate the PMF and derive the CDF from it, representing ordinal data through the CDF is often more convenient as it this form that is used to compute metrics for location and dispersion.

- **Nominal data:** Since the categories of nominal variables do not possess a meaningful order, statistical procedures cannot make use of cumulative structure. Therefore the marginal distribution is best represented by the probability mass function (PMF):

$$\boldsymbol{p} = (p_0, \ldots, p_m)^\top, \qquad p_i := P(X_t = s_i).$$

1.4. Bivariate Probabilities at Lag h

Definition 1.3 (Bivariate lag-h probabilities)**.** Let (X_t) be a stationary categorical time series. The bivariate lag-h probabilities are defined as follows.

(i) **Ordinal series:**

$$f_{ij}(h) := P(X_t \leq s_i,\, X_{t+h} \leq s_j), \qquad i, j = 0, \ldots, m-1.$$

(ii) **Nominal series:**

$$p_{ij}(h) := P(X_t = s_i,\, X_{t+h} = s_j), \qquad i, j = 0, \ldots, m.$$

Under serial independence, the bivariate probabilities factorize as $f_{ij}(h) = f_i f_j$ and $p_{ij}(h) = p_i p_j$ respectively.

1.5. Location

For a nominal process, any measure that relies on ordering is inappropriate. Instead, a location measure must depend only on the category probabilities. The mode, defined as the most probable category,

$$\mathrm{Mod}(X_t) = \arg\max_{s_i \in S} P(X_t = s_i),$$

is therefore the natural choice. It provides a meaningful summary of the process by identifying the most frequent category.
For an ordinal process, the natural ordering of the categories allows for additional measures of central tendency. While the mode can still be informative as the most frequent category, the median respects the ordering of categories and is often more useful for describing the central location:

$$\mathrm{Med}(X_t) = \min\Big\{\, s_i \in S : F(s_i) \geq \tfrac{1}{2} \,\Big\}.$$

Thus, for ordinal data, both the mode and the median carry information, but they capture slightly different aspects of location: the mode identifies the most typical category, whereas the median indicates the midpoint in the ordered scale.

1.6. Measures of Dispersion

Since the concept of dispersion differs between ordinal and nominal processes, two separate measures are employed: the *Index of Ordinal Variation* (IOV) for ordinal data and the *Index of Qualitative Variation* (IQV) for nominal data, defined as

$$\mathrm{IOV}(\boldsymbol{f}) = \frac{4}{m} \sum_{i=0}^{m-1} f_i(1 - f_i), \qquad \mathrm{IQV}(\boldsymbol{p}) = \frac{m+1}{m} \left(1 - \sum_{i=0}^{m} p_i^2\right). \tag{1.2}$$

Both indices are scaled to $[0, 1]$, with 0 indicating a degenerate (one-point) distribution and 1 indicating maximal dispersion. However, what constitutes maximal dispersion differs: for ordinal data, dispersion is maximal when all probability mass is concentrated in the two outermost categories, whereas for nominal data it corresponds to the uniform distribution over all $m + 1$ categories.

Proposition 1.4 (Dispersion Bounds)**.** *Both* IOV *and* IQV *satisfy:*

(i) $\mathrm{IOV} = 0$ *(resp.* $\mathrm{IQV} = 0$*) if and only if the distribution is degenerate.*

(ii) $\mathrm{IOV} = 1$ *iff* $\boldsymbol{f} = (\frac{1}{2}, \ldots, \frac{1}{2})^\top$*;* $\mathrm{IQV} = 1$ *iff* $p_i = \frac{1}{m+1}$ *for all* i*.*

Proof. Part (i), ordinal case. A one-point distribution assigns all probability mass to a single category $s_j \in S$. For ordinal data, this corresponds to a CDF vector $\mathbf{f}$ of the form

$$\mathbf{f} \in \left\{ \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 1 \end{pmatrix}, \ldots, \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix} \right\} := \{\boldsymbol{c}_0, \ldots, \boldsymbol{c}_m\} \subset [0, 1]^{m+1}.$$

In each of these cases, every component f_i satisfies $f_i \in \{0, 1\}$, and thus

$$f_i(1 - f_i) = 0 \qquad \forall i.$$

Therefore,

$$\mathrm{IOV} = \frac{4}{m} \sum_{i=0}^{m-1} f_i(1 - f_i) = 0.$$

Conversely, if $\mathrm{IOV} = 0$, then

$$\sum_{i=0}^{m-1} f_i(1 - f_i) = 0,$$

and since each term is nonnegative,

$$f_i(1 - f_i) \stackrel{!}{=} 0 \quad \forall i.$$

This is only possible if $f_i \in \{0, 1\}$ for all i, which implies $\mathbf{f} \in \{\boldsymbol{e}_0, \ldots, \boldsymbol{e}_m\}$. Hence, $\mathrm{IOV} = 0$ holds *exactly* for a degenerate one-point distribution.

Part (i), nominal case. A degenerate (one–point) distribution assigns all probability mass to a single category $s_j \in S$, which corresponds to a PMF vector of the form

$$\mathbf{p} \in \left\{ \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \ldots, \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix} \right\} := \{\boldsymbol{e}_0, \ldots, \boldsymbol{e}_m\} \subset [0, 1]^{m+1}.$$

In each of these cases, exactly one component equals 1 and all others equal 0, so

$$\sum_{i=0}^{m} p_i^2 = 1.$$

Thus,

$$\mathrm{IQV} = \frac{m+1}{m}\Big(1 - \sum_{i=0}^{m} p_i^2\Big) = \frac{m+1}{m}(1 - 1) = 0.$$

Conversely, if $\mathrm{IQV} = 0$, then

$$\frac{m+1}{m}\Big(1 - \sum_{i=0}^{m} p_i^2\Big) = 0 \quad \Longrightarrow \quad \sum_{i=0}^{m} p_i^2 = 1.$$

As all probabilities satisfy $0 \le p_i \le 1$, it follows that

$$p_i^2 \le p_i \quad \forall p_i, \text{ and } p_i^2 < p_i \quad \forall p_i \notin \{0, 1\}.$$

Since $\sum_{i=1}^{m} p_i = 1$, it follows that the only way $\sum_i p_i^2 = 1$ can hold is if exactly one p_i equals 1 and all others are 0. In other words, the distribution is degenerate and $\boldsymbol{p} \in \{\boldsymbol{e}_0, \ldots, \boldsymbol{e}_m\}$.

Part (ii), ordinal case. Dispersion of an ordinal variable is maximal when half of the probability mass lies in the lowest category and half in the highest category; with all intermediate categories having probability zero. This results in a CDF vector,

$$\mathbf{f} = \left(\tfrac{1}{2}, \tfrac{1}{2}, \ldots, \tfrac{1}{2}\right)^\top.$$

Then

$$f_i(1-f_i) = \frac{1}{2}\left(1-\frac{1}{2}\right) = \frac{1}{4} \qquad \forall i,$$

and thus

$$\text{IOV} = \frac{4}{m} \cdot \sum_{i=0}^{m-1} \frac{1}{4} = \frac{4}{m} \cdot \frac{m}{4} = 1.$$

To show that IOV $= 1 \Rightarrow \boldsymbol{f} = (\frac{1}{2}, \ldots, \frac{1}{2})^T$, recall that the IOV only depends on $\boldsymbol{f}$ through the sum

$$\sum_{i=0}^{m-1} f_i(1-f_i).$$

Taking the derivative of $g(f_i) := f_i(1-f_i)$ and setting it to zero yields:

$$\begin{aligned} \frac{\partial}{\partial f_i} g(f_i) &= 1 - 2f_i, \\ 1 - 2f_i &\overset{!}{=} 0, \\ f_i &= \frac{1}{2}. \end{aligned}$$

Since $g(f_i = 0) = g(f_i = 1) = 0$, the function is concave and this critical point is indeed the global maximum. Since it has been already shown that $\text{IOV}(\boldsymbol{f} = (\frac{1}{2}, \ldots, \frac{1}{2})^T) = 1$, it directly follows that

$$\text{IOV}(\boldsymbol{f}) = 1 \Leftrightarrow \boldsymbol{f} = (\frac{1}{2}, \ldots, \frac{1}{2})^\top$$

Part (ii), nominal case. For nominal data, maximal dispersion is attained when probability mass is distributed uniformly across all $m+1$ categories, i.e.

$$p_i = \frac{1}{m+1}, \qquad i = 0, \ldots, m.$$

In this case,

$$\sum_{i=0}^{m} p_i^2 = (m+1)\frac{1}{(m+1)^2} = \frac{1}{m+1},$$

and hence

$$\text{IQV} = \frac{m+1}{m}\left(1 - \frac{1}{m+1}\right) = 1.$$

Conversely, suppose that IQV $= 1$. Then

$$\frac{m+1}{m}\left(1 - \sum_{i=0}^{m} p_i^2\right) = 1 \quad \Longrightarrow \quad \sum_{i=0}^{m} p_i^2 = \frac{1}{m+1}.$$

Thus, maximizing IQV is equivalent to minimizing

$$f(\boldsymbol{p}) := \sum_{i=0}^{m} p_i^2 \quad \text{subject to} \quad \sum_{i=0}^{m} p_i = 1.$$

Introducing the Lagrangian (with Lagrange multiplier λ)

$$\mathcal{L}(\boldsymbol{p}, \lambda) := \sum_{i=0}^{m} p_i^2 - \lambda \left(\sum_{i=0}^{m} p_i - 1 \right),$$

the first-order conditions are

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial p_j} &= 2p_j - \lambda = 0 \\ &\Rightarrow \quad 2p_j = \lambda, \qquad j = 0, \ldots, m, \end{aligned}$$

together with the constraint $\sum_{i=0}^{m} p_i = 1$. Since the same multiplier λ applies to all components, it follows that

$$p_0 = p_1 = \cdots = p_m.$$

Imposing the constraint yields

$$(m+1)p_j = 1 \quad \Longrightarrow \quad p_j = \frac{1}{m+1} \quad \forall j \in 0, \ldots, m.$$

Hence, $f(\boldsymbol{p})$ is uniquely minimized at

$$\boldsymbol{p} = \left(\tfrac{1}{m+1}, \ldots, \tfrac{1}{m+1} \right)^{\top},$$

and therefore IQV attains its maximal value at the uniform distribution.

$\square$

1.7. Measures of serial dependence

The marginal distribution of a categorical time series captures the long-run frequencies of the individual categories, but contains no information about the temporal structure of the process. To describe how the process evolves over time, measures of serial dependence are required.

For real-valued processes, the autocorrelation function (ACF) is the standard tool for quantifying temporal dependence (Brockwell and Davis, 2016, see Section 1.4). However, the ACF is not directly applicable to categorical data, since it relies on the existence of meaningful arithmetic operations on the observed values. For nominal data, such operations are undefined, and for ordinal data they are questionable, as the categories carry no numerical meaning beyond their ordering.

Dependence measures for categorical time series are typically constructed from the bivariate lag-h probabilities introduced in Definition 1.3. Under serial independence, these would factorize as $f_{ij}(h) = f_i f_j$ and $p_{ij}(h) = p_i p_j$ respectively. Measures for serial dependence are naturally constructed by quantifying the deviation of the bivariate probabilities from this independence benchmark.

Natural and widely used measures of serial dependence are the ordinal and nominal versions of Cohen's κ (Weiß, 2018, p. 130), which compare the joint probabilities to those expected under independence. For lag h, they are defined as

Definition 1.5 (Cohen's κ)**.**

$$\kappa_{\mathrm{ord}}(h) = \frac{\sum_{j=0}^{m-1}\Big(f_{jj}(h) - f_j^2\Big)}{\sum_{i=0}^{m-1} f_i(1-f_i)} \quad \text{and} \quad \kappa_{\mathrm{nom}}(h) = \frac{\sum_{j=0}^{m}\Big(p_{jj}(h) - p_j^2\Big)}{1-\sum_{i=0}^{m} p_i^2}.$$

The numerator of each measure sums the diagonal deviations $f_{jj}(h) - f_j^2$ (resp. $p_{jj}(h) - p_j^2$), which capture the excess probability of observing the same category at times t and $t+h$ relative to what would be expected under independence. The denominator normalizes the numerator by a measure of marginal dispersion: for the ordinal version it equals $\frac{m}{4}\cdot\mathrm{IOV}$, and for the nominal version it is equal to $\frac{m}{m+1}\cdot\mathrm{IQV}$. This ensures that $\kappa(h)$ is invariant to the overall level of dispersion in the marginal distribution. For both measures, positive values indicate that the process tends to remain in the same category across lag h, while negative values indicate a tendency to switch categories.
Under perfect positive dependence, i.e. $P(X_{t+h} = s_i \mid X_t = s_i) = 1$ for all $s_i \in S$, we have $f_{jj}(h) = f_j$ and $p_{jj}(h) = p_j$, and both measures attain their upper bound:

$$\kappa_{\mathrm{ord}}(h) = \frac{\sum_{j=0}^{m-1}(f_j - f_j^2)}{\sum_{i=0}^{m-1}(f_i - f_i^2)} = 1,$$

$$\kappa_{\mathrm{nom}}(h) = \frac{\sum_{j=0}^{m}(p_j - p_j^2)}{1-\sum_{i=0}^{m} p_i^2} = \frac{1-\sum_{j=0}^{m} p_j^2}{1-\sum_{i=0}^{m} p_i^2} = 1,$$

where we used that $\sum_{j=0}^{m} p_j = 1$, so that $\sum_{j=0}^{m}(p_j - p_j^2) = 1 - \sum_{j=0}^{m} p_j^2$. Under serial independence, $f_{jj}(h) = f_j^2$ and $p_{jj}(h) = p_j p_j$, so that the numerator vanishes and $\kappa(h) = 0$.
For the nominal version, the lower bound is attained when $p_{jj}(h) = 0$ for all j, i.e. when the process never remains in the same category across lag h:

$$\kappa_{\mathrm{nom}}(h) = \frac{-\sum_{j=0}^{m} p_j^2}{1-\sum_{i=0}^{m} p_i^2},$$

which depends on the marginal distribution and equals -1 only in the special case of a uniform marginal. For the ordinal version, the lower bound takes an analogous form but involves the cumulative probabilities f_j.

2. Samples Measures

2.1. Estimation of Marginal Distributions

To estimate the marginal (cumulative) probabilities, we express the relative frequencies through a binarization of the original process (X_t). For ordinal data, define the indicator vector

$$\boldsymbol{Z}_t := (Z_{t,0}, \dots, Z_{t,m-1})^\top, \qquad Z_{t,i} := \mathbb{I}\{X_t \leq s_i\},$$

and for nominal data analogously

$$\boldsymbol{Y}_t := (Y_{t,0}, \dots, Y_{t,m})^\top, \qquad Y_{t,i} := \mathbb{I}\{X_t = s_i\}.$$

Here, $\mathbb{I}\{A\}$ denotes the indicator function, which equals 1 if the event A is true and 0 otherwise. The marginal CDF and PMF vectors are estimated by the corresponding sample means:

Definition 2.1 (Empirical CDF and PMF estimators)**.**

$$\hat{\boldsymbol{f}} := \frac{1}{n} \sum_{t=1}^{n} \boldsymbol{Z}_t, \qquad \hat{\boldsymbol{p}} := \frac{1}{n} \sum_{t=1}^{n} \boldsymbol{Y}_t.$$

2.2. Asymptotics of the marginal estimator

To derive the asymptotic distribution of $\hat{\boldsymbol{f}}$ and $\hat{\boldsymbol{p}}$, a central limit theorem for dependent data is required. Since independence is not assumed, we impose weak dependence conditions on the original process (X_t).
Specifically, $(X_t)_{t\in\mathbb{Z}}$ is assumed to be strictly stationary (recall Definition 1.2) and α-mixing with exponentially decaying coefficients.

Definition 2.2 (α-Mixing with Exponential Decay)**.** Following Bradley (2005, Section 1.2.), let

$$\mathcal{F}_J^L := \sigma(X_t : J \leq t \leq L)$$

denote the σ-field generated by $\{X_t : J \leq t \leq L\}$. The α-mixing coefficients of (X_t), given that it's strictly stationary, are defined as

$$\alpha_{X_t}(n) := \sup_{A\in\mathcal{F}_{-\infty}^0,\, B\in\mathcal{F}_n^\infty} \Big| P(A\cap B) - P(A)P(B) \Big|.$$

The process is called α-mixing if $\alpha(n) \to 0$ as $n \to \infty$.
We additionally assume exponential decay of the mixing coefficients, i.e., there exist constants $C > 0$ and $\rho \in (0,1)$ such that

$$\alpha(n) \le C\rho^n \qquad \text{for all } n \in \mathbb{N}. \tag{2.1}$$

The indicator mappings

$$g_i(X_t) := \mathbb{I}\{X_t \le s_i\}, \qquad h_i(X_t) := \mathbb{I}\{X_t = s_i\},$$

are measurable functions of X_t (Klenke, 2005, p. 36). Since measurability preserves strict stationarity, the binarized processes $(g_i(X_t))$ and $(h_i(X_t))$ are strictly stationary whenever (X_t) is strictly stationary (Kallenberg, 2021, p. 157). Consequently, the vector-valued processes $\boldsymbol{Z}_t$ and $\boldsymbol{Y}_t$ are strictly stationary as well.
Moreover, each indicator variable $Z_{t,i}$ is defined on the same probability space as X_t, and the σ-field it generates satisfies

$$\sigma(Z_{t,i} : J \le t \le L) \subset \sigma(X_t : J \le t \le L), \qquad J \le L,$$

which implies $\alpha_{Z_{t,i}}(n) \le \alpha_{X_t}(n)$ for all i and n. Hence, $\boldsymbol{Z}_t$ and $\boldsymbol{Y}_t$ are also α-mixing with exponentially decaying coefficients.
To derive the asymptotic distribution of $\hat{\boldsymbol{f}}$, we apply the Cramér–Wold theorem (Kallenberg, 2021, Corollary 4.5). Specifically, $\hat{\boldsymbol{f}}$ converges in distribution to a multivariate normal vector if and only if the univariate central limit theorem holds for all linear combinations of $\boldsymbol{Z}_t$ with nonzero coefficients, that is,

$$U_t := \boldsymbol{a}^\top \boldsymbol{Z}_t, \qquad \boldsymbol{a} \in \mathbb{R}^m \setminus \{\boldsymbol{0}\}.$$

To see that U_t is measurable, let

$$L : \mathbb{R}^{m+1} \to \mathbb{R}, \quad L(\boldsymbol{z}) = \boldsymbol{a}^\top \boldsymbol{z}.$$

Since L is linear on a finite-dimensional space, it is continuous and hence Borel-measurable (Klenke, 2005, Theorem 1.88). The measurability of $g(X_t) = \boldsymbol{Z}_t$ follows from the measurability of its components, and the composition $U_t = L \circ g \circ X_t$ is measurable by Klenke (2005, Theorem 1.80). Hence U_t inherits strict stationarity and α-mixing with $\alpha_{U_t}(n) \le \alpha_{X_t}(n) \le C\rho^n$ from (X_t). We apply the CLT of Ibragimov (1962, Theorem 7), which requires four conditions:

(i) **Strict stationarity.** Verified above, as U_t is a measurable function of the strictly stationary process (X_t).

(ii) **α-mixing.** Verified above, with $\alpha_{U_t}(n) \le C\rho^n$.

(iii) **Summability.** The mixing coefficients are required to satisfy

$$\sum_{n=1}^{\infty} \alpha(n)^{\lambda/(2+\lambda)} < \infty \quad \text{for some } \lambda > 0.$$

Under exponential decay $\alpha(n) \le C\rho^n$, this follows from

$$\sum_{n=1}^{\infty} \alpha(n)^{\lambda/(2+\lambda)} \le C^{\lambda/(2+\lambda)} \sum_{n=1}^{\infty} \rho^{n\lambda/(2+\lambda)}$$
$$= C^{\lambda/(2+\lambda)} \frac{\rho^{\lambda/(2+\lambda)}}{1-\rho^{\lambda/(2+\lambda)}} < \infty,$$

since $0 < \rho^{\lambda/(2+\lambda)} < 1$ for all $\lambda > 0$.

(iv) **Moment condition.** The theorem requires $\mathrm{E}|U_t|^{2+\lambda} < \infty$ for some $\lambda > 0$. Since each $Z_{t,i} \in \{0,1\}$, the projection U_t is bounded by $M := \sum_{i=0}^{m-1} |a_i| < \infty$, and hence $\mathrm{E}|U_t|^{2+\lambda} \le M^{2+\lambda} < \infty$ for all $\lambda > 0$.

All four conditions being satisfied, Ibragimov's CLT applies to (U_t), and by the Cramér–Wold device, the asymptotic variance matrix $\boldsymbol{\Sigma}_Z = (\sigma_{Z;i,j})$ is determined by

$$\boldsymbol{a}^\top \boldsymbol{\Sigma}_Z \boldsymbol{a} = \mathrm{Var}(U_t) + 2\sum_{h=1}^{\infty} \mathrm{Cov}(U_t, U_{t+h}),$$

which gives

$$\sigma_{Z;i,j} = \mathrm{Cov}(Z_{t,i}, Z_{t,j}) + \sum_{h=1}^{\infty} \Big[\mathrm{Cov}(Z_{t,i}, Z_{t+h,j}) + \mathrm{Cov}(Z_{t,j}, Z_{t+h,i})\Big].$$

Using $\mathrm{Cov}(Z_{t,i}, Z_{t,j}) = f_{\min\{i,j\}} - f_i f_j$ and $\mathrm{Cov}(Z_{t,i}, Z_{t+h,j}) = f_{ij}(h) - f_i f_j$, the elements of $\boldsymbol{\Sigma}_Z$ take the explicit form

$$\sigma_{Z;i,j} = \Big(f_{\min\{i,j\}} - f_i f_j\Big) + \sum_{h=1}^{\infty} \Big(f_{ij}(h) + f_{ji}(h) - 2 f_i f_j\Big).$$

Theorem 2.3 (Asymptotic Normality of $\hat{\boldsymbol{f}}$)**.** *Let (X_t) be strictly stationary and α-mixing with exponentially decaying coefficients. Then*

$$\sqrt{n}\Big(\hat{\boldsymbol{f}} - \boldsymbol{f}\Big) \xrightarrow[n\to\infty]{d} \mathcal{N}\Big(\mathbf{0},\, \boldsymbol{\Sigma}_Z\Big),$$

where $\boldsymbol{\Sigma}_Z = (\sigma_{Z;i,j})_{i,j=0,\ldots,m-1}$ is given by

$$\sigma_{Z;i,j} = \Big(f_{\min\{i,j\}} - f_i f_j\Big) + \sum_{h=1}^{\infty} \Big(f_{ij}(h) + f_{ji}(h) - 2 f_i f_j\Big). \tag{2.2}$$

2.3. Asymptotic Properties of Sample Dispersion Measures

IOV

We can now use this information of the asymptotic distribution of $\hat{\boldsymbol{f}}$ to compute the asymptotic properties of sample measures that are based on $\hat{\boldsymbol{f}}$. According to the multivariate delta method (van der Vaart, 1998, Chapter 3) a random variable $G^* := g^*(\boldsymbol{\Theta}^*)$, where $\mathbf{\Theta}^* \in \mathbb{R}^m$, $\sqrt{n}(\mathbf{\Theta}^* - \mathbf{\Theta}) \overset{asy.}{\sim} \mathcal{N}(0, \mathbf{\Sigma}^*)$ and $g^* : \mathbb{R}^m \to \mathbb{R}$ is a continuously differentiable function, satisfies

$$\sqrt{n}(G^* - g^*(\mathbf{\Theta})) \overset{asy.}{\sim} \mathcal{N}(0, \boldsymbol{j}\mathbf{\Sigma}^*\boldsymbol{j}^\top)$$

where $\boldsymbol{j}$ is the Jacobian and where $j_i = \frac{\partial g^*(\mathbf{\Theta})}{\partial \theta_i}$.

Since we have shown that $\hat{\boldsymbol{f}}$ is asymptotically normal we can use this to derive the asymptotic properties of sample measures like the IOV. The respective Jacobian is given by:

$$\boldsymbol{j}_{\text{IOV}} = \left(\frac{\partial \text{IOV}(\boldsymbol{f})}{\partial f_0}, \ldots, \frac{\partial \text{IOV}(\boldsymbol{f})}{\partial f_{m-1}}\right) = \frac{4}{m}\left((1-2f_0), \ldots, (1-2f_{m-1})\right)$$

which leads to an asymptotic variance of the IOV:

$$\begin{aligned}\sqrt{n}\text{Var}(\widehat{IOV(\boldsymbol{f})}) &\overset{asy.}{=} \boldsymbol{j}_{\text{IOV}}\mathbf{\Sigma}_Z\boldsymbol{j}_{\text{IOV}}^\top = \sum_{i,\ j=0}^{m-1} \frac{4(1-2f_i)}{m}\frac{4(1-2f_j)}{m}\sigma_{\boldsymbol{Z};i,j} \\ &= \frac{16}{m^2}\sum_{i,j=0}^{m-1}(1-2f_i)(1-2f_j)\sigma_{\boldsymbol{Z};i,j}\end{aligned}$$

However, since IOV is not a linear function, the first-order delta method only provides information about the asymptotic distribution and variance, not on the finite-sample bias:

$$(\widehat{\text{IOV}} \overset{p}{\to} \text{IOV}(\boldsymbol{f})) \not\Rightarrow (\text{E}[\widehat{\text{IOV}}] = \text{IOV}(\boldsymbol{f})).$$

For nonlinear transformations, the curvature induces a bias of order n^{-1}. Since $\text{IOV}(\boldsymbol{f}) = \frac{4}{m}\sum_{i=0}^{m-1} f_i(1-f_i)$ is quadratic in $\boldsymbol{f}$, all derivatives of order three and higher are equal to 0. Consequently, the second-order Taylor expansion is not merely an approximation but exact:

$$\text{IOV}(\hat{\boldsymbol{f}}) = \text{IOV}(\boldsymbol{f}) + \boldsymbol{j}_{\text{IOV}}(\hat{\boldsymbol{f}} - \boldsymbol{f}) + \frac{1}{2}(\hat{\boldsymbol{f}} - \boldsymbol{f})^\top \boldsymbol{H}_{\text{IOV}}(\boldsymbol{f})(\hat{\boldsymbol{f}} - \boldsymbol{f}),$$

with no remainder term. $\boldsymbol{H}_{\text{IOV}}(\boldsymbol{f})$ is the Hessian matrix such that

$$\boldsymbol{H}_{\text{IOV}}(\boldsymbol{f}) = \left(\frac{\partial^2 \text{IOV}(\boldsymbol{f})}{\partial f_i f_j}\right)_{i,j=0,\ldots,m-1} = \begin{cases} -\frac{8}{m} & \text{if } i = j \\ 0 & \text{else} \end{cases} = -\frac{8}{m}\boldsymbol{I}_m.$$

To take the expectation of the quadratic term, we use the following lemma:

Lemma 2.4. Let $\boldsymbol{u} \in \mathbb{R}^m$ be a random vector with $\text{E}[\boldsymbol{u}] = \mathbf{0}$ and let $\boldsymbol{A} \in \mathbb{R}^{m\times m}$ be deterministic. Then

$$\text{E}[\boldsymbol{u}^\top \boldsymbol{A}\boldsymbol{u}] = \text{tr}(\boldsymbol{A}\,\text{Cov}(\boldsymbol{u})).$$

Proof. Since $\boldsymbol{u}^\top\boldsymbol{A}\boldsymbol{u}$ is a scalar, it follows from the cyclic property of the trace (Banerjee and Roy, 2014, Theorem 1.5) that $\boldsymbol{u}^\top\boldsymbol{A}\boldsymbol{u} = \text{tr}(\boldsymbol{A}\boldsymbol{u}\boldsymbol{u}^\top)$. Because expectation and trace are both linear and $\text{E}[\boldsymbol{u}] = \mathbf{0}$, it follows that

$$\text{E}[\boldsymbol{u}^\top\boldsymbol{A}\boldsymbol{u}] = \text{tr}\Big(\boldsymbol{A}\,\text{E}[\boldsymbol{u}\boldsymbol{u}^\top]\Big) = \text{tr}(\boldsymbol{A}\,\text{Cov}(\boldsymbol{u})). \qquad \square$$

Since $\mathrm{E}(\boldsymbol{j}_{\mathrm{IOV}}(\hat{\boldsymbol{f}} - \boldsymbol{f})) = 0$, the finite-sample bias is given by

$$\begin{aligned}
\mathrm{Bias}(\mathrm{IOV}(\hat{\boldsymbol{f}})) &= \mathrm{E}\Big[\frac{1}{2}(\hat{\boldsymbol{f}} - \boldsymbol{f})^\top \boldsymbol{H}_{\mathrm{IOV}}(\boldsymbol{f})(\hat{\boldsymbol{f}} - \boldsymbol{f})\Big] && \text{(Lemma 2.4)} \\
&= \frac{1}{2}\,\mathrm{tr}\Big(\boldsymbol{H}_{\mathrm{IOV}}\,\mathrm{Cov}(\hat{\boldsymbol{f}})\Big) \\
&= \frac{1}{2}\,\mathrm{tr}\Big(-\frac{8}{m}\boldsymbol{I}_m \cdot \frac{1}{n}\boldsymbol{\Sigma}_Z\Big) \\
&= -\frac{4}{mn}\sum_{i=0}^{m-1} \sigma_{\boldsymbol{Z};\,i,i}.
\end{aligned}$$

This implies that the bias is strictly negative and only depends on the marginal variances $\sigma_{\boldsymbol{Z};i,i}$ of the individual estimators f_i, not on the covariances. The results are summarized in the following theorem:

Theorem 2.5 (Asymptotic Distribution of $\widehat{\mathrm{IOV}}$)**.** *Under the assumptions of Theorem 2.3:*

(i) ***Asymptotic variance:***

$$\sqrt{n}\mathrm{Var}(\widehat{\mathrm{IOV}}) \overset{asy.}{=} \frac{16}{m^2}\sum_{i,j=0}^{m-1}(1-2f_i)(1-2f_j)\,\sigma_{Z;i,j}. \tag{2.3}$$

(ii) ***Bias to order*** n^{-1}***:***

$$\mathrm{Bias}(\widehat{\mathrm{IOV}}) = -\frac{4}{m\,n}\sum_{i=0}^{m-1}\sigma_{Z;i,i}. \tag{2.4}$$

Skewness

Similarly, the asymptotic properties of other dispersion metrics such as the skew can be analyzed. A simple measure for the skew of an ordinal process is given by:

$$\mathrm{skew}(\boldsymbol{f}) = \frac{2}{m}\left(\sum_{i=0}^{m-1} f_i\right) - 1.$$

This measure captures asymmetry through the average level of the CDF. A symmetric distribution is taken as a baseline, and deviations from this baseline reflect asymmetry in the distribution. The measure is scaled to $[-1, 1]$. For a more detailed discussion of skewness measures for ordinal data, see Klein and Doll (2021). The three extreme cases illustrate its behavior clearly:

- **Left-skewed** (skew = 1): all probability mass is concentrated in the lowest category s_0, so $f_i = 1$ for all i and skew $= \frac{2}{m} \cdot m - 1 = 1$.

- **Symmetric** (skew $=0$): the distribution is symmetric around the midpoint, which implies $f_i + f_{m-1-i} = 1$ for all i. It follows that $\sum_{i=0}^{m-1} f_i = m/2$ and skew $= \frac{2}{m} \cdot \frac{m}{2} - 1 = 0$.
- **Right-skewed** (skew $=-1$): all probability mass is concentrated in the highest category s_m, so $f_i = 0$ for all $i < m$ and skew $= \frac{2}{m} \cdot 0 - 1 = -1$.

The Jacobian for the skew is given by

$$\boldsymbol{j}_{skew} = \left(\frac{\partial \text{ skew}(\boldsymbol{f})}{\partial f_0}, \dots, \frac{\partial \text{ skew}(\boldsymbol{f})}{\partial f_{m-1}}\right) = \left(\frac{2}{m}, \dots, \frac{2}{m}\right)$$

such that we end up with the following results:

Proposition 2.6 (Asymptotic Properties of $\widehat{\text{skew}}$)**.** *Under the assumptions of Theorem 2.3:*

$$\sqrt{n}\text{Var}(\widehat{\text{skew}}) \overset{asy.}{=} \frac{4}{m^2} \sum_{i,j=0}^{m-1} \sigma_{Z;ij}. \tag{2.5}$$

Since skew *is linear in* $\boldsymbol{f}$*, the Hessian vanishes and the estimator is exactly unbiased for all* n*.*

3. Introducing Missingness

3.1. Missingness Mechanisms

In many applied settings, categorical time series are not fully observed. Individual time points may be missing due to sensor failures, non-response in survey data, or dropout in longitudinal studies. Before modelling the statistical consequences of missingness, it is useful to distinguish between different mechanisms that may give rise to it, as these have fundamentally different implications for estimation.
The classical taxonomy of missing data mechanisms goes back to Rubin (1976) and Little and Rubin (2002), who distinguished between three types based on the dependence structure between the missingness indicator and the underlying data. Let $\boldsymbol{X}^* = (\boldsymbol{X}^{(o)}, \boldsymbol{X}^{(m)})$ denote the complete data, partitioned into observed and missing components, and let $\boldsymbol{O} = (O_1, \ldots, O_n)^\top$ be the vector of missingness indicators. A probability model for the missingness mechanism specifies the distribution of $\boldsymbol{O}$ conditional on $\boldsymbol{X}^*$. Following Diggle et al. (2002, Section 13), the three mechanisms are classified as follows.

Definition 3.1 (Missingness Mechanisms)**.**

1. **Missing Completely at Random (MCAR):** $\boldsymbol{O}$ is independent of both the observed and the missing data:

$$f(\boldsymbol{o} \mid \boldsymbol{x}^{(o)}, \boldsymbol{x}^{(m)}) = f(\boldsymbol{o}).$$

2. **Missing at Random (MAR):** $\boldsymbol{O}$ is independent of the missing data, given the observed data:

$$f(\boldsymbol{o} \mid \boldsymbol{x}^{(o)}, \boldsymbol{x}^{(m)}) = f(\boldsymbol{o} \mid \boldsymbol{x}^{(o)}).$$

3. **Informative/ Missing Not at Random (MNAR):** The distribution of $\boldsymbol{O}$ depends on the missing data $\boldsymbol{x}^{(m)}$, even after conditioning on the observed data $\boldsymbol{x}^{(o)}$.

Remark 3.2. While this classification was originally developed for multivariate cross-sectional data (Rubin, 1976), it extends naturally to the time series setting. In the sequential context considered here, the conditioning on observed data is more naturally expressed through the history $\mathcal{F}_{t-1} := \sigma(X_s, O_s : s < t)$, leading to the time series formulation of Definition 3.1: MCAR corresponds to $P(O_t = 1 \mid X_t, \mathcal{F}_{t-1}) = \pi$, MAR to $P(O_t = 1 \mid X_t, \mathcal{F}_{t-1}) = P(O_t = 1 \mid \mathcal{F}_{t-1})$, and MNAR to the case where this equality fails.

The assumption under which the subsequent results are derived is that (O_t) and (X_t) are *independent*, which corresponds to the MCAR case. This is the strongest of the three assumptions and implies in particular that the corrected estimator $\hat{\boldsymbol{f}}^*$ introduced in Section 3.2 is consistent. Under MAR or MNAR, the independence assumption fails and estimators may be biased and inconsistent — as illustrated later in Section 4.2.3.

3.2. Amplitude Modulation

Definition of the observation process

To model the missingness, the concept of **amplitude modulation** is employed. This approach models missingness through a binary stochastic process $(O_t)_t$ that runs concurrently with the main process $(X_t)_t$ and modulates its "amplitude" in a binary manner:

$$O_t \in \{0, 1\} \quad \forall t$$

where $O_t = 1$ indicates that X_t is fully observed (amplitude = 1), and $O_t = 0$ indicates that X_t is missing (amplitude = 0).
The amplitude modulating process O_t can be:

- **Deterministic**: For each t, $O_t = o_t$ where $o_t \in \{0, 1\}$ is a known, fixed value.
- **Stochastic**: O_t follows a probability distribution with $\pi_t := P(O_t = 1)$, possibly dependent on the history of X_t, O_t, or external factors.

Bias of the naive estimator

The amplitude modulation is applied to the binarization vector $\boldsymbol{Z}_t$ to obtain the observable process $(O_t\boldsymbol{Z}_t)$. We then define the naive estimator of $\boldsymbol{f}$ as:

$$\hat{\mathfrak{f}} := \frac{1}{n}\sum_{t=1}^{n} O_t\boldsymbol{Z}_t.$$

To assess the performance of this naive estimator we compute its expectation

$$\mathrm{E}[\hat{\mathfrak{f}}] = \frac{1}{n}\sum_{t=1}^{n} \mathrm{E}[O_t\boldsymbol{Z_t}] = \frac{1}{n}\sum_{t=1}^{n} \mathrm{E}[\boldsymbol{Z_t}\mathrm{E}[O_t|\boldsymbol{Z}_t]],$$

where the last equality follows from the law of iterated expectations.
The dependence structure between $(O_t)_t$ and $(\boldsymbol{Z}_t)_t$ (or equivalently $(X_t)_t$) is crucial. Under the assumption of independence (see Definition 3.1, part MCAR), the expression simplifies to

$$\frac{1}{n}\sum_{t=1}^{n} \mathrm{E}[\boldsymbol{Z_t}\mathrm{E}[O_t|\boldsymbol{Z}_t]] = \frac{1}{n}\sum_{t=1}^{n} \mathrm{E}[\boldsymbol{Z}_t]\mathrm{E}[O_t] = \left(\frac{1}{n}\sum_{t=1}^{n} \mathrm{E}[O_t]\right)\boldsymbol{f}$$

The missingness corrected estimator

Definition 3.3 (Corrected Estimator of $\boldsymbol{f}$)**.** Since $\mathrm{E}[\hat{\mathfrak{f}}]$ deviates from $\boldsymbol{f}$ by the factor $\frac{1}{n}\sum_{t=1}^{n}\mathrm{E}[O_t]$, the corrected estimator of $\boldsymbol{f}$ is defined as

$$\hat{\boldsymbol{f}}^* := \frac{\frac{1}{n}\sum_{t=1}^{n} O_t \boldsymbol{Z}_t}{\frac{1}{n}\sum_{t=1}^{n} O_t} =: \frac{\hat{\mathfrak{f}}}{\bar{O}}.$$

Under the assumption of the independence of (O_t) and (X_t), the stochastic properties of the corrected estimator $\hat{\boldsymbol{f}}^*$ can be studied. If (O_t) is deterministic, then $\mathrm{E}[O_t] = o_t \in \{0, 1\}$ for any given t. It follows that

$$\frac{1}{n}\sum_{t=1}^{n}\mathrm{E}[O_t] = \frac{1}{n}\sum_{t=1}^{n} O_t = \bar{O},$$

and therefore, here $\hat{\boldsymbol{f}}^* = \hat{\mathfrak{f}}/\bar{O}$ is a perfectly unbiased estimator of $\boldsymbol{f}$. However, if O_t is a stochastic process then $\mathrm{E}[O_t] = \pi_t \in (0; 1]$ and

$$\frac{1}{n}\sum_{t=1}^{n}\mathrm{E}[O_t] = \frac{1}{n}\sum_{t=1}^{n}\pi_t \neq \frac{1}{n}\sum_{t=1}^{n} O_t = \bar{O},$$

to where $\hat{\boldsymbol{f}}^*$ is generally biased.

3.3. Asymptotic Theory for the corrected estimator

The extended process - stationarity and α-mixing

To study this bias we assume (O_t) and (X_t) to be stationary and α-mixing with exponentially decaying coefficients (see Definition 2.2), with $\mathrm{E}[O_t] = \pi$, an autocovariance function $\gamma_O(h) := \mathrm{Cov}[O_h, O_0]$ and joint expectations $\pi(h) := \mathrm{E}[O_h O_0] = \gamma_O(h) + \pi^2$. To study the joint asymptotic distribution of the estimator $\hat{\boldsymbol{f}}^*$, we introduce the extended binary vectors in order to derive the joint asymptotic distribution of all sample averages entering its definition:

$$\boldsymbol{Z}_t^* := (O_t, O_t Z_{t,0}, \ldots, O_t Z_{t,m-1})^\top = O_t(1, \boldsymbol{Z}_t^\top)^\top,$$

where the elements of $\boldsymbol{Z}_t^*$ are numbered as $Z_{t,-1}^*, Z_{t,0}^*, \ldots, Z_{t,m-1}^*$. The expectation of $\boldsymbol{Z}_t^*$ is given by

$$\begin{aligned}
\mathrm{E}[\boldsymbol{Z}_t^*] &= (\mathrm{E}[O_t], \mathrm{E}[O_t Z_{t,0}], \ldots, \mathrm{E}[O_t Z_{t,m-1}])^\top \\
&= (\mathrm{E}[O_t], \mathrm{E}[O_t]\mathrm{E}[Z_{t,0}], \ldots, \mathrm{E}[O_t]\mathrm{E}[Z_{t,m-1}])^\top && \text{(Independence)} \\
&= (\pi, \pi f_0, \ldots, \pi f_{m-1})^\top \\
&= \pi(1, \boldsymbol{f}^\top)^\top =: \boldsymbol{\mu}_{\boldsymbol{Z}_t^*}.
\end{aligned}$$

Since the processes (O_t) and $(\boldsymbol{Z}_t)$ are independent and stationary, the joint process $((O_t, \boldsymbol{Z}_t)^\top)$ is also stationary. Moreover, if both processes are α-mixing, then the joint process is α-mixing. As $\boldsymbol{Z}_t^*$ is a measurable function of $(O_t, \boldsymbol{Z})^\top$ it follows that $\boldsymbol{Z}_t^*$ is stationary and α-mixing (with exponentially decaying coefficients).

Multivariate CLT

Proposition 3.4 (Asymptotic distribution of $\bar{\boldsymbol{Z}}_n^*$). *Let (X_t) and (O_t) be independent, strictly stationary and α-mixing with exponentially decaying coefficients, with* $\mathrm{E}[O_t] = \pi \in (0,1]$. *Define the extended process*

$$\boldsymbol{Z}_t^* := O_t(1, \boldsymbol{Z}_t^\top)^\top \in \mathbb{R}^{m+1},$$

with sample mean $\bar{\boldsymbol{Z}}_n^* := \frac{1}{n}\sum_{t=1}^n \boldsymbol{Z}_t^*$. Then

$$\sqrt{n}\Big(\bar{\boldsymbol{Z}}_n^* - \boldsymbol{\mu}_{\boldsymbol{Z}^*}\Big) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{\boldsymbol{Z}^*}),$$

where $\boldsymbol{\mu}_{\boldsymbol{Z}^*} = \pi(1, \boldsymbol{f}^\top)^\top$ and

$$\boldsymbol{\Sigma}_{\boldsymbol{Z}^*} = \mathrm{Var}(\boldsymbol{Z}_t^*) + 2\sum_{h=1}^{\infty} \mathrm{Cov}(\boldsymbol{Z}_t^*, \boldsymbol{Z}_{t+h}^*),$$

with elements $\sigma_{\boldsymbol{Z}^*;i,j}$ as given in cases (i)–(iii) *below.*

Proof. Since $U_t^* = \boldsymbol{a}^\top \boldsymbol{Z}_t^*$ is a measurable function of the strictly stationary and α-mixing process $\boldsymbol{Z}_t^*$, it inherits both properties, with $\alpha_{U_t^*}(n) \leq \alpha_{\boldsymbol{Z}_t^*}(n) \leq C\rho^n$ following from the inclusion $\sigma(U_t^*) \subset \sigma(\boldsymbol{Z}_t^*)$. The summability condition therefore holds by the same geometric series argument as in Theorem 2.3. Since each $Z_{t,i}^* \in \{0,1\}$, the projection U_t^* is bounded and all moments are finite. Ibragimov's CLT (Ibragimov, 1962, Theorem 7) therefore applies to U_t^*, and the Cramér–Wold theorem (Kallenberg, 2021, Corollary 4.5) yields the multivariate result. □

Remark 3.5 (Elements of $\boldsymbol{\Sigma}_{\boldsymbol{Z}^*}$). Note that since $O_t \in \{0,1\}$ with $\mathrm{P}(O_t = 1) = \pi$, it follows that $O_t \sim \mathrm{Bernoulli}(\pi)$, so that $\mathrm{E}[O_t] = \pi$, $\mathrm{Var}(O_t) = \pi(1-\pi)$, and $\mathrm{E}[O_t^2] = \pi$. Recall also that $\mathrm{E}[O_t O_{t+h}] = \pi(h) = \gamma_O(h) + \pi^2$ for all $h \in \mathbb{N}$. The individual elements of $\boldsymbol{\Sigma}_{\boldsymbol{Z}^*}$ are then given by

(i) $i = j = -1$:

$$\begin{aligned}&\mathrm{Var}(O_t) + 2\sum_{h=1}^{\infty}\mathrm{Cov}(O_t, O_{t+h})\\&= \pi(1-\pi) + 2\sum_{h=1}^{\infty}\gamma_O(h) =: \sigma_{\boldsymbol{Z}^*;-1,-1}\end{aligned}$$

(ii) $j > i = -1$:

$$\begin{aligned}&\mathrm{Cov}(O_t, O_t Z_{t,j}) + 2\sum_{h=1}^{\infty}\mathrm{Cov}(O_t, O_{t+h}Z_{t+h,j})\\&= f_j\, \sigma_{\boldsymbol{Z}^*;-1,-1}\end{aligned}$$

(iii) $i, j \geq 0$:

$$\begin{aligned}&\mathrm{Cov}(O_t Z_{t,i}, O_t Z_{t,j}) + \sum_{h=1}^{\infty}[\mathrm{Cov}(O_t Z_{t,i}, O_{t+h}Z_{t+h,j}) + \mathrm{Cov}(O_{t+h}Z_{t+h,i}, O_t Z_{t,j})]\\&= f_i f_j\, \sigma_{\boldsymbol{Z}^*;-1,-1} + \pi(f_{\min\{i,j\}} - f_i f_j) + \sum_{h=1}^{\infty}\pi(h)(f_{ij}(h) + f_{ji}(h) - 2f_i f_j)\end{aligned}$$

The detailed derivations of these expressions are provided in Appendix A.1.

To now derive the asymptotic properties of $\hat{\boldsymbol{f}}^*$ we define the vector-valued function $\boldsymbol{g} := [0,1]^{m+1} \to [0,1]^m$ as

$$g_j(x_{-1}, x_0, \ldots, x_{m-1}) = x_j / x_{-1}, \quad \text{for } j = 0\ldots, m-1,$$

such that

$$\begin{aligned}\boldsymbol{g}(\bar{\boldsymbol{Z}}_t^*) &= (g_0(\bar{\boldsymbol{Z}}_t^*), \ldots, g_{m-1}(\bar{\boldsymbol{Z}}_t^*))^\top = (\hat{f}_0 O_t/O_t, \ldots, \hat{f}_{m-1}O_t/O_t)^\top\\&= \hat{\boldsymbol{f}}^*\end{aligned}$$

and accordingly

$$g(\bar{\boldsymbol{\mu}}_{\boldsymbol{Z}^*}^*) = \boldsymbol{f}.$$

The Jacobian of $\boldsymbol{g}(\boldsymbol{x})$ evaluated at $\boldsymbol{x} = \boldsymbol{\mu}_{\boldsymbol{Z}^*}$ is given by

$$\boldsymbol{G}_{m\times(m+1)} = \begin{pmatrix} \frac{\partial g_0(\boldsymbol{\mu}_{\boldsymbol{Z}^*})}{\partial x_{-1}} & \ldots & \frac{\partial g_0(\boldsymbol{\mu}_{\boldsymbol{Z}^*})}{\partial x_{m-1}} \\ \vdots & \ldots & \vdots \\ \frac{\partial g_{m-1}(\boldsymbol{\mu}_{\boldsymbol{Z}^*})}{\partial x_{-1}} & \ldots & \frac{\partial g_{m-1}(\boldsymbol{\mu}_{\boldsymbol{Z}^*})}{\partial x_{m-1}} \end{pmatrix} = \begin{pmatrix} -\frac{f_0}{\pi} & \frac{1}{\pi} & 0 & \ldots & 0 \\ -\frac{f_1}{\pi} & 0 & \frac{1}{\pi} & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & 0 \\ -\frac{f_{m-1}}{\pi} & 0 & \ldots & 0 & \frac{1}{\pi} \end{pmatrix}$$

Then, following the delta method, we have the following result

Theorem 3.6 (Asymptotic Normality of $\hat{\boldsymbol{f}}^*$)**.** *Let (X_t) be a strictly stationary, ordinal process that is α-mixing with exponentially decaying coefficients. Let the modulating process (O_t) also be strictly stationary and α-mixing with exponentially decaying coefficients, with moments $\mathrm{E}[O_t] = \pi > 0$ and $\mathrm{E}[O_h O_0] = \pi(h) > 0$. Furthermore, let (O_t) and (X_t) be independent, where X_t is observed if $O_t = 1$. Then the corrected estimator from Definition 3.3 satisfies the following multivariate CLT:*

$$\sqrt{n}\Big(\hat{\boldsymbol{f}}^* - \boldsymbol{f}\Big) \xrightarrow[n\to\infty]{d} \mathcal{N}\Big(\mathbf{0}, \boldsymbol{\Sigma}_{\hat{\boldsymbol{f}}^*}\Big), \quad \textit{with } \boldsymbol{\Sigma}_{\hat{\boldsymbol{f}}^*} = (\sigma_{\hat{\boldsymbol{f}}^*;i,j})_{i,j=0,\dots,m-1}, \tag{3.1}$$

where

$$\sigma_{\hat{\boldsymbol{f}}^*;i,j} = \frac{1}{\pi}(f_{\min\{i,j\}} - f_i f_j) + \frac{1}{\pi^2}\sum_{h=1}^{\infty} \pi(h)(f_{ij}(h) + f_{ji}(h) - 2f_i f_j). \tag{3.2}$$

The structure of $\boldsymbol{\Sigma}_{\hat{\boldsymbol{f}}^*}$ can be computed as follows

$$\begin{aligned}
\boldsymbol{\Sigma}_{\hat{\boldsymbol{f}}^*} &= \boldsymbol{G}\boldsymbol{\Sigma}_{\boldsymbol{Z}^*}\boldsymbol{G}^\top \\
&= \begin{pmatrix} -\frac{f_0}{\pi} & \frac{1}{\pi} & 0 & \dots & 0 \\ -\frac{f_1}{\pi} & 0 & \frac{1}{\pi} & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & 0 \\ -\frac{f_{m-1}}{\pi} & 0 & \dots & 0 & \frac{1}{\pi} \end{pmatrix} \begin{pmatrix} \sigma_{\boldsymbol{Z}^*;-1,-1} & \dots & \sigma_{\boldsymbol{Z}^*;-1,m-1} \\ \vdots & \ddots & \vdots \\ \sigma_{\boldsymbol{Z}^*;m-1,-1} & \dots & \sigma_{\boldsymbol{Z}^*;m-1,m-1} \end{pmatrix} \boldsymbol{G}^\top \\
&= \begin{pmatrix} -\frac{f_0}{\pi}\sigma_{\boldsymbol{Z}^*;-1,-1} + \frac{1}{\pi}\sigma_{\boldsymbol{Z}^*;0,-1} & \dots & -\frac{f_0}{\pi}\sigma_{\boldsymbol{Z}^*;-1,m-1} + \frac{1}{\pi}\sigma_{\boldsymbol{Z}^*;0,m-1} \\ \vdots & \ddots & \vdots \\ -\frac{f_{m-1}}{\pi}\sigma_{\boldsymbol{Z}^*;-1,-1} + \frac{1}{\pi}\sigma_{\boldsymbol{Z}^*;m-1,-1} & \dots & -\frac{f_{m-1}}{\pi}\sigma_{\boldsymbol{Z}^*;-1,m-1} + \frac{1}{\pi}\sigma_{\boldsymbol{Z}^*;m-1,m-1} \end{pmatrix} \boldsymbol{G}^\top.
\end{aligned}$$

Such that the individual elements look like this:

$$\begin{aligned}
\sigma_{\hat{\boldsymbol{f}}^*;i,j} &= (-\frac{f_j}{\pi})(-\frac{f_i}{\pi}\sigma_{\boldsymbol{Z}^*;-1,-1} + \frac{1}{\pi}\sigma_{\boldsymbol{Z}^*;i,-1}) + \frac{1}{\pi}(-\frac{f_i}{\pi}\sigma_{\boldsymbol{Z}^*;-1,j} + \frac{1}{\pi}\sigma_{\boldsymbol{Z}^*;i,j}) \\
&= \frac{f_j f_i}{\pi^2}\sigma_{\boldsymbol{Z}^*;-1,-1} - \frac{f_j}{\pi^2}\sigma_{\boldsymbol{Z}^*;i,-1} - \frac{f_i}{\pi^2}\sigma_{\boldsymbol{Z}^*;-1,j} + \frac{1}{\pi^2}\sigma_{\boldsymbol{Z}^*;i,j} \\
&= \frac{1}{\pi^2}\left(f_i f_j \sigma_{\boldsymbol{Z}^*;-1,-1} - f_i f_j \sigma_{\boldsymbol{Z}^*;-1,-1} - f_i f_j \sigma_{\boldsymbol{Z}^*;-1,-1} + \sigma_{\boldsymbol{Z}^*;i,j}\right) = \frac{1}{\pi^2}(\sigma_{\boldsymbol{Z}^*;i,j} - f_i f_j \sigma_{\boldsymbol{Z}^*;-1,-1}) \\
&= \frac{1}{\pi^2}\left(\left[f_i f_j \sigma_{\boldsymbol{Z}^*;-1,-1} + \pi(f_{\min\{i,j\}} - f_i f_j) + \sum_{h=1}^{\infty}\pi(h) f_{ij}(h) + f_{ji}(h) - 2f_i f_j\right] - f_i f_j \sigma_{\boldsymbol{Z}^*;-1,-1}\right) \\
&= \underbrace{\frac{1}{\pi}(f_{\min\{i,j\}} - f_i f_j)}_{\mathrm{lag}(0)} + \frac{1}{\pi^2}\sum_{h=1}^{\infty}\underbrace{\pi(h)(f_{ij}(h) + f_{ji}(h) - 2f_i f_j)}_{\mathrm{lag}(h)}
\end{aligned}$$

Here, the lag-0 term captures the contemporaneous covariance structure of the observed process, while the lag-h terms accumulate the serial dependence across all lags $h \geq 1$.

Bias analysis

Since g_j is not a linear function of $\boldsymbol{Z}_t^*$, the bias of order $O(n^{-1})$ is computable using a second-order Taylor expansion. The Hessian of g_j evaluated in $\boldsymbol{\mu}_{\boldsymbol{Z}^*}$ is given by

$$\boldsymbol{H}_{g_j} = \left(\frac{\partial^2 g_j(\mu_{\boldsymbol{Z}^*})}{\partial x_i \partial x_k}\right)_{i,k=-1,\dots,m-1} = \begin{cases} \frac{2f_j}{\pi^2} & \text{if } i = k = -1 \\ -\frac{1}{\pi^2} & \text{if } i = -1,\ k = j \\ 0 & \text{else.} \end{cases}$$

Using the symmetry $\sigma_{\boldsymbol{Z}^*;-1,j} = \sigma_{\boldsymbol{Z}^*;j,-1}$ and the equality from Lemma 2.4 the bias simplifies to

$$\begin{aligned} \mathrm{Bias}(\hat{f}_j) &= \frac{1}{2}\mathrm{tr}(\boldsymbol{H}_{g_j}\boldsymbol{\Sigma}_{\boldsymbol{Z}^*}) \\ &= \frac{1}{2}h_{g_j;-1,-1}\,\sigma_{\boldsymbol{Z}^*;-1,-1} + h_{g_j;-1,j}\,\sigma_{\boldsymbol{Z}^*;-1,j} \\ &= \frac{1}{2}\cdot\frac{2f_j}{\pi^2}\,\sigma_{\boldsymbol{Z}^*;-1,-1} - \frac{1}{\pi^2}\cdot f_j\,\sigma_{\boldsymbol{Z}^*;-1,-1} \\ &= \frac{f_j}{\pi^2}\,\sigma_{\boldsymbol{Z}^*;-1,-1} - \frac{f_j}{\pi^2}\,\sigma_{\boldsymbol{Z}^*;-1,-1} = 0, \end{aligned}$$

where the last step uses $\sigma_{\boldsymbol{Z}^*;-1,j} = f_j\,\sigma_{\boldsymbol{Z}^*;-1,-1}$ from case **(ii)** of $\boldsymbol{\Sigma}_{\boldsymbol{Z}^*}$.

Special Cases

To further study the asymptotic distribution of $\hat{\boldsymbol{f}}^*$, we consider three special cases of the general result.

Corollary 3.7 (Full Observation)**.** *If the process is fully observed, i.e. $\pi = 1$, then*

$$\sigma_{\hat{\boldsymbol{f}}^*;i,j} = \left(f_{\min\{i,j\}} - f_i f_j\right) + \sum_{h=1}^{\infty} f_{ij}(h) + f_{ji}(h) - 2f_i f_j = \sigma_{\boldsymbol{Z};i,j}.$$

That is, the asymptotic covariance of $\hat{\boldsymbol{f}}^*$ reduces to that of the complete-data estimator $\hat{\boldsymbol{f}}$, as one would expect.

Corollary 3.8 (i.i.d. Observation Process)**.** *If (O_t) is i.i.d., then $\pi(h) = \pi^2$ for all $h \geq 1$, and the asymptotic covariance simplifies to*

$$\sigma_{\hat{\boldsymbol{f}}^*;i,j} = \frac{1}{\pi}\left(f_{\min\{i,j\}} - f_i f_j\right) + \sum_{h=1}^{\infty} f_{ij}(h) + f_{ji}(h) - 2f_i f_j.$$

Compared to the complete-data case, the lag-0 term is inflated by a factor of $1/\pi \geq 1$, reflecting the additional uncertainty introduced by the random missingness. The lag-h terms for $h \geq 1$, however, are unaffected, since i.i.d. missingness introduces no additional serial dependence into the observation process.

Corollary 3.9 (i.i.d. Main Process)**.** *If (X_t) is i.i.d., then $f_{ij}(h) = f_i f_j$ for all $i, j = 0, \dots, m-1$ and $h \geq 1$, so that the asymptotic covariance reduces to*

$$\sigma_{\hat{\boldsymbol{f}}^*;i,j} = \frac{1}{\pi}\Big(f_{\min\{i,j\}} - f_i f_j\Big) + \frac{1}{\pi^2}\sum_{h=1}^{\infty} \pi(h)\, \overbrace{\big(f_{ij}(h) + f_{ji}(h) - 2 f_i f_j\big)}^{=0} = \frac{1}{\pi}\Big(f_{\min\{i,j\}} - f_i f_j\Big).$$

When the main process is serially independent, all temporal dependence in $\hat{\boldsymbol{f}}^*$ vanishes and the asymptotic covariance only depends on the lag-0 term.

3.4. Asymptotic marginal properties

The asymptotic properties of marginal statistics based on $\hat{\boldsymbol{f}}^*$ follow directly from the delta method, in exact analogy to the complete-data case. The following results are for the IOV and the skew. However, the derivations for other marginal statistics are analogous.

Corollary 3.10 (Asymptotic distribution of $\mathrm{IOV}(\hat{\boldsymbol{f}}^*)$)**.** *Under the conditions of Theorem 3.6, the delta method argument from Theorem 2.5 carries over to the missing-data case with $\boldsymbol{\Sigma}_Z$ replaced by $\boldsymbol{\Sigma}_{\hat{\boldsymbol{f}}^*}$, yielding*

$$\sqrt{n}\mathrm{Var}\Big(\mathrm{IOV}(\hat{\boldsymbol{f}}^*)\Big) \overset{\text{asy.}}{=} \frac{16}{m^2}\sum_{i,j=0}^{m-1} (1-2f_i)(1-2f_j)\,\sigma_{\hat{\boldsymbol{f}}^*;i,j}.$$

Moreover, since $\boldsymbol{H}_{\mathrm{IOV}} = -\frac{8}{m}\boldsymbol{I}_m$ remains diagonal, the bias to order n^{-1} depends only on the marginal variances of $\hat{\boldsymbol{f}}^*$:

$$\mathrm{Bias}\Big(\mathrm{IOV}(\hat{\boldsymbol{f}}^*)\Big) = -\frac{4}{mn}\sum_{i=0}^{m-1} \sigma_{\hat{\boldsymbol{f}}^*;i,i}.$$

Remark 3.11 (Skew under missingness)**.** The analogous result for the skew follows immediately, since the skew is linear in $\boldsymbol{f}$ and hence $\boldsymbol{H}_{\text{skew}} = 0$. In particular, the estimator is unbiased, and its asymptotic variance is given by

$$\sqrt{n}\mathrm{Var}\Big(\mathrm{skew}(\hat{\boldsymbol{f}}^*)\Big) \overset{\text{asy.}}{=} \frac{4}{m^2}\sum_{i,j=0}^{m-1} \sigma_{\hat{\boldsymbol{f}}^*;i,j}.$$

3.5. Serial Dependence

So far, we have only examined aspects of the marginal distribution of the categorical process (X_t). However, an essential component of time series analysis is the study of serial dependence. Recall that the most widely used measures to describe such dependence are the ordinal and nominal Cohen's κ (Definition 1.5). To estimate these measures, we again rely on relative frequencies:

$$\hat{\kappa}_{\text{ord}}(h) = \frac{\sum_{j=0}^{m-1}\left(\hat{f}^*_{jj}(h) - \left(\hat{f}^*_j\right)^2\right)}{\sum_{i=0}^{m-1}\hat{f}^*_i\left(1-\hat{f}^*_i\right)}.$$

Hence, it remains to define the estimator $\hat{f}^*_{ij}(h)$ for the bivariate lag-h probabilities $f_{ij}(h)$. As before, we begin with the naive estimator given by the bivariate relative frequency of the amplitude-modulated binarizations,

$$\hat{\mathfrak{f}}_{ij}(h) = \frac{1}{n-h}\sum_{t=h+1}^{n} O_t\, O_{t-h}\, Z_{t,i}\, Z_{t-h,j},$$

and compute its expectation:

$$\begin{aligned}
\mathrm{E}\Big[\hat{\mathfrak{f}}_{ij}(h)\Big] &= \frac{1}{n-h}\sum_{t=h+1}^{n} \mathrm{E}[O_t O_{t-h} Z_{t,i} Z_{t-h,j}] \\
&= \frac{1}{n-h}\sum_{t=h+1}^{n} \mathrm{E}[O_t O_{t-h}]\cdot \mathrm{E}[Z_{t,i} Z_{t-h,j}] \\
&= \left(\frac{1}{n-h}\sum_{t=h+1}^{n} \mathrm{E}[O_t O_{t-h}]\right) f_{ij}(h),
\end{aligned}$$

where the second equality follows from the independence of (O_t) and (X_t), and the third from the stationarity of (X_t). This motivates the corrected estimator

$$\hat{f}^*_{ij}(h) = \frac{\dfrac{1}{n-h}\displaystyle\sum_{t=h+1}^{n} O_t\, O_{t-h}\, Z_{t,i}\, Z_{t-h,j}}{\dfrac{1}{n-h}\displaystyle\sum_{t=h+1}^{n} O_t\, O_{t-h}} =: \hat{\mathfrak{f}}_{ij}(h)\Big/\overline{OO}_h.$$

As in Section 3.2, the properties of this estimator depend on whether (O_t) is deterministic or stochastic. If (O_t) is deterministic, the corrected estimator is exactly unbiased. In the stochastic case, both numerator and denominator are random and depend on the same process (O_t), so that in general

$$\mathrm{E}\left[\frac{\hat{\mathfrak{f}}_{ij}(h)}{\overline{OO}_h}\right] \neq \frac{\mathrm{E}\Big[\hat{\mathfrak{f}}_{ij}(h)\Big]}{\mathrm{E}\Big[\overline{OO}_h\Big]},$$

and hence $\hat{f}^*_{ij}(h)$ is generally biased.

3.5.1. Asymptotic distribution under the general model

To derive the asymptotic distribution of $\hat{f}^*_{ij}(h)$, we proceed analogously to Section 3.2 and apply the Cramér–Wold device to the sample mean of a suitably defined binary vector:

$$\boldsymbol{Z}_t^{(h)*} = \Bigg(Z_{t,-2}^{(h)*},\ Z_{t,-1}^{(h)*},\ \overbrace{\ldots,\ Z_{t,i}^{(h)*},\ \ldots,}^{i=0,\ldots,m-1}\ \overbrace{\ldots,\ Z_{t,m+j}^{(h)*},\ \ldots}^{j=0,\ldots,m-1} \Bigg)^\top$$

$$:= \Bigg(O_t O_{t-h},\ \underbrace{O_t,\ \ldots,\ O_t Z_{t,i},\ \ldots,}_{=\boldsymbol{Z}_t^*}\ \ldots,\ O_t O_{t-h} Z_{t,j} Z_{t-h,j},\ \ldots \Bigg)^\top .$$

The components of $\boldsymbol{Z}_t^{(h)*}$ are indexed from -2 to $2m-1$: the component at index -2 captures the product $O_t O_{t-h}$, the components at indices -1 through $m-1$ coincide with $\boldsymbol{Z}_t^*$ from Section 3.2, and the remaining components at indices m through $2m-1$ capture the lag-h interaction terms $O_t O_{t-h} Z_{t,j} Z_{t-h,j}$. Since κ_{ord} depends only on the diagonal frequencies $\hat{f}^*_{jj}(h)$, it suffices to include the corresponding diagonal components in $\boldsymbol{Z}_t^{(h)*}$. Should one wish to study a dependence measure that also involves off-diagonal frequencies $\hat{f}^*_{ij}(h)$ with $i \neq j$, the vector can straightforwardly be extended by the additional components $O_t O_{t-h} Z_{t,i} Z_{t-h,j}$.

Proposition 3.12 (Asymptotic distribution of $\bar{\boldsymbol{Z}}_n^{(h)*}$)**.** *Let the assumptions of Theorem 3.6 hold. Then*

$$\sqrt{n}\Big(\bar{\boldsymbol{Z}}_n^{(h)*} - \boldsymbol{\mu}_{\boldsymbol{Z}^{(h)*}} \Big) \xrightarrow{d} \mathcal{N}(\boldsymbol{0},\ \boldsymbol{\Sigma}_{\boldsymbol{Z}^{(h)*}}),$$

where $\bar{\boldsymbol{Z}}_n^{(h)*} := \frac{1}{n} \sum_{t=1}^n \boldsymbol{Z}_t^{(h)*}$,

$$\boldsymbol{\mu}_{\boldsymbol{Z}^{(h)*}} = \Big(\pi(h),\ \pi,\ \pi \boldsymbol{f}^\top,\ \pi(h) f_{00}(h),\ \ldots,\ \pi(h) f_{m-1,m-1}(h) \Big)^\top,$$

and $\boldsymbol{\Sigma}_{\boldsymbol{Z}^{(h)*}}$ *has elements*

$$\sigma_{i,j}^{(h)} = \mathrm{Cov}(Z_{t,i}^{(h)*}, Z_{t,j}^{(h)*}) + \sum_{k=1}^{\infty} \Big[\mathrm{Cov}(Z_{t,i}^{(h)*}, Z_{t+k,j}^{(h)*}) + \mathrm{Cov}(Z_{t+k,i}^{(h)*}, Z_{t,j}^{(h)*}) \Big].$$

Proof. Since $\boldsymbol{Z}_t^{(h)*}$ is a measurable function of the bounded indicator variables $(O_t, O_{t-h}, X_t, X_{t-h})$, it inherits strict stationarity and α-mixing with exponentially decaying coefficients from the underlying processes by the same argument as in Proposition 3.4, and all moment conditions are trivially satisfied. The result then follows from the Cramér–Wold theorem (Kallenberg, 2021, Corollary 4.5) and Ibragimov's CLT (Ibragimov, 1962, Theorem 7). □

Using the independence of (O_t) and (X_t), the elements $\sigma_{i,j}^{(h)}$ can be computed explicitly. We record the following notation: since $O_t \sim \text{Bernoulli}(\pi)$, we have $\mathrm{E}[O_t] = \pi$, $\mathrm{Var}(O_t) = \pi(1-\pi)$, $\mathrm{E}[O_t^2] = \pi$, and we write $\pi(h) := \mathrm{E}[O_t O_{t-h}]$ and $\gamma_O(h) := \mathrm{Cov}(O_t, O_{t-h}) = \pi(h) - \pi^2$ for all $h \geq 1$. The elements of $\mathbf{\Sigma}_{\boldsymbol{Z}^{(h)*}}$ are then given as follows:

(i) $i = j = -2:$

$$\begin{aligned}
&\mathrm{Var}(O_t O_{t-h}) + 2\sum_{k=1}^{\infty} \mathrm{Cov}(O_t O_{t-h}, O_{t+k} O_{t+k-h}) \\
&= \pi(h)(1-\pi(h)) + 2\sum_{k=1}^{\infty} (\mathrm{E}[O_t O_{t-h} O_{t+k} O_{t+k-h}] - \pi(h)^2)
\end{aligned}$$

(ii) $i = -2,\ j = -1:$

$$\begin{aligned}
&\mathrm{Cov}(O_t O_{t-h}, O_t) + \sum_{k=1}^{\infty} [\mathrm{Cov}(O_t O_{t-h}, O_{t+k}) + \mathrm{Cov}(O_{t+k} O_{t+k-h}, O_t)] \\
&= (1-\pi)\pi(h) + \sum_{k=1}^{\infty} [\mathrm{E}[O_t O_{t-h} O_{t+k}] + \mathrm{E}[O_{t+k} O_{t+k-h} O_t] - 2\pi\pi(h)] =: \sigma_{-2,-1}^{(h)}
\end{aligned}$$

(iii) $i = -2,\ j = 0, \ldots, m-1:$

$$\begin{aligned}
&\mathrm{Cov}(O_t O_{t-h}, O_t Z_{t,j}) + \sum_{k=1}^{\infty} [\mathrm{Cov}(O_t O_{t-h}, O_{t+k} Z_{t+k,j}) + \mathrm{Cov}(O_{t+k} O_{t+k-h}, O_t Z_{t,j})] \\
&\overset{\perp}{=} \mathrm{E}[O_t^2 O_{t-h}]\,\mathrm{E}[Z_{t,j}] - \pi(h)\,\pi\,\mathrm{E}[Z_{t,j}] \\
&\quad + \sum_{k=1}^{\infty} [\mathrm{E}[O_t O_{t-h} O_{t+k}]\,\mathrm{E}[Z_{t+k,j}] + \mathrm{E}[O_{t+k} O_{t+k-h} O_t]\,\mathrm{E}[Z_{t,j}] - 2\pi\pi(h)\,\mathrm{E}[Z_{t,j}]] \\
&= f_j \Big((1-\pi)\pi(h) + \sum_{k=1}^{\infty} [\mathrm{E}[O_t O_{t-h} O_{t+k}] + \mathrm{E}[O_t O_{t+k} O_{t+k-h}] - 2\pi\pi(h)] \Big) = f_j\, \sigma_{-2,-1}^{(h)}
\end{aligned}$$

(iv) $i = -2,\ j = m, \ldots, 2m-1:$

$$\begin{aligned}
&\mathrm{Cov}(O_t O_{t-h},\ O_t O_{t-h} Z_{t,(j-m)} Z_{t-h,(j-m)}) \\
&\quad + \sum_{k=1}^{\infty} [\mathrm{Cov}(O_t O_{t-h},\ O_{t+k} O_{t+k-h} Z_{t+k,(j-m)} Z_{t+k-h,(j-m)}) \\
&\quad + \mathrm{Cov}(O_{t+k} O_{t+k-h},\ O_t O_{t-h} Z_{t,(j-m)} Z_{t-h,(j-m)})] \\
&\overset{\perp}{=} \mathrm{E}[O_t^2 O_{t-h}^2]\,\mathrm{E}[Z_{t,(j-m)} Z_{t-h,(j-m)}] - \pi(h)^2\, f_{(j-m),(j-m)}(h) \\
&\quad + \sum_{k=1}^{\infty} \mathrm{E}[O_t O_{t-h} O_{t+k} O_{t+k-h}]\,[\mathrm{E}[Z_{t+k,(j-m)} Z_{t+k-h,(j-m)}] + \mathrm{E}[Z_{t,(j-m)} Z_{t-h,(j-m)}]] - 2\pi(h)^2\, f_{(j-m),(j-m)}(h \\
&= f_{(j-m),(j-m)}(h) \Big(\pi(h)(1-\pi(h)) + 2\sum_{k=1}^{\infty} [\mathrm{E}[O_t O_{t-h} O_{t+k} O_{t+k-h}] - \pi(h)^2] \Big) =: f_{(j-m),(j-m)}(h)\, \sigma_{-2,-2}^{(h)}
\end{aligned}$$

(v) $i = j = -1:$

$$\mathrm{Var}(O_t) + 2\sum_{k=1}^{\infty} \mathrm{Cov}(O_t, O_{t+k}) = \pi(1-\pi) + 2\sum_{k=1}^{\infty} \gamma_O(k) = \sigma_{\boldsymbol{Z}^*;\,-1,-1}$$

(vi) $i = -1,\ j = 0, \dots, m-1:$

$$\mathrm{Cov}(O_t, O_t Z_{t,j}) + \sum_{k=1}^{\infty} \big[\mathrm{Cov}(O_t, O_{t+k} Z_{t+k,j}) + \mathrm{Cov}(O_{t+k}, O_t Z_{t,j})\big] = f_j\, \sigma_{\boldsymbol{Z}^*;\,-1,-1}$$

(vii) $i = -1,\ j = m, \dots, 2m-1:$

$$\begin{aligned}
&\mathrm{Cov}(O_t,\ O_t O_{t-h} Z_{t,(j-m)} Z_{t-h,(j-m)}) \\
&\quad + \sum_{k=1}^{\infty} \big[\mathrm{Cov}(O_t,\ O_{t+k} O_{t+k-h} Z_{t+k,(j-m)} Z_{t+k-h,(j-m)}) + \mathrm{Cov}(O_{t+k},\ O_t O_{t-h} Z_{t,(j-m)} Z_{t-h,(j-m)})\big] \\
&\overset{\perp}{=} f_{(j-m),(j-m)}(h) \Big((1-\pi)\pi(h) + \sum_{k=1}^{\infty} \big[\mathrm{E}[O_t O_{t+k} O_{t+k-h}] + \mathrm{E}[O_t O_{t+k} O_{t-h}] - 2\pi\pi(h)\big] \Big) \\
&= f_{(j-m),(j-m)}(h)\, \sigma^{(h)}_{-2,-1}
\end{aligned}$$

(viii) $i, j = 0, \dots, m-1:$

$$\begin{aligned}
&\mathrm{Cov}(O_t Z_{t,i}, O_t Z_{t,j}) + \sum_{k=1}^{\infty} \big[\mathrm{Cov}(O_t Z_{t,i}, O_{t+k} Z_{t+k,j}) + \mathrm{Cov}(O_{t+k} Z_{t+k,i}, O_t Z_{t,j})\big] \\
&= \sigma_{\boldsymbol{Z}^*;\,i,j} = f_i f_j\, \sigma_{\boldsymbol{Z}^*;\,-1,-1} + \pi(f_{\min\{i,j\}} - f_i f_j) + \sum_{k=1}^{\infty} \pi(k)\big(f_{ij}(k) + f_{ji}(k) - 2 f_i f_j\big)
\end{aligned}$$

(ix) $i = 0, \dots, m-1,\ j = m, \dots, 2m-1:$

$$\begin{aligned}
&\mathrm{Cov}(O_t Z_{t,i},\ O_t O_{t-h} Z_{t,(j-m)} Z_{t-h,(j-m)}) \\
&\quad + \sum_{k=1}^{\infty} \big[\mathrm{Cov}(O_t Z_{t,i},\ O_{t+k} O_{t+k-h} Z_{t+k,(j-m)} Z_{t+k-h,(j-m)}) \\
&\quad + \mathrm{Cov}(O_{t+k} Z_{t+k,i},\ O_t O_{t-h} Z_{t,(j-m)} Z_{t-h,(j-m)})\big] \\
&\overset{\perp}{=} \pi(h)\, \mathrm{E}[Z_{t,i} Z_{t,(j-m)} Z_{t-h,(j-m)}] - \pi\pi(h)\, f_i\, f_{(j-m),(j-m)}(h) \\
&\quad + \sum_{k=1}^{\infty} \big[\mathrm{E}[O_t O_{t+k} O_{t+k-h}]\, \mathrm{E}[Z_{t,i} Z_{t+k,(j-m)} Z_{t+k-h,(j-m)}] \\
&\quad + \mathrm{E}[O_{t+k} O_t O_{t-h}]\, \mathrm{E}[Z_{t+k,i} Z_{t,(j-m)} Z_{t-h,(j-m)}] - 2\pi\pi(h)\, f_i\, f_{(j-m),(j-m)}(h)\big]
\end{aligned}$$

(x) $i, j = m, \dots, 2m-1:$

$$\begin{aligned}
&\mathrm{Cov}(O_t O_{t-h} Z_{t,(i-m)} Z_{t-h,(i-m)},\ O_t O_{t-h} Z_{t,(j-m)} Z_{t-h,(j-m)}) \\
&\quad + \sum_{k=1}^{\infty} \big[\mathrm{Cov}(O_t O_{t-h} Z_{t,(i-m)} Z_{t-h,(i-m)},\ O_{t+k} O_{t+k-h} Z_{t+k,(j-m)} Z_{t+k-h,(j-m)}) \\
&\quad + \mathrm{Cov}(O_{t+k} O_{t+k-h} Z_{t+k,(i-m)} Z_{t+k-h,(i-m)},\ O_t O_{t-h} Z_{t,(j-m)} Z_{t-h,(j-m)})\big] \\
&\overset{\perp}{=} \pi(h)\, \mathrm{E}[Z_{t,(i-m)} Z_{t-h,(i-m)} Z_{t,(j-m)} Z_{t-h,(j-m)}] - \pi(h)^2\, f_{(i-m),(i-m)}(h)\, f_{(j-m),(j-m)}(h) \\
&\quad + \sum_{k=1}^{\infty} \mathrm{E}[O_t O_{t-h} O_{t+k} O_{t+k-h}]\, \big[\mathrm{E}[Z_{t,(i-m)} Z_{t-h,(i-m)} Z_{t+k,(j-m)} Z_{t+k-h,(j-m)}] \\
&\quad + \mathrm{E}[Z_{t+k,(i-m)} Z_{t+k-h,(i-m)} Z_{t,(j-m)} Z_{t-h,(j-m)}] - 2 f_{(i-m),(i-m)}(h)\, f_{(j-m),(j-m)}(h)\big]
\end{aligned}$$

Note that cases **(v)**, **(vi)** and **(viii)** coincide with cases **(i)**-**(iii)** from $\boldsymbol{\Sigma}_{\boldsymbol{Z}^*}$ of Proposition 3.4. Also note that cases **(i)–(viii)** admit fully explicit closed-form expressions, whereas cases **(ix)** and **(x)** involve third- and fourth-order mixed moments of (X_t) — such as $\mathrm{E}[Z_{t,i}Z_{t,(j-m)}Z_{t-h,(j-m)}]$ — that do not simplify further without additional assumptions on the dependence structure of (X_t).

To obtain the joint asymptotic distribution of $\hat{f}_i^*$ and $\hat{f}_{jj}^*(h)$ for all $i, j = 0, \ldots, m-1$, we define the vector-valued function $\boldsymbol{g}\colon [0,1]^{2(m+1)} \to [0,1]^{2m}$ by

$$g_j(\boldsymbol{x}) = \begin{cases} x_j/x_{-1} & \text{for } j = 0, \ldots, m-1, \\ x_j/x_{-2} & \text{for } j = m, \ldots, 2m-1, \end{cases}$$

so that $\hat{\boldsymbol{f}}^{(h)*} := \boldsymbol{g}(\bar{\boldsymbol{Z}}_n^{(h)*})$ and $\boldsymbol{\mu}^{(h)*} := \boldsymbol{g}(\boldsymbol{\mu}_{\boldsymbol{Z}^{(h)*}}) = (f_0, \ldots, f_{m-1}, f_{00}(h), \ldots, f_{m-1,m-1}(h))^\top$. Since $\boldsymbol{g}$ is continuous at $\boldsymbol{\mu}_{\boldsymbol{Z}^{(h)*}}$, consistency of $\hat{\boldsymbol{f}}^{(h)*}$ follows immediately from Theorem 3.12 and the continuous mapping theorem (Kallenberg, 2021, Theorem 3.27). Asymptotic normality then follows by the multivariate delta method (van der Vaart, 1998, Chapter 3):

Theorem 3.13 (Asymptotic normality of $\hat{\boldsymbol{f}}^{(h)*}$)**.** *Let the assumptions of Theorem 3.6 hold. Then $\hat{\boldsymbol{f}}^{(h)*}$ is consistent for $\boldsymbol{\mu}^{(h)*}$, and*

$$\sqrt{n}\Big(\hat{\boldsymbol{f}}^{(h)*} - \boldsymbol{\mu}^{(h)*}\Big) \xrightarrow{d} \mathcal{N}\Big(\boldsymbol{0},\, \boldsymbol{\Sigma}_{\boldsymbol{f}^{(h)*}}\Big), \qquad \boldsymbol{\Sigma}_{\boldsymbol{f}^{(h)*}} = \boldsymbol{G}\,\boldsymbol{\Sigma}_{\boldsymbol{Z}^{(h)*}}\,\boldsymbol{G}^\top,$$

where the Jacobian $\boldsymbol{G}_{2m\times 2(m+1)}$, evaluated at $\boldsymbol{\mu}_{\boldsymbol{Z}^{(h)*}}$, is given by

$$\boldsymbol{G} = \begin{pmatrix} 0 & \frac{-f_0}{\pi} & \frac{1}{\pi} & 0 & \cdots & & & \cdots & 0 \\ \vdots & \vdots & 0 & \ddots & \ddots & & & & \vdots \\ 0 & \frac{-f_{m-1}}{\pi} & \vdots & \ddots & \frac{1}{\pi} & & & & \vdots \\ \frac{-f_{00}(h)}{\pi(h)} & 0 & & & & \frac{1}{\pi(h)} & \ddots & & \vdots \\ \vdots & \vdots & \vdots & & & \ddots & \ddots & & 0 \\ \frac{-f_{m-1,m-1}(h)}{\pi(h)} & 0 & 0 & \cdots & & \cdots & & 0 & \frac{1}{\pi(h)} \end{pmatrix}.$$

3.5.2. Hypothesis Testing for Serial Dependence

The asymptotic results derived above provide the foundation for a formal test of serial independence based on $\hat{\kappa}_{\mathrm{ord}}(h)$. The null and alternative hypotheses are

$$H_0 : \kappa_{\mathrm{ord}}(h) = 0 \qquad \text{vs.} \qquad H_A : \kappa_{\mathrm{ord}}(h) \neq 0.$$

A key asymmetry between H_0 and H_A shapes the subsequent analysis: the asymptotic variance σ_κ^2 of $\sqrt{n}\,\hat{\kappa}_{\mathrm{ord}}(h)$ is only available in closed form under H_0, where the i.i.d. assumption on (X_t) eliminates the higher-order mixed moments of (X_t) that appear in $\boldsymbol{\Sigma}_{\boldsymbol{f}^{(h)*}}$ (cases **(ix)** and **(x)** of Section 3.5.1). Under H_A, the general expression for σ_κ^2 is intractable without further assumptions on the dependence structure of (X_t). Consequently, confidence intervals for the true value $\kappa_{\mathrm{ord}}(h)$ cannot be constructed under H_A in the present framework. What can be established under H_A is the consistency of the test, which is treated in Section 3.5.3.

Asymptotic null distribution, test statistic, and confidence intervals under H_0

Under the null hypothesis H_0 that (X_t) is i.i.d., the condition $f_{ij}(h) = f_i f_j$ holds for all $h \geq 1$. This considerably simplifies the third- and fourth-order mixed moments of $\boldsymbol{Z}_t^{(h)*}$ appearing in cases **(ix)** and **(x)** of $\boldsymbol{\Sigma}_{\boldsymbol{Z}^{(h)*}}$. Before carrying out the simplifications, we collect the relevant moments under H_0, which will be used repeatedly below:

$$\mathrm{E}[Z_{t,i}Z_{t-h,i}Z_{t-k,j}Z_{t-k-h,j}] - f_{ii}(h)f_{jj}(h) = \begin{cases} f_{\min\{i,j\}}^2 - f_i^2 f_j^2 & \text{if } k=0, \\ f_i f_j f_{\min\{i,j\}} - f_i f_j^2 & \text{if } k=h, \\ 0 & \text{else,} \end{cases}$$

$$\mathrm{E}[Z_{t+k,i}Z_{t,j}Z_{t-h,j}] - f_i f_{jj}(h) = \begin{cases} f_j f_{\min\{i,j\}} - f_i f_j^2 & \text{if } k=0, \\ 0 & \text{else,} \end{cases}$$

$$\mathrm{E}[Z_{t,i}Z_{t+k,j}Z_{t+k-h,j}] - f_i f_{jj}(h) = \begin{cases} f_j f_{\min\{i,j\}} - f_i f_j^2 & \text{if } k=0 \text{ or } k=h, \\ 0 & \text{else.} \end{cases}$$

Applying these simplifications to cases **(ix)** and **(x)** of $\boldsymbol{\Sigma}_{\boldsymbol{Z}^{(h)*}}$ yields:

(ix) $i = 0,\ldots,m-1,\ j = m,\ldots,2m-1$ (continued from above) :

$$\begin{aligned}
&\overset{H_0}{=} \pi(h)\, f_{j-m} f_{\min\{i,j-m\}} - \pi\pi(h) f_i f_{j-m}^2 \\
&\quad + \underbrace{\mathrm{E}[O_t^2 O_{t+h}]\, f_{j-m} f_{\min\{i,j-m\}} + \mathrm{E}[O_{t+h}O_tO_{t-h}]\, f_i f_{j-m}^2 - 2\pi\pi(h) f_i f_{j-m}^2}_{k=h} \\
&\quad + f_i f_{(j-m)}^2 \mathrm{E}[O_t^2 O_{t-h}] - f_i f_{(j-m)}^2 \mathrm{E}[O_t^2 O_{t-h}] \\
&\quad + \sum_{\substack{k=1 \\ k\neq h}}^{\infty} \left[\mathrm{E}[O_tO_{t+k}O_{t+k-h}] + \mathrm{E}[O_{t+k}O_tO_{t-h}] - 2\pi\pi(h)\right] f_i f_{j-m}^2 \\
&= 2\pi(h) f_{j-m} f_{\min\{i,j-m\}} - \pi\pi(h) f_i f_{j-m}^2 - \pi(h) f_i f_{j-m}^2 \\
&\quad + (1-\pi)\pi(h) f_i f_{j-m}^2 - (1-\pi)\pi(h) f_i f_{j-m}^2 \\
&\quad + f_i f_{j-m}^2 \sum_{k=1}^{\infty} \left[\mathrm{E}[O_tO_{t+k}O_{t+k-h}] + \mathrm{E}[O_{t+k}O_tO_{t-h}] - 2\pi\pi(h)\right] \\
&= 2\pi(h)(f_{(j-m)} f_{\min\{i,j-m\}} - f_i f_{j-m}^2) + f_i f_{j-m}^2 \sigma_{-1,-2}^{(h)}
\end{aligned}$$

(x) $i,j = m,\ldots,2m-1$ (continued from above) :

$$\begin{aligned}
&\overset{H_0}{=} \pi(h) f_{\min\{i,j\}}^2 - \pi(h)^2 f_{(i-m)}^2 f_{(j-m)}^2 + 2\mathrm{E}[O_tO_{t+h}O_{t-h}] f_i f_j f_{\min\{i,j\}} - 2\pi(h)^2 f_{i-m}^2 f_{j-m}^2 \\
&\quad + \sum_{\substack{k=1 \\ k\neq h}}^{\infty} 2 f_i^2 f_j^2 \left[\mathrm{E}[O_tO_{t-h}O_{t+k}O_{t+k-h}] - \pi(h)^2\right] \\
&= \pi(h) f_{\min\{i-m,j-m\}}^2 - \pi(h)^2 f_{(i-m)}^2 f_{(j-m)}^2 + 2\mathrm{E}[O_tO_{t+h}O_{t-h}] f_{i-m} f_{j-m} f_{\min\{i-m,j-m\}} \\
&\quad - 2 f_{i-m}^2 f_{j-m}^2 E[O_tO_{t+h}O_{t-h}] + \pi(h) f_{i-m}^2 f_{j-m}^2 - \pi(h) f_{i-m}^2 f_{j-m}^2 + \\
&\quad + 2 f_{i-m}^2 f_{j-m}^2 \sum_{k=1}^{\infty} \left[\mathrm{E}[O_tO_{t-h}O_{t+k}O_{t+k-h}] - \pi(h)^2\right]
\end{aligned}$$

$$= \pi(h)(f^2_{\min\{i-m,j-m\}} - f^2_{i-m}f^2_{j-m}) + 2\pi(h,2h)f_{i-m}f_{j-m}(f_{\min\{i-m,j-m\}} - f_{i-m}f_{j-m})$$
$$+ f^2_{i-m}f^2_{j-m}\sigma^{(h)}_{-2,-2},$$

where $\pi(h,2h) := \mathrm{E}[O_t O_{t+h} O_{t-h}]$. Substituting all elements of $\boldsymbol{\Sigma}_{\boldsymbol{Z}^{(h)*}}$ into $\boldsymbol{\Sigma}_{\boldsymbol{f}^{(h)*}} = \boldsymbol{G}\boldsymbol{\Sigma}_{\boldsymbol{Z}^{(h)*}}\boldsymbol{G}^\top$ and carrying out the matrix multiplication under H_0, the elements $\sigma^{(h)*}_{i,j}$ simplify to (see Appendix A.2):

(i) $i,j = 0,\dots,m-1:$

$$\sigma^{(h)*}_{i,j} \overset{H_0}{=} \frac{1}{\pi}\Big(f_{\min\{i,j\}} - f_i f_j\Big)$$

(ii) $i = 0,\dots,m-1,\ j = m,\dots,2m-1\ (j' = j-m):$

$$\sigma^{(h)*}_{i,j} \overset{H_0}{=} \frac{2}{\pi}\, f_{j'}\Big(f_{\min\{i,j'\}} - f_i f_{j'}\Big) = 2 f_{j'}\,\sigma^{(h)*}_{i,j'}$$

(iii) $i,j = m,\dots,2m-1\ (i' = i-m,\ j' = j-m):$

$$\sigma^{(h)*}_{i,j} \overset{H_0}{=} \frac{1}{\pi(h)^2}(f_{\min\{i',j'\}} - f_{i'}f_{j'})\Big(\pi(h)(f_{\min\{i',j'\}} - f_{i'}f_{j'}) + 2\pi(h,2h)f_{i'}f_{j'}\Big)$$

Applying the delta method to $\kappa_{\mathrm{ord}} = A/B$, where $A = \sum_{j=0}^{m-1}(f_{jj}(h) - f_j^2)$ and $B = \sum_{j=0}^{m-1} f_j(1-f_j)$, the relevant Jacobian is

$$\boldsymbol{D}_\kappa := (d_{1,0},\dots,d_{1,m-1},\, d_{2,0},\dots,d_{2,m-1})\,,$$

where for $i = 0,\dots,m-1$:

$$\textbf{(i)}\ d_{1;i} = \frac{\partial \kappa_{\mathrm{ord}}}{\partial f_i} = \frac{-2f_i}{B} - \frac{(1-2f_i)A}{B^2},$$
$$\textbf{(ii)}\ d_{2;i} = \frac{\partial \kappa_{\mathrm{ord}}}{\partial f_{ii}(h)} = \frac{1}{B}.$$

Plugging the derivatives and the elements of $\boldsymbol{\Sigma}_{\boldsymbol{f}^{(h)*}}$ into the delta method formula and evaluating under H_0 yields (see Appendix A.3 for the detailed calculation)

$$\sigma^2_\kappa = \boldsymbol{D}_\kappa \boldsymbol{\Sigma}_{\boldsymbol{f}^{(h)*}} \boldsymbol{D}_\kappa^\top = \frac{\frac{1}{\pi(h)}\sum\limits_{i,j=0}^{m-1}\Big(f_{\min\{i,j\}} - f_i f_j\Big)\Big(f_{\min\{i,j\}} + \Big(1 + \frac{2\pi(h,2h)}{\pi(h)} - \frac{4\pi(h)}{\pi}\Big)f_i f_j\Big)}{\Big(\sum\limits_{k=0}^{m-1} f_k(1-f_k)\Big)^2}.$$

Building on this, we now investigate the finite-sample bias by means of a second-order Taylor expansion. Since the linear term $\boldsymbol{D}_\kappa(\hat{\boldsymbol{f}}^{(h)*} - \boldsymbol{\mu}^{(h)*})$ has expectation of order $o(n^{-1})$, it does not contribute to the leading bias term, and the $O(n^{-1})$ bias reduces by Lemma 2.4 to

$$\mathrm{Bias}(n\hat{\kappa}_{\mathrm{ord}}) = \frac{1}{2}\,\mathrm{tr}\Big(\boldsymbol{H}_\kappa\, \boldsymbol{\Sigma}_{\boldsymbol{f}^{(h)*}}\Big) + o(n^{-1}).$$

The elements of the Hessian $\boldsymbol{H}_\kappa$ are given by

$$\frac{\partial d_{1,i}}{\partial f_j} = \begin{cases} 2\left(\dfrac{2f_i - 4f_i^2}{B^2} - \dfrac{1}{B}\right) & \text{if } i = j, \\ 2\,\dfrac{f_i + f_j - 4f_i f_j}{B^2} & \text{if } i \neq j, \end{cases}$$

$$\frac{\partial d_{1,i}}{\partial f_{jj}(h)} = \frac{-(1-2f_i)}{B^2}, \qquad \frac{\partial d_{2,i}}{\partial f_j} = \frac{-(1-2f_j)}{B^2}, \qquad \frac{\partial d_{2,i}}{\partial f_{jj}(h)} = 0,$$

with detailed derivations given in Appendix A.4. Evaluating $\frac{1}{2}\mathrm{tr}(\boldsymbol{H}_\kappa \boldsymbol{\Sigma}_{\boldsymbol{f}^{(h)*}})$ under H_0, using $\sigma_{i,m+l}^{(h)*} = 2f_l \sigma_{i,l}^{(h)*}$ from case **(ii)** of $\boldsymbol{\Sigma}_{\boldsymbol{f}^{(h)*}}$ and symmetry $\sigma_{i,l}^{(h)*} = \sigma_{l,i}^{(h)*}$, yields

$$\mathrm{Bias}(\hat{\kappa}_{\mathrm{ord}}(h)) \stackrel{H_0}{=} -\frac{1}{n\pi}.$$

The detailed calculation is carried out in Appendix A.5. The results are summarized in the following theorem.

Theorem 3.14 (Asymptotic distribution and bias of $\hat{\kappa}_{\mathrm{ord}}(h)$)**.** *Let the assumptions of Theorem 3.6 hold and denote*

$$\pi(h_1, \ldots, h_r) := \mathrm{E}[O_0 \cdot O_{h_1} \cdots O_{h_r}].$$

Then, under the null hypothesis H_0 that (X_t) is i.i.d., the following hold as $n \to \infty$:

1. ***Asymptotic normality.*** *The estimator $\hat{\kappa}_{\mathrm{ord}}(h)$ satisfies*

$$\sqrt{n}\,\hat{\kappa}_{\mathrm{ord}}(h) \xrightarrow{d} \mathcal{N}(0, \sigma_\kappa^2),$$

where

$$\sigma_\kappa^2 = \frac{\frac{1}{\pi(h)} \sum_{i,j=0}^{m-1} \Big(f_{\min\{i,j\}} - f_i f_j\Big)\Big(f_{\min\{i,j\}} + \Big(1 + \frac{2\pi(h,2h)}{\pi(h)} - \frac{4\pi(h)}{\pi}\Big) f_i f_j\Big)}{\left(\sum_{k=0}^{m-1} f_k(1-f_k)\right)^2}.$$

2. ***Finite-sample bias.*** *The bias of $\hat{\kappa}_{\mathrm{ord}}(h)$ under H_0 is*

$$\mathrm{Bias}\Big(\hat{\kappa}_{\mathrm{ord}}(h)\Big) = -\frac{1}{n\pi} + o(n^{-1}),$$

so that the bias-corrected estimator is

$$\hat{\kappa}_{\mathrm{ord}}^{\mathrm{bc}}(h) := \hat{\kappa}_{\mathrm{ord}}(h) + \frac{1}{n\pi}.$$

Remark 3.15 (Simplification under i.i.d. observation process)**.** Suppose in addition that the observation process (O_t) is i.i.d. Then the joint moments factorize as

$$\pi(h) = \mathrm{E}[O_t O_{t-h}] = \pi^2, \qquad \pi(h, 2h) = \mathrm{E}[O_t O_{t-h} O_{t-2h}] = \pi^3.$$

Substituting into σ_κ^2 yields (see Appendix A.6 for the detailed calculation)

$$\sigma_\kappa^2 = \frac{1}{\pi^2} \frac{\sum_{i,j=0}^{m-1} \Big(f_{\min\{i,j\}} - f_i f_j\Big)\Big(f_{\min\{i,j\}} + (1-2\pi) f_i f_j\Big)}{\left(\sum_{k=0}^{m-1} f_k(1-f_k)\right)^2}.$$

Test statistic and confidence intervals. Under H_0, Theorem 3.14 and Slutsky's theorem (Klenke, 2005, Theorem 13.18) immediately yield a consistent plug-in test. Replacing all unknown quantities in σ_κ^2 by their sample counterparts, specifically f_i by $\hat{f}_i^*$, the bivariate probabilities $f_{ij}(h)$ by $\hat{f}_{ij}^*(h)$, and the observation-process moments π, $\pi(h)$, and $\pi(h, 2h)$ by

$$\hat{\pi} := \bar{O} = \frac{1}{n} \sum_{t=1}^{n} O_t, \quad \widehat{\pi(h)} := \overline{OO}_h = \frac{1}{n-h} \sum_{t=h+1}^{n} O_t O_{t-h},$$
$$\widehat{\pi(h, 2h)} := \frac{1}{n-2h} \sum_{t=2h+1}^{n} O_t O_{t+h} O_{t-h},$$

yields a consistent estimator $\hat{\sigma}_\kappa^2$ of σ_κ^2. Consistency follows from the continuous mapping theorem together with the consistency of $\hat{f}_i^*$, $\hat{f}_{ij}^*(h)$ (Theorems 3.6 and 3.13) and the law of large numbers applied to the observation-process moments.

Alternatively, one may test for serial independence of the observation process (O_t) using, for example, a portmanteau test such as the Ljung–Box test (see (Brockwell and Davis, 2016, Chapter 1.6)). Note that the Ljung-Box test only tests for linear dependence structures and may fail to detect nonlinear dependence in (O_t). Since higher moments of O_t are directly determined by its first moment (since $O_t \in \{0, 1\}$), the implied limitations of the Ljung-Box test for binary time series are less severe than for continuous-valued time series. However, since there also may be dependence involving products like $O_t O_{t-h} O_{t+k}$ that is not captured by the Ljung-Box test, the assumption of i.i.d. O_t should be adopted with care. If the null hypothesis of no serial dependence cannot be rejected, it is justified to assume that (O_t) is i.i.d., in which case the simplified variance expression in Remark 3.15 can be used. In this situation, only π needs to be estimated, which can improve numerical stability of the plug-in estimator. If, however, serial dependence is detected, the general estimator based on $\hat{\pi}(h)$ and $\hat{\pi}(h, 2h)$ should be preferred.

The test statistic is

$$T_n(h) := \frac{\sqrt{n}\, \hat{\kappa}_{\mathrm{ord}}(h)}{\hat{\sigma}_\kappa},$$

and a test of level α rejects H_0 whenever $|T_n(h)| > z_{\alpha/2}$, where $z_{\alpha/2}$ denotes the $(1-\alpha/2)$-quantile of the standard normal distribution. Since $T_n(h) \xrightarrow{d} \mathcal{N}(0,1)$ under H_0 by Theorem 3.14 and Slutsky's theorem, this test has asymptotic level α.
The test can equivalently be expressed through a confidence interval: H_0 is rejected at level α if and only if 0 lies outside

$$\hat{\kappa}_{\mathrm{ord}}(h) \ \pm \ z_{\alpha/2}\,\frac{\hat{\sigma}_\kappa}{\sqrt{n}}. \tag{3.3}$$

It is important to note that (3.3) is an interval for the purpose of testing $H_0 : \kappa_{\mathrm{ord}}(h) = 0$; it does *not* serve as a confidence interval for the true value $\kappa_{\mathrm{ord}}(h)$ under H_A, as the asymptotic variance of $\hat{\kappa}_{\mathrm{ord}}(h)$ differs from σ_κ^2 when (X_t) is not i.i.d.
Also note that $\hat{\kappa}_{\mathrm{ord}}(h)$ for a fixed lag h constitutes a necessary but not sufficient condition for the i.i.d. null hypothesis: rejection at any lag h implies departure from i.i.d., but non-rejection at a single lag does not rule out dependence at other lags or through off-diagonal bivariate probabilities. In practice, the test is therefore applied simultaneously for several lags $h = 1, \ldots, H$.
When testing simultaneously for $h = 1, \ldots, H$, a multiple testing problem arises: even under the i.i.d. null hypothesis, the probability of falsely rejecting at least one individual test exceeds the nominal level α. This quantity, known as the *familywise error rate*, grows with H. There are various more or less conservative methods to control it, such as the Bonferroni correction, which rejects if $|T_n(h)| > z_{\alpha/(2H)}$ for any h, or the less conservative Holm procedure, which orders the H p-values as $p_{(1)} \leq \ldots \leq p_{(H)}$ and rejects the hypothesis associated with $p_{(k)}$ if $p_{(l)} \leq \alpha/(H+1-l)$ for all $l \leq k$. For a comprehensive treatment of these and further procedures, see Lehmann and Romano (2005, pp. 348–351).

3.5.3. Consistency under H_A

Under the alternative hypothesis H_A, the process (X_t) is not i.i.d., so that there exists at least one lag $h \geq 1$ with $\kappa_{\mathrm{ord}}(h) \neq 0$.
The estimator $\hat{\kappa}_{\mathrm{ord}}(h)$ is a continuous function of the empirical frequencies $\hat{f}_i^*$ and $\hat{f}_{ii}^*(h)$. Since these are consistent estimators of f_i and $f_{ii}(h)$ (see Theorem 3.6 and Theorem 3.13), the continuous mapping theorem (see, e.g., Kallenberg, 2021, Theorem 3.27) implies

$$\hat{\kappa}_{\mathrm{ord}}(h) \xrightarrow{p} \kappa_{\mathrm{ord}}(h) \neq 0.$$

It follows that

$$\sqrt{n}\,\hat{\kappa}_{\mathrm{ord}}(h) \xrightarrow{p} \infty,$$

since $\hat{\kappa}_{\mathrm{ord}}(h)$ converges to a nonzero constant.
Therefore, the test statistic diverges under H_A, implying that the rejection probability converges to one. This establishes the consistency of the test.

4. Simulation Study

The theoretical results derived in the preceding chapters characterize the asymptotic behavior of the proposed estimators, but leave open the question of how well these approximations perform in finite samples. In particular, the bias corrections derived for the IOV estimator (Corollary 3.10) and for $\hat{\kappa}_{\text{ord}}(h)$ (Theorem 3.14) are asymptotic results, and it is not a priori clear at what sample sizes they provide reliable guidance. This chapter therefore complements the theoretical analysis with a Monte Carlo simulation study, pursuing three goals: first, to assess the finite-sample accuracy of the asymptotic bias formulas; second, to investigate the rate of convergence of the estimators to their asymptotic normal distributions; and third, to examine the sensitivity of the estimators to violations of the MCAR assumption.

4.1. Setup and Design

Data generating process

All simulations are based on the BinAR(1) process as described in Weiß (2018, Section 3.3), which serves as a flexible and tractable model for ordinal categorical time series with controllable serial dependence.

Model definition. The BinAR(1) process is parameterized by the number of categories $m \in \mathbb{N}$, the marginal probability $p \in (0, 1)$, and the thinning parameter

$$r \in \left(\max \left\{ -\frac{p}{1-p}, -\frac{1-p}{p} \right\}, 1 \right).$$

It is defined recursively by

$$X_t = B(X_{t-1},\, \alpha) + B(m - X_{t-1},\, \beta),$$

where $B(k, q)$ denotes a binomial random variable with parameters k and q, and where $\alpha = p(1-r)+r$ and $\beta = p(1-r)$. This representation has a simple interpretation: units currently in state “1” persist with probability α, while units in state “0” switch to state “1” with probability β. The parameter r controls the strength and direction of serial dependence: $r = 0$ yields an i.i.d. process, positive values imply positive autocorrelation, and negative values imply negative autocorrelation.

Distributional properties. The stationary marginal distribution of (X_t) is Binomial(m,p), so that the process is strictly stationary. In particular, the true CDF vector $\boldsymbol{f}$ and the index of ordinal variation (IOV) are available in closed form as

$$f_i = P(X_t \le i) = \sum_{k=0}^{i} \binom{m}{k} p^k (1-p)^{m-k}, \qquad i = 0, \dots, m-1,$$

$$\mathrm{IOV} = \frac{4}{m} \sum_{i=0}^{m-1} f_i(1-f_i).$$

The one-step-ahead transition probabilities

$$p_{i|j}(1) := P(X_t = i \mid X_{t-1} = j)$$

are given by (Weiß, 2018, p. 60):

$$p_{i|j}(1) = \sum_{q=\max\{0,i+j-m\}}^{\min\{i,j\}} \binom{j}{q} \binom{m-j}{i-q} \alpha^q (1-\alpha)^{j-q} \beta^{i-q} (1-\beta)^{m-j+q-i}.$$

The h-step-ahead transition probabilities are obtained by replacing α and β with $\alpha_h = p(1-r^h) + r^h$ and $\beta_h = p(1-r^h)$. The joint probabilities

$$p_{ij}(h) = P(X_t = i, X_{t-h} = j)$$

are then given by

$$p_{ij}(h) = p_{i|j}(h) \cdot p_j,$$

where $p_j = P(X_t = j)$. The corresponding joint CDF components are

$$f_{ij}(h) = \sum_{k=0}^{i} \sum_{l=0}^{j} p_{kl}(h).$$

Hence, $\kappa_{\mathrm{ord}}(h)$ is available in closed form by plugging the relevant $f_{ij}(h)$ and f_i into Definition 1.5. Figure 4.1 illustrates the serial dependence structure of the BinAR(1) process by displaying the analytically computed values of $\kappa_{\mathrm{ord}}(h)$ as a function of the lag h for Scenario B with $r \in \{-0.5, -0.2, 0.2, 0.40.6, 0.8, 0.95\}$. For $r > 0$, the curves decay exponentially with h, with the rate of decay decreasing for higher values of r — reflecting the stronger persistence of the process. For $r < 0$, $\kappa_{\mathrm{ord}}(h)$ alternates in sign: negative at odd lags and positive at even lags, while its absolute value decays exponentially to zero as $h \to \infty$.

Mixing properties. Finally, we verify that the BinAR(1) process satisfies the dependence assumptions required for the asymptotic theory. Since the BinAR(1) process is an ergodic Markov chain on the finite state space $\{0, \dots, m\}$ (Weiß, 2018, p. 60), it follows that the process is ϕ-mixing with exponentially decaying coefficients, i.e. there

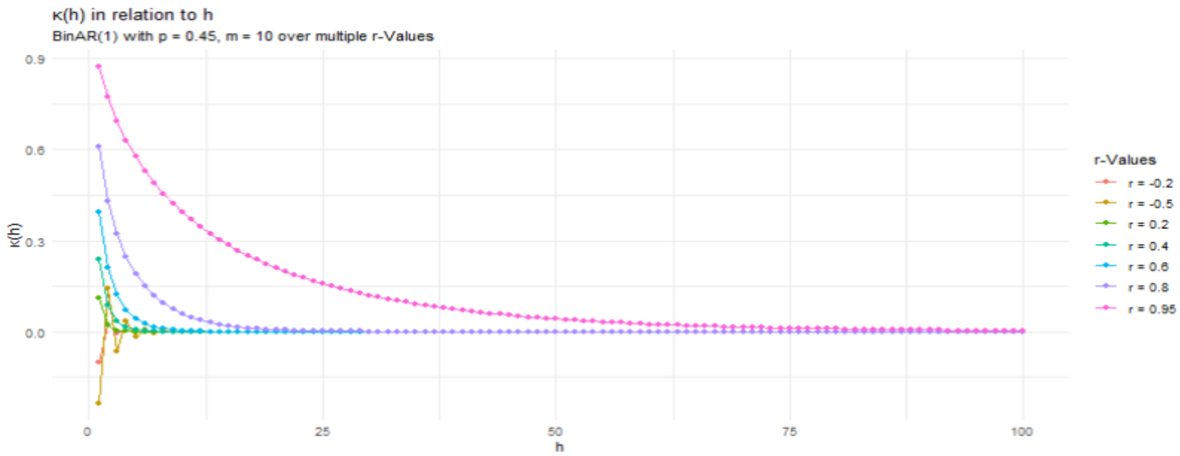

Figure 4.1.: Cohen's $\kappa_{\text{ord}}(h)$ for different values of r.

exist constants $C > 0$ and $\rho \in (0, 1)$ such that $\phi(n) \leq C\rho^n$ for all n (Weiß, 2018, Appendix B.2.2).
Moreover, the mixing coefficients satisfy the inequality $2\alpha(n) \leq \phi(n)$ for all n (Bradley, 2005, Eq. (1.11)). In particular, this also implies

$$\alpha(n) \leq \phi(n) \leq C\rho^n \text{ for some } C > 0, \rho \in (0, 1),$$

showing that the BinAR(1) process satisfies the mixing-assumptions of Theorem 3.6.

Missingness mechanism. Missingness is introduced via the amplitude modulation framework of Section 3.2. We consider three different data-generating mechanisms for the observation process (O_t).
First, as a baseline, we assume independent missingness, where (O_t) is i.i.d. Bernoulli distributed with constant observation probability $\pi \in (0, 1]$, i.e.

$$O_t \sim \text{Bernoulli}(\pi).$$

Second, we consider a serially dependent observation process, where (O_t) is a stationary binary Markov chain on $\{0, 1\}$ with marginal observation probability $\mathrm{E}[O_t] = \pi$ and autocorrelation parameter $r_\pi \in (0, 1)$, with transition probabilities

$$P(O_t = 1 \mid O_{t-1} = 1) = (1 - r_\pi)\pi + r_\pi, \qquad P(O_t = 1 \mid O_{t-1} = 0) = \pi(1 - r_\pi).$$

This mechanism preserves the marginal observation probability π while inducing serial dependence in (O_t), with $\mathrm{Cov}(O_t, O_{t+h}) = r_\pi^h\, \pi(1 - \pi)$ for all $h \geq 1$ (Weiß, 2018, p. 139). Notably, this process is equal to a BinAR(1)-process with parameters $m = 1,\ p = \pi$ and $r = r_h$, which implies that the process satisfies the relevant mixing conditions. Third, we consider observation schemes that depend on the current state of the underlying process (X_t), corresponding to MNAR-type mechanisms. In the first variant, the observation probability increases linearly with X_t, i.e.

$$\pi_t = \pi_{\text{low}} + (\pi_{\text{high}} - \pi_{\text{low}})\frac{X_t}{m},$$

while in the second variant it decreases linearly with X_t, i.e.

$$\pi_t = \pi_{\text{high}} - (\pi_{\text{high}} - \pi_{\text{low}})\frac{X_t}{m}.$$

The parameters π_{low} and π_{high} define the minimum and maximum observation probabilities, respectively, and therefore determine the range in which π_t can vary. In both cases,

$$O_t \mid X_t \sim \text{Bernoulli}(\pi_t).$$

These three mechanisms allow us to study independent, serially dependent, and state-dependent missingness within a unified simulation framework.

Simulation scenarios

Two combinations of (m, p, r) are considered, chosen to represent qualitatively different settings:

- **Scenario A:** $m = 3$, $p = 0.20$, $r = 0.35$ — a coarse ordinal scale with few categories, a skewed marginal distribution, and moderate serial dependence.
- **Scenario B:** $m = 10$, $p = 0.45$, $r = 0.50$ — a finer ordinal scale, a near-symmetric marginal distribution, and stronger serial dependence.

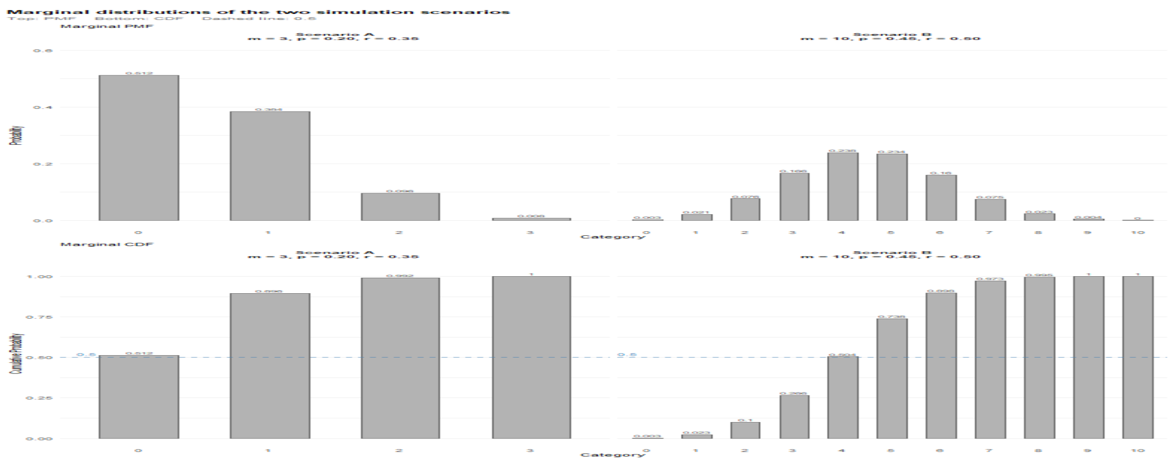

Figure 4.2.: Marginal CDFS of the two simulation scenarios.

The PMF's and CDFs for both scenarios are shown in Figure 4.2. Depending on the experiment, the serial dependence parameter r may be varied to illustrate certain results. For each scenario, two observation probabilities are considered: $\pi = 1$ (no missingness) and $\pi = 0.75$ (25% missing at random).
In addition, for we examine serially dependent missingness, with $\pi = 0.75$ and $r_\pi \in \{0.2, 0.75\}$, as well as the two MNAR-type mechanisms described above with $\pi_{\text{low}} = 0.5$ and $\pi_{\text{high}} = 1$.

The sample size n and the number of Monte Carlo replications B vary across experiments and are stated explicitly for each figure. As a default, results are based on $n \in \{50, 100, 250, 500, 1000\}$ and $B = 1{,}000$ replications.

Estimators and performance measures

In each replication, the following quantities are computed from the observed data $(O_t X_t)_{t=1}^n$: the corrected CDF estimator $\hat{\boldsymbol{f}}^*$ (Definition 3.3), the IOV estimator $\widehat{\mathrm{IOV}}$, the skewness estimator $\widehat{\mathrm{skew}}$, and the ordinal Cohen's κ estimator $\hat{\kappa}_{\mathrm{ord}}(h)$ for lags $h \in \{1, 2\}$. Performance is assessed through the following measures, averaged across replications:

- **Mean bias:** $\frac{1}{B}\sum_{b=1}^{B}(\hat{\theta}_b - \theta)$, where θ denotes the true parameter value.
- **n-scaled bias:** $n \cdot \frac{1}{B}\sum_{b=1}^{B}(\hat{\theta}_b - \theta)$, used to assess convergence to the theoretical $O(n^{-1})$ bias limit.
- **Standard deviation** across replications, compared against the asymptotic standard deviation $\sqrt{\sigma_\theta^2/n}$ derived in the theoretical chapters.

The theoretical bias limit for the IOV estimator under MCAR is given by $-\frac{4}{mn}\mathrm{tr}(\boldsymbol{\Sigma}_Z)$ (Corollary 3.10), and for $\hat{\kappa}_{\mathrm{ord}}(h)$ under H_0 by $-\frac{1}{n\pi}$ (Theorem 3.14). These serve as benchmarks against which the simulated bias curves are compared.

4.2. Results

4.2.1. Marginal Estimation and Dispersion Measures

Convergence to the asymptotic distribution. Figure 4.3 illustrates the convergence of $\sqrt{n}(\widehat{\mathrm{IOV}} - \mathrm{IOV})$ to its asymptotic normal distribution. The left panel shows a kernel density plot of the raw quantity, while the right panel shows the same quantity after multiplication by $\sqrt{n}$. Both panels display results for $n \in \{50, 1000, 5000\}$ under Scenario B with $\pi = 0.75$.

In the left panel, the residual leftward shift at $n = 50$ relative to the asymptotic limit reflects the negative finite-sample bias of $\widehat{\mathrm{IOV}}$ established in Corollary 3.10. Crucially, this shift is visible at $n = 50$ but has mostly vanished by $n = 1000$, consistent with the $O(n^{-1})$ order of the bias. The right panel reveals a complementary picture: after $\sqrt{n}$-rescaling, the three distributions collapse onto the same spread, directly confirming that the standard deviation converges at rate $O(n^{-1/2})$ as implied by the CLT. At the same time, a residual leftward shift remains visible for $n = 50$ and even $n = 1000$, indicating that the bias does not vanish under $\sqrt{n}$-scaling. This is consistent with the fact that the bias is of order $O(n^{-1})$, so that $\sqrt{n}$-rescaling yields a remaining shift of order $O(n^{-1/2})$.

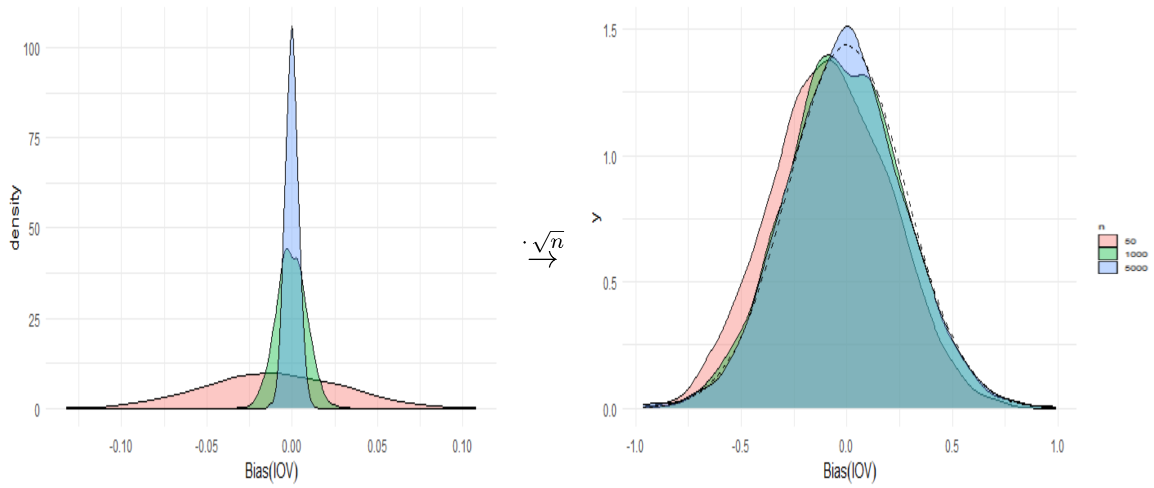

Figure 4.3.: Left: simulated density of $(\widehat{\mathrm{IOV}} - \mathrm{IOV})$ for $n \in \{50, 1000, 5000\}$. Right: the same quantity rescaled by a factor of $\sqrt{n}$, i.e. $\sqrt{n}(\widehat{\mathrm{IOV}} - \mathrm{IOV})$, overlaid with the asymptotic $\mathcal{N}(0, \sigma^2_{\mathrm{IOV}})$ limit (dashed). Scenario B with $\pi = 0.75$. Based on $B = 1{,}000$ replications.

Finite-sample bias and its order. Figure 4.4 summarizes the finite-sample bias of $\widehat{\mathrm{IOV}}$ as a function of n. The left panel shows the mean bias $\frac{1}{B}\sum_{b=1}^{B}(\widehat{\mathrm{IOV}}_b - \mathrm{IOV})$, together with the theoretical approximation $-\frac{4}{mn}\sum_{i=0}^{m-1}\sigma_{\hat{\boldsymbol{f}}^*;i,i}$ from Corollary 3.10 (dashed). The simulated and theoretical curves are nearly indistinguishable across the entire range of n, confirming the accuracy of the $O(n^{-1})$ approximation even at small sample sizes.
The right panel displays the n-scaled bias $n \cdot \frac{1}{B}\sum_{b=1}^{B}(\widehat{\mathrm{IOV}}_b - \mathrm{IOV})$, along with the corresponding theoretical limit $-\frac{4}{m}\sum_{i=0}^{m-1}\sigma_{\hat{\boldsymbol{f}}^*;i,i}$ (dashed). If the bias were exactly of order n^{-1}, this quantity would be constant in n. The figure shows that it is indeed nearly constant, with only minor fluctuations around the limiting value.
The pointwise 95% confidence bands (vertical lines) widen with n. Since $\widehat{\mathrm{IOV}}$ has standard deviation of order $n^{-1/2}$, its Monte Carlo mean based on B replications has standard error of order $(Bn)^{-1/2}$. After multiplication by n, this becomes $\sqrt{n/B}$, explaining the increasing width of the bands. This effect is therefore purely induced by the scaling and should not be interpreted as a deterioration of the asymptotic approximation.

Asymptotic standard deviation: simulation vs. theory. Figure 4.5 compares the simulated standard deviation of $\left(\widehat{\mathrm{IOV}} - \mathrm{IOV}\right)$ against the asymptotic approximation σ_{IOV} across all combinations of $n \in \{50, 100, 250, 500, 1000\}$ and $\pi \in \{0.75, 1\}$, for Scenario A on the left panel and Scenario B on the right. Several observations are worth highlighting.
First, the asymptotic approximation tracks the simulated standard deviation closely across all sample sizes, with the agreement improving monotonically as n grows. At $n = 50$, small discrepancies are visible, reflecting the finite-sample correction terms omitted by the first-order delta method. By $n = 250$, the approximation is already very accurate in both scenarios.

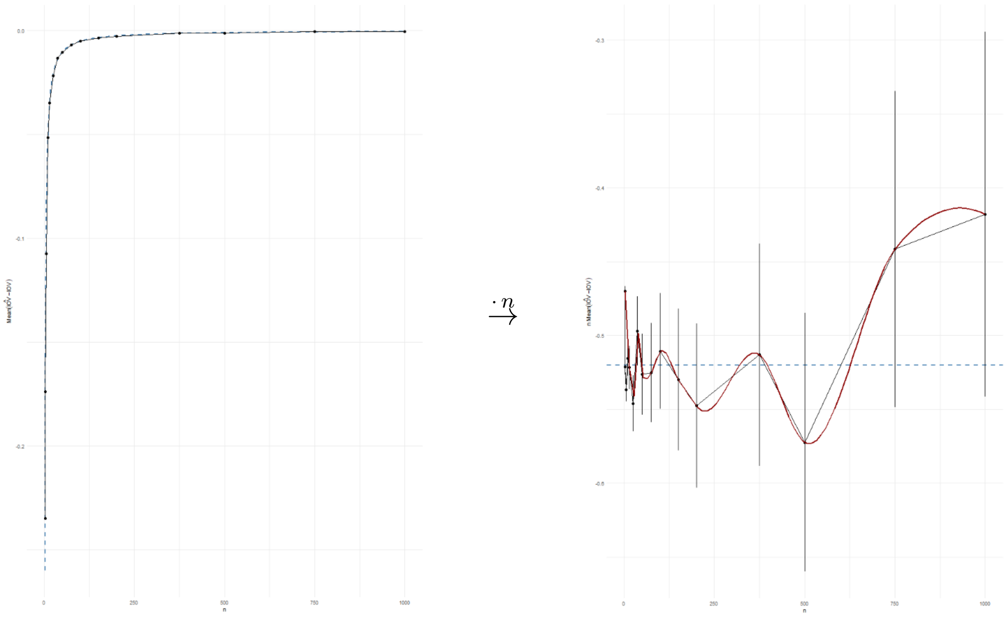

Figure 4.4.: Left: Mean from simulation (solid) and theoretical approximation (dashed) of $(\widehat{\mathrm{IOV}} - \mathrm{IOV})$. Right: the same quantities rescaled by a factor of n. For $n \in N_{\mathrm{fine}}$ and Scenario B with $\pi = 0.75$. Based on $B = 20{,}000$ replications.

Second, the effect of missingness is clearly visible: for both scenarios, the standard deviation under $\pi = 0.75$ is uniformly larger than under $\pi = 1$, consistent with the $1/\pi$ inflation of the lag-0 term in $\boldsymbol{\Sigma}_{\hat{\boldsymbol{f}}^*}$ established in Theorem 3.6.
Third, the convergence behavior differs subtly between the two scenarios. In Scenario A ($m = 3$, skewed marginal distribution), the standard deviation is larger in absolute terms. This indicates that the effect of marginal dispersion on $\mathrm{Var}(\widehat{\mathrm{IOV}}(\boldsymbol{f}^*))$ dominates the effect of serial dependence: the former is more pronounced in Scenario A, whereas the latter is stronger in Scenario B.
Although less immediately visible in the figure, this also affects the difference between the cases $\pi = 1$ and $\pi = 0.75$. The covariance terms $\sigma_{\hat{\boldsymbol{f}}^*;i,j}$ follow the forms given in Corollaries 3.7 and 3.8, respectively. Since here missingness only inflates the lag-0 component of the covariance structure, whereas serial dependence contributes through the higher-lag terms, the relative impact of missingness is smaller in Scenario B, where serial dependence is stronger. Consequently, the gap between $\pi = 1$ and $\pi = 0.75$ is more pronounced in Scenario A than in Scenario B.

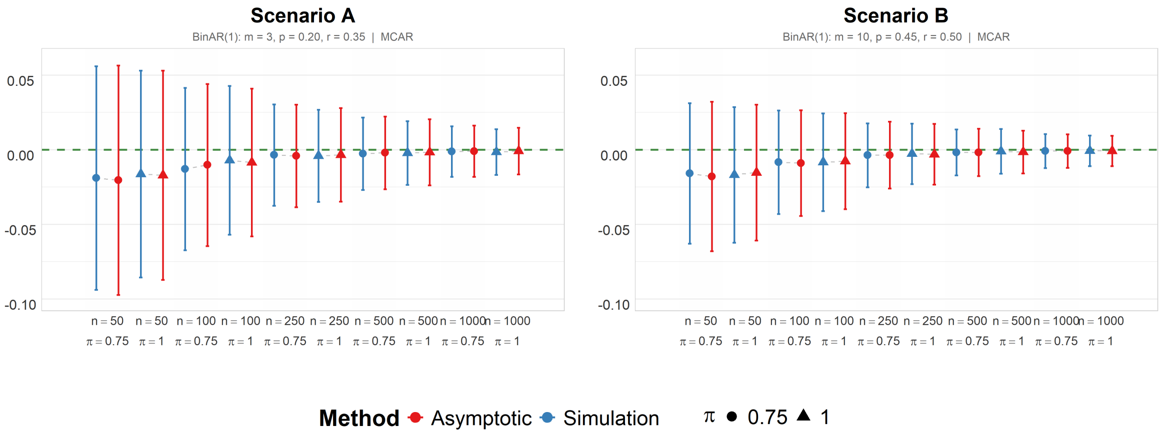

Figure 4.5.: Simulated vs. asymptotic standard deviation of $\left(\widehat{\mathrm{IOV}} - \mathrm{IOV}\right)$ for $n \in \{50, 100, 250, 500, 1000\}$, with $O_t \overset{i.i.d.}{\sim} Bernoulli(\pi)$ and $\pi \in \{0.75, 1\}$.

Serially dependent observation process. When (O_t) is a stationary binary Markov chain with autocorrelation parameter $r_\pi \in \{0.2, 0.75\}$ (see Section 4.1), the marginal observation probability $\pi = 0.75$ is preserved, but (O_t) is no longer i.i.d. Figure 4.6 presents the same comparison as Figure 4.5, now varying the serial dependence parameter r_π while keeping the marginal observation probability constant. A comparison of Figure 4.6 and Figure 4.5 reveals that the difference between i.i.d. O_t and weak serial dependence ($r_\pi = 0.2$) is not substantial. Only upon closer inspection can a slight increase in the standard deviation be observed. Even under much stronger serial dependence ($r_\pi = 0.75$), the effect remains limited, with only a slight increase in the standard deviations. This serves to illustrate that the dispersion of $\widehat{IOV}$ is primarily driven by the marginal distributions of X_t and O_T, and not an any serial dependence structure.

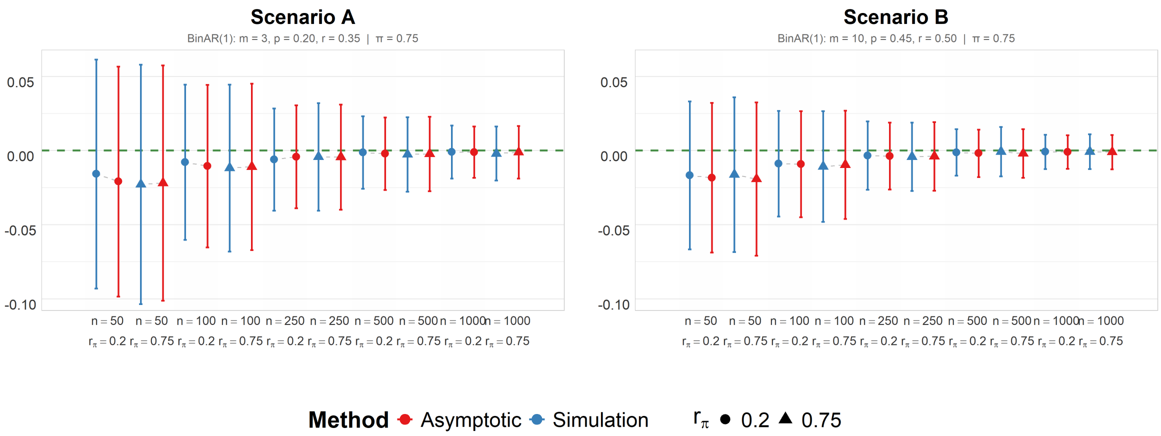

Figure 4.6.: Simulated vs. asymptotic standard deviation of $\left(\widehat{\mathrm{IOV}} - \mathrm{IOV}\right)$ for both scenarios across $n \in \{50, 100, 250, 500, 1000\}$, with $\pi = 0.75$ and $r_\pi \in \{0.2, 0.75\}$.

Skewness. Since $\widehat{\text{skew}}$ is exactly unbiased for all n (Proposition 2.6), we focus only on the standard deviation. Figure 4.7 compares the simulated standard deviation of $\widehat{\text{skew}}$ against the asymptotic approximation $\hat{\sigma}_{\text{skew}}/\sqrt{n}$ for both scenarios. Agreement is close even at $n = 50$, reflecting the faster convergence of the linear estimator. Beyond that, the same observations and reasoning regarding the differences between the two scenarios hold as for the IOV.

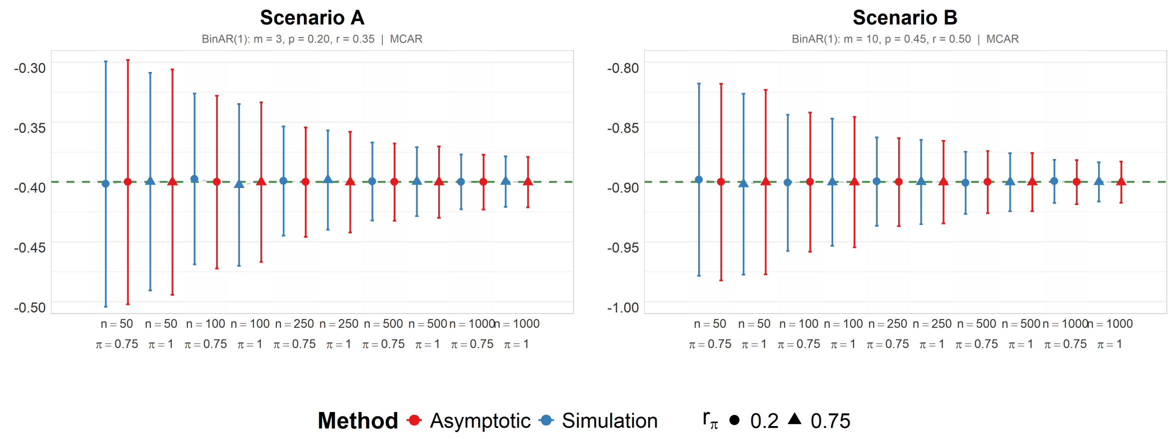

Figure 4.7.: Simulated vs. asymptotic standard deviation of $\widehat{\text{skew}}$ for both scenarios, across $n \in \{50, 100, 250, 500, 1000\}$ and $\pi \in \{0.75, 1\}$.

4.2.2. Serial Dependence: Cohen's κ

Distribution under H_0. Under the null hypothesis ($r = 0$, i.i.d. process), Figure 4.8 compares simulated and asymptotic results for $\hat{\kappa}_{\text{ord}}(1)$ as a function of n for both scenarios (with r set to 0) and $\pi \in \{1, 0.75\}$. As before, the agreement between simulation and asymptotic theory is close even for small sample sizes and improves further as n increases.
The bias under $\pi = 0.75$ is substantially larger than under $\pi = 1$, consistent with the $-1/\pi$ limit established in Theorem 3.14. Moreover, the increase in standard error induced by missingness is considerably more pronounced for $\hat{\kappa}_{\text{ord}}(1)$ than for the marginal estimators. As before, Scenario A exhibits larger absolute standard errors due to its greater marginal dispersion.

Asymptotic level of the test. To assess whether the test based on $T_n(h)$ achieves its nominal level $\alpha = 0.05$, Figure 4.9 reports the empirical rejection frequency under H_0 as a function of n. Both the uncorrected and bias-corrected test statistics are considered, using the plug-in estimator $\hat{\sigma}_\kappa$ in both cases. Both tests exhibit slight over-rejection at small sample sizes, but are already very close to the nominal level by $n = 100$. This is consistent with the fact that $\kappa_{\text{ord}}(h)$ is not a purely quadratic functional and therefore exhibits higher-order bias terms beyond the leading $O(n^{-1})$ component, that is corrected

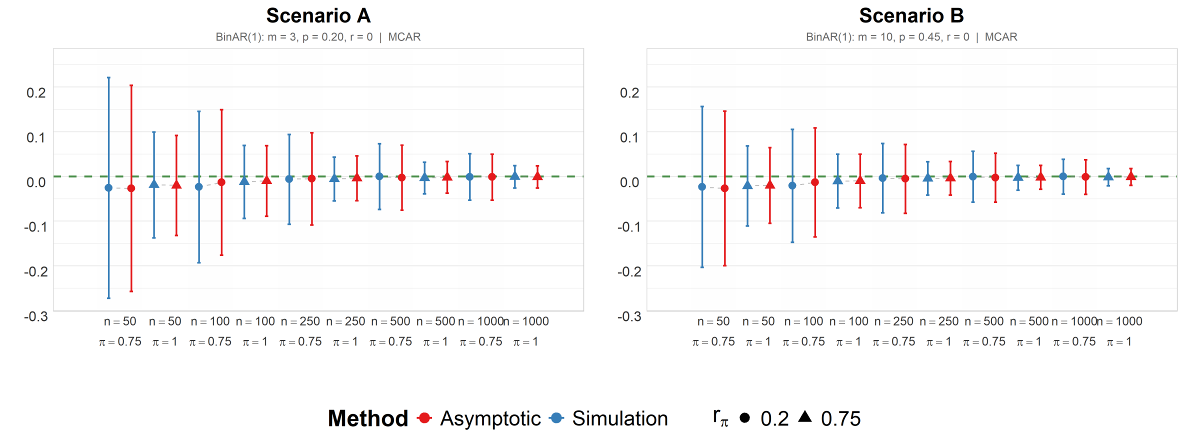

Figure 4.8.: Simulated vs. asymptotic standard deviation of $\widehat{\kappa}(1)$ for both scenarios across $n \in \{50, 100, 250, 500, 1000\}$ and $\pi \in \{0.75, 1\}$.

for in the bias-corrected version. In both scenarios, the test based on the biased corrected estimator performs better for small n, however the difference decays quite quickly and is hardly visible already for $n = 50$. However, the difference seems to be more pronounced in scenario B.

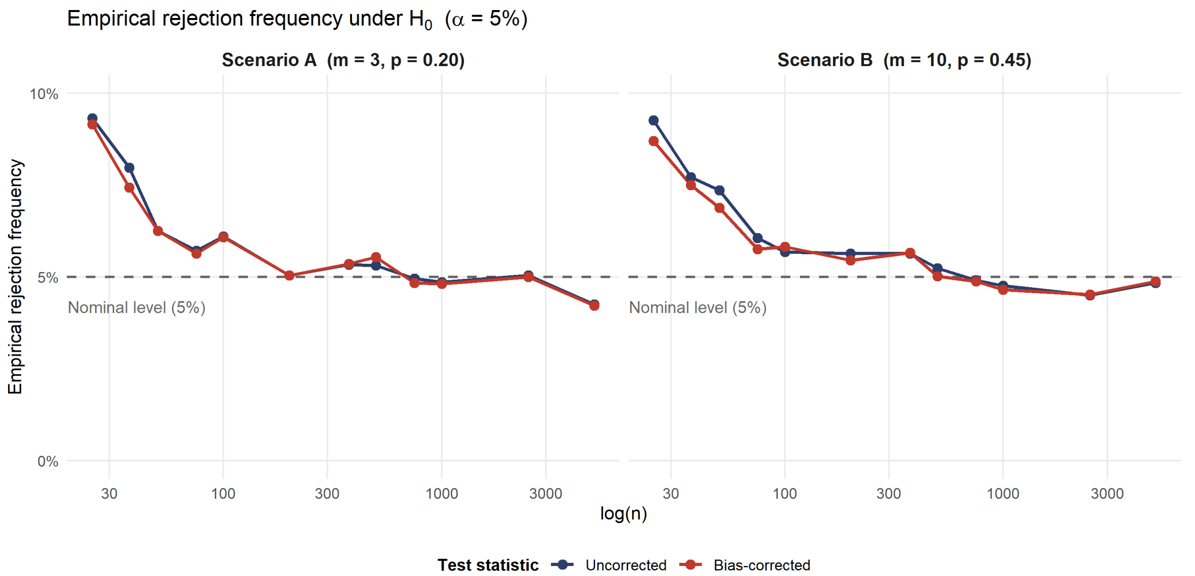

Figure 4.9.: Rejection rates of $\hat{\kappa}_{\text{ord}}(1)$ and $\hat{\kappa}^{\text{bc}}_{\text{ord}}(1)$ for both scenarios under H_0 (i.e. $r = 0$), across $n \in \{25, 37, 50, 75, 100, 200, 375, 500, 750, 1000, 2500, 5000\}$, using a nominal significance level of $\alpha = 5\%$ for $B = 5000$ iterations.

Power under H_A. To illustrate the power of the test under violation of the i.i.d. assumption of X_t, Figure 4.10 displays two panels. The left panel shows the mean estimate of $\kappa_{\text{ord}}(1)$ (black dots) for Scenario A with $\pi = 0.75$, together with ± 1.96 standard errors (grey ribbon) and the true value of $\kappa_{\text{ord}}(1)$ (red dashed line), compared to the 95% confidence intervals (blue ribbon) computed using Theorem 3.14. Asymptotically, the test is consistent: the two ribbons monotonically get narrower, and at around 500 observations they become non-overlapping, indicating that less than 2.5% of samples under the given scenario would fall inside the 95% confidence intervals. However, for smaller n, the two ribbons are clearly overlapping, indicating that the test would fail to reject H_0, even though there is serial dependence in the underlying process. To show which factors determine the convergence speed of the test, the right panel shows rejection rates for Scenario A with varying serial dependence parameter $r \in \{0.15, 0.35, 0.6\}$ and marginal observation probability $\pi \in \{0.75, 1\}$. The test shows that the convergence rate increases strongly with r: the scenario with $r = 0.6$ already achieves a near 100% rejection rate at $n = 100$, whereas the scenario with $r = 0.15$ (for $\pi = 1$) only reaches this level at $n = 2000$. The marginal observation probability is also important: in all scenarios, the speed of convergence is substantially slower for $\pi = 0.75$ compared to $\pi = 1$.

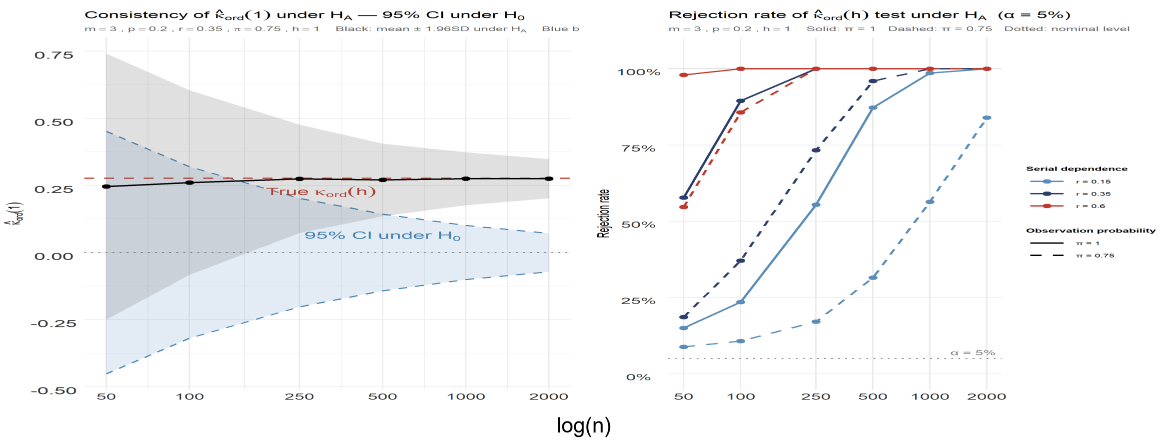

Figure 4.10.: Left: Mean estimate of $\hat{\kappa}(1)$ together with ± 1.96 standard deviations (gray ribbon) for Scenario A, compared to the 95% confidence intervals under H_0. Right: Rejection rate under H_A for Scenario A with varying serial dependence parameter $r \in \{0.15, 0.35, 0.6\}$ and marginal observation probability $\pi \in \{0.75, 1\}$. For both panels, $n \in \{50, 100, 250, 500, 1000, 2000\}$ and $B = 1000$ for all scenarios.

4.2.3. Robustness under MCAR Violations

The preceding results were obtained under the MCAR assumption. We now examine what happens when this assumption is violated.

IOV under MNAR. Under the two MNAR mechanisms described in Section 4.1, Figure 4.11 shows the simulated bias of $\widehat{\text{IOV}}$ for Scenario A across $n \in \{50, 100, 250, 500, 1000\}$ with marginal observation probability $\pi = 0.75$. In both settings, the bias of $\hat{\boldsymbol{f}}^*$ is non-negligible and does not vanish as $n \to \infty$, confirming that the corrected estimator is not consistent under MNAR. Since $\widehat{\text{IOV}}$ is a plug-in estimator based on $\hat{\boldsymbol{f}}^*$, it inherits this inconsistency through the delta method — as is clearly visible in Figure 4.11 from the narrowing ribbons and the absence of any convergence of the bias curves with growing

n. The direction of the bias depends on the particular dependence structure between X_t and O_T together with features of the marginal distribution of X_t. Here, when higher categories are more likely to be observed (blue ribbon), the upper tail of the distribution is over-represented. Given the right-skewed marginal distribution of Scenario A, this artificially inflates the estimated dispersion, resulting in a persistent positive bias of $\widehat{\text{IOV}}$. The inverse holds when lower categories are more likely to be observed (red ribbon): the lower tail is over-represented relative to the true distribution, which compresses the estimated spread and produces a persistent negative bias.

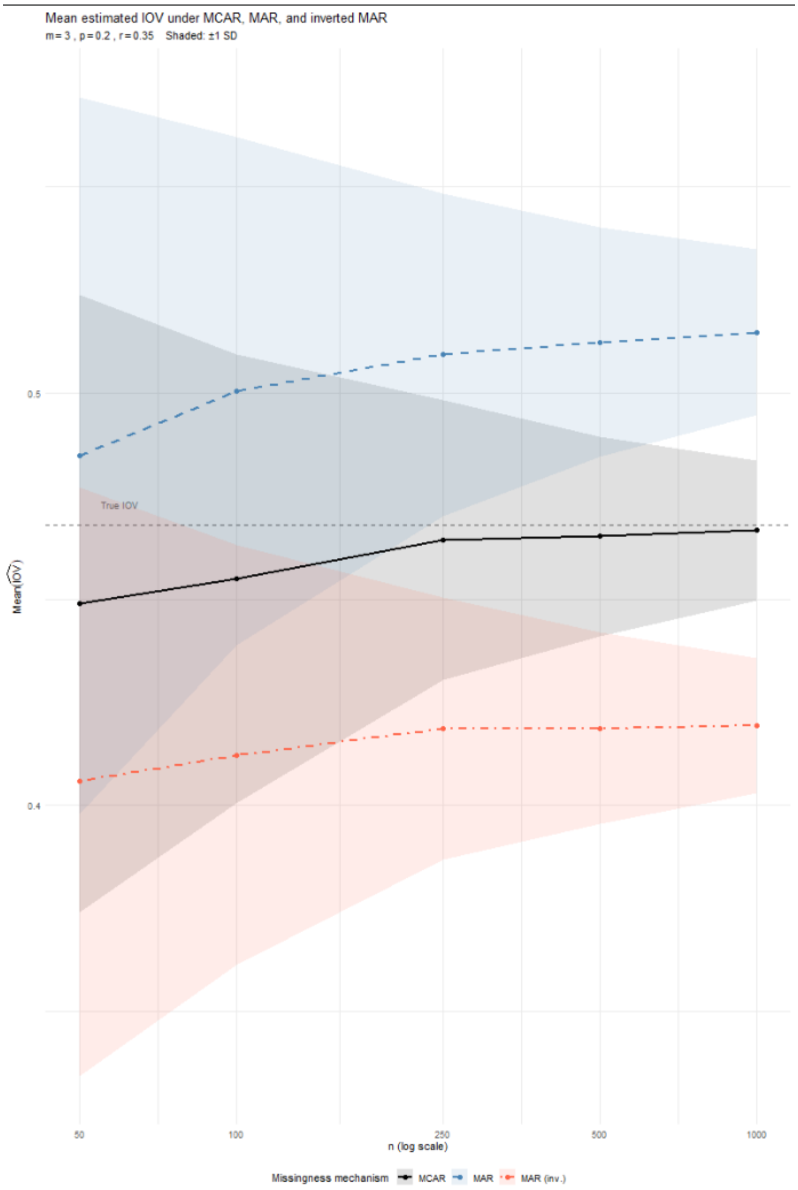

Figure 4.11.: Simulated vs. asymptotic standard deviation of Simulated bias of $I\hat{O}V$ under MNAR missingness for Scenario A, across $n \in \{50, 100, 250, 500, 1000\}$ and $\pi = 0.75$. O_t and X_t are related as described in Section 4.1.

Cohen's κ under MNAR. To illustrate the effect of MNAR missingness on the test for serial dependence based on $\hat{\kappa}_{\text{ord}}(h)$, Figure **??** uses the same two-panel format as in Figure 4.10, with the key differences being that we included the simulated results for both MNAR mechanisms (red and blue ribbons) in the left panel, and that we now show rejection rates under H_A for both MNAR mechanisms in the right panel, while keeping the marginal observation probability $\pi = 0.75$ constant. We also included the rejection rates under H_0 (black line) to illustrate the effect of MNAR missingness on the level of the test. Otherwise, the same specifications for the scenarios, sample sizes, and number of replications are used as in Figure 4.10.

A few observations are worth highlighting. First, the left panel seemingly implies that $\hat{\kappa}_{\text{ord}}(1)$ is not consistent under MNAR missingness: for both mechanisms, the average estimate of $\hat{\kappa}_{\text{ord}}(1)$ for $n \geq 250$ does not further approach the true value $\kappa_{\text{ord}}(1)$ (dashed line). Second, the direction of the asymptotic bias differs between the two MNAR mechanisms. This becomes apparent in the right panel: under the mechanism where $P(O_t = 1) \propto -X_t$, the test statistic converges considerably faster than under $P(O_t = 1) \propto X_t$. The precise mechanism driving this difference likely depends on an interplay between the marginal distribution of X_t and the particular missingness mechanism, though the exact nature of this dependence warrants further investigation. Third, both panels suggest that the test based on $\hat{\kappa}_{\text{ord}}(1)$ remains consistent under MNAR missingness. In the left panel, for $n > 1000$ the two ribbons lie outside the 95% confidence band under H_0, implying that H_0 would be rejected in more than 97.5% of samples under the given scenario. In the right panel, rejection rates under H_A approach 100% for $n > 1000$ across both mechanisms, with the exception of the scenario with $r = 0.15$ and $P(O_t = 1) \propto X_t$, where the rejection rate remains around 35% even at $n = 2000$. This demonstrates that a combination of weak serial dependence and an unfavorable MNAR mechanism can cause a substantial loss of power even at large sample sizes. Nevertheless, the power curve for this scenario remains strictly increasing, suggesting that the test would eventually achieve consistency as n grows, albeit at a considerably slower rate than in the remaining scenarios. Under H_0 the difference between the two missingness mechanisms is even more pronounced: for the $P(O_t = 1) \propto -X_t$ mechanism, the rejection rate for $n = 50$ is already below the 5% nominal level, and it further decreases with n, whereas for the $P(O_t = 1) \propto X_t$ mechanism, the rejection rate is around 20% at $n = 50$ and only approaches the nominal level very slowly, remaining above 15% even at $n = 2000$. This illustrates that MNAR missingness can lead to severe over-rejection of the test under H_0, and that the severity of this effect depends on the particular MNAR mechanism at play.

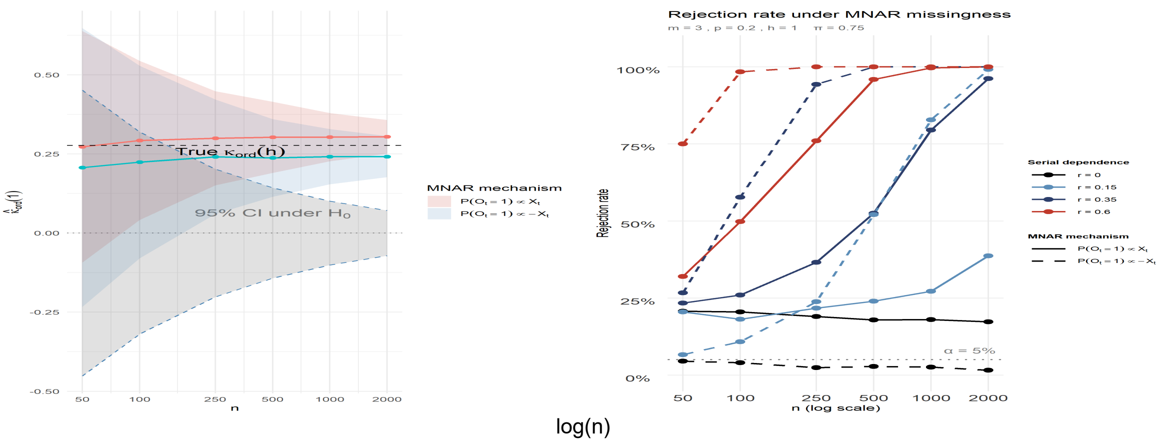

Figure 4.12.: Left: Mean estimateS of $\hat{\kappa}(1)$ together with ± 1.96 standard deviations for Scenario A with MNAR missingness, with the red (increasing) and blue (decreasing) ribbons corresponding to the two MNAR mechanisms described in Section 4.1, compared to the 95% confidence intervals under H_0. Right: Rejection rate under H_0 (black) and H_A (colours) for Scenario A with varying serial dependence parameter $r \in \{0.15, 0.35, 0.6\}$, marginal observation probability $\pi = 0.75$ and both MNAR mechanisms. For both panels, $n \in \{50, 100, 250, 500, 1000, 2000\}$ and $B = 1000$ for all scenarios.

5. Extension to Nominal Processes

The theoretical framework developed in the preceding chapters was presented for ordinal categorical processes, where the natural ordering of the categories allows the marginal distribution to be represented through the CDF vector $\boldsymbol{f}$ and its sample analogue $\hat{\boldsymbol{f}}$. For nominal processes, no such ordering exists, and the marginal distribution is instead characterized by the PMF vector $\boldsymbol{p}$ and its estimator $\hat{\boldsymbol{p}}$ (recall Definition 2.1). Since both estimators are sample means of bounded indicator processes, and since we again assume (X_t) to be strictly stationary and α-mixing with exponentially decaying coefficients (Definition 2.2), the asymptotic theory carries over almost entirely. This chapter makes this correspondence precise and focuses on the points where the nominal case genuinely differs: the choice of dispersion measure and the form of the serial dependence measure.

5.1. Asymptotic Distribution of $\hat{\boldsymbol{p}}$

The structure of $\boldsymbol{\Sigma}_Y$ differs from that of $\boldsymbol{\Sigma}_Z$, however, in the lag-0 term. For ordinal data, the joint expectation $\mathrm{E}[Z_{t,i}Z_{t,j}] = P(X_t \le s_i, X_t \le s_j) = f_{\min\{i,j\}}$ exploits the cumulative structure of the binarization. For nominal data, no such ordering exists, and instead $\mathrm{E}[Y_{t,i}Y_{t,j}] = P(X_t = s_i, X_t = s_j)$, which equals p_i if $i = j$ and zero otherwise. Using the Kronecker delta $\delta_{i,j} := \mathbb{I}\{i = j\}$, this can be written compactly as $\mathrm{E}[Y_{t,i}Y_{t,j}] = \delta_{i,j}p_i$, so that $\mathrm{Cov}(Y_{t,i}, Y_{t,j}) = p_j(\delta_{i,j} - p_i)$, yielding the following result.

Theorem 5.1 (Asymptotic Normality of $\hat{\boldsymbol{p}}$)**.** *Let (X_t) be strictly stationary and α-mixing with exponentially decaying coefficients. Then*

$$\sqrt{n}\Big(\hat{\boldsymbol{p}} - \boldsymbol{p}\Big) \xrightarrow[n\to\infty]{d} \mathcal{N}\Big(\boldsymbol{0},\, \boldsymbol{\Sigma}_Y\Big),$$

where $\boldsymbol{\Sigma}_Y = (\sigma_{Y;i,j})_{i,j=0,\dots,m}$ is given by

$$\sigma_{Y;i,j} = p_j(\delta_{i,j} - p_i) + \sum_{h=1}^{\infty}\Big(p_{ij}(h) + p_{ji}(h) - 2p_ip_j\Big).$$

5.2. Asymptotic Properties of the IQV

Since $\mathrm{IQV}(\boldsymbol{p}) = \frac{m+1}{m}(1 - \sum_{i=0}^{m} p_i^2)$ is quadratic in $\boldsymbol{p}$, all derivatives of order three and higher vanish identically, and the second-order Taylor expansion is exact. The Jacobian and Hessian are

$$\boldsymbol{j}_{\mathrm{IQV}} = -\frac{2(m+1)}{m}(p_0, \dots, p_m), \qquad \boldsymbol{H}_{\mathrm{IQV}} = -\frac{2(m+1)}{m}\boldsymbol{I}_{m+1},$$

and the following result is obtained by the same argument as Theorem 2.5, with $\boldsymbol{\Sigma}_Z$ replaced by $\boldsymbol{\Sigma}_Y$ throughout:

Proposition 5.2 (Asymptotic Distribution of $\widehat{\mathrm{IQV}}$)**.** *Under the assumptions of Theorem 5.1:*

(i) ***Asymptotic variance:***

$$\sqrt{n}\,\mathrm{Var}\Big(\widehat{\mathrm{IQV}}\Big) \overset{\text{asy.}}{=} \frac{4(m+1)^2}{m^2} \sum_{i,j=0}^{m} p_i p_j\, \sigma_{Y;i,j}.$$

(ii) ***Bias to order*** n^{-1}***:***

$$\mathrm{Bias}\Big(\widehat{\mathrm{IQV}}\Big) = -\frac{2(m+1)}{mn} \sum_{i=0}^{m} \sigma_{Y;i,i}.$$

5.3. Missingness

Marginal measures The amplitude modulation framework of Section 3.2 extends to nominal processes without modification: the observation indicator O_t multiplies the binarization vector $\boldsymbol{Y}_t$ rather than $\boldsymbol{Z}_t$, and the corrected estimator of $\boldsymbol{p}$ is defined analogously to Definition 3.3 as

Definition 5.3 (Corrected estimator $\hat{\boldsymbol{p}}^*$)**.**

$$\hat{\boldsymbol{p}}^* := \frac{\frac{1}{n}\sum_{t=1}^{n} O_t \boldsymbol{Y}_t}{\frac{1}{n}\sum_{t=1}^{n} O_t} = \frac{\hat{\mathbf{p}}}{\bar{O}}.$$

Since $O_t \boldsymbol{Y}_t$ is a bounded indicator process that inherits stationarity and α-mixing from (O_t) and (X_t) by the same argument as in Section 3.2, the asymptotic normality of $\hat{\boldsymbol{p}}^*$ follows from Theorem 3.6 with $\boldsymbol{Z}_t^*$ replaced by $\boldsymbol{Y}_t^* := O_t(1, \boldsymbol{Y}_t^\top)^\top$. The asymptotic covariance matrix $\boldsymbol{\Sigma}_{\hat{\boldsymbol{p}}^*}$ takes the same form as $\boldsymbol{\Sigma}_{\hat{\boldsymbol{f}}^*}$ with f_i, $f_{ij}(h)$ replaced by p_i, $p_{ij}(h)$ and $(f_{\min\{i,j\}} - f_i f_j)$ by $p_j(\delta_{i,j} - p_i)$ throughout. The results are summarized in the following theorem:

Theorem 5.4 (Asymptotic Normality of $\hat{\boldsymbol{p}}$)**.** *Let the same assumptions regarding X_t and O_t hold as in Theorem 3.6, except that X_t is nominal instead of ordinal. Then the corrected estimator $\hat{\boldsymbol{p}}^*$ (Definition 5.3) satisifies the following multivariate CLT:*

$$\sqrt{n}\Big(\hat{\boldsymbol{p}}^* - \boldsymbol{p}\Big) \xrightarrow[n\to\infty]{d} \mathcal{N}\Big(\mathbf{0},\, \boldsymbol{\Sigma}_{\boldsymbol{p}^*}\Big),$$

where $\boldsymbol{\Sigma}_{\boldsymbol{p}^*} = (\sigma_{\boldsymbol{p}^*;i,j})_{i,j=0,\dots,m}$ is given by

$$\sigma_{\boldsymbol{p}^*;i,j} = \frac{1}{\pi} p_j(\delta_{i,j} - p_i) + \frac{1}{\pi^2} \sum_{h=1}^{\infty} \Big(p_{ij}(h) + p_{ji}(h) - 2p_i p_j\Big).$$

As for $\hat{\boldsymbol{f}}^*$, the second-order term in the Taylor expansion vanishes identically, implying that no second-order asymptotic bias occurs. The detailed calculations for the asymptotic variance and bias are provided in Appendix A.7.

Serial dependence The serial dependence measure κ_{nom} (recall Definition 1.5) depends on the bivariate probabilities $p_{ij}(h)$. Therefore, the corrected estimator $\hat{p}^*_{ij}(h)$ is defined in direct analogy to $\hat{f}^*_{ij}(h)$ from Section 3.5 as

$$\hat{p}^*_{ij}(h) = \frac{\sum_{t=h+1}^{n} O_t O_{t-h} Y_{t,i} Y_{t-h,j}}{\sum_{t=h+1}^{n} O_t O_{t-h}}.$$

We define $\boldsymbol{Y}_n^{(h)*}$ in direct analogy to $\boldsymbol{Z}_n^{(h)*}$ from Section 3.5 with $Z_{t,i}$ replaced by $Y_{t,i}$ and $Z_{t-h,j}$ replaced by $Y^*_{t-h,j}$. The elements of the asymptotic covariance matrix $\boldsymbol{\Sigma}_{\boldsymbol{Y}^{(h)*}}$ are computed in direct analogy to those of $\boldsymbol{\Sigma}_{\boldsymbol{Z}^{(h)*}}$ from Section 3.5.1 with $Z_{t,i,}$, f_i, $f_{ij}(h)$ replaced by $Y_{t,i}$, p_i, $p_{ij}(h)$ and $(f_{\min\{i,j\}} - f_i f_j)$ by $p_j(\delta_{i,j} - p_i)$ throughout. From here, the asymptotic normality of

$$\hat{\boldsymbol{p}}^{(h)*} := (p_0, \ldots, p_m, p_{00}(h), \ldots, p_{mm}(h))^\top$$

follows from the same argument delta method argument as in Theorem 3.13, with the results summarized in the following theorem:

Theorem 5.5 (Asymptotic Normality of $\hat{\boldsymbol{p}}$)**.** *Let the same assumptions regarding X_t and O_t hold as in Theorem 3.13, except that X_t is nominal instead of ordinal. Then $\hat{\boldsymbol{p}}^{(h)*}$ is consistent for $\boldsymbol{\mu}^{(h)*} := (p_0, \ldots, p_m, p_{00}(h), \ldots, p_{mm}(h))^\top$, and:*

$$\sqrt{n}\Big(\hat{\boldsymbol{p}}^{(h)*} - \boldsymbol{\mu}^{(h)*}\Big) \xrightarrow{d} \mathcal{N}\Big(\boldsymbol{0},\, \boldsymbol{\Sigma}_{\boldsymbol{p}^{(h)*}}\Big), \qquad \boldsymbol{\Sigma}_{\boldsymbol{p}^{(h)*}} = \boldsymbol{G}\,\boldsymbol{\Sigma}_{\boldsymbol{Y}^{(h)*}}\,\boldsymbol{G}^\top,$$

where the Jacobian $\boldsymbol{G}_{2m+2\times 2(m+2)}$, evaluated at $\boldsymbol{\mu}_{\boldsymbol{Y}^{(h)*}}$, is given by

$$\boldsymbol{G} = \begin{pmatrix} 0 & \frac{-p_0}{\pi} & \frac{1}{\pi} & 0 & \cdots & & & \cdots & 0 \\ \vdots & \vdots & 0 & \ddots & \ddots & & & & \vdots \\ 0 & \frac{-p_m}{\pi} & \vdots & \ddots & \frac{1}{\pi} & & & & \vdots \\ \frac{-p_{00}(h)}{\pi(h)} & 0 & & & & \frac{1}{\pi(h)} & \ddots & & \vdots \\ \vdots & \vdots & \vdots & & & \ddots & \ddots & & 0 \\ \frac{-p_{m,m}(h)}{\pi(h)} & 0 & 0 & \cdots & & \cdots & 0 & & \frac{1}{\pi(h)} \end{pmatrix}.$$

Once again, higher-order mixed moments appear in $\boldsymbol{\Sigma}_{\boldsymbol{Y}^{(h)*}}$, such that closed form expression for the elements of $\boldsymbol{\Sigma}_{\hat{\boldsymbol{p}}^{(h)*}}$ and consequently $\sigma^2_{\hat{\kappa}_{\text{nom}}(h)}$ are not without further assumptions. However, under H_0 that (X_t) is i.i.d., the bivariate probabilities factorize as $p_{ij}(h) = p_i p_j$. The relevant higher-order mixed moments thus simplify as follows:

$$\mathrm{E}[Y_{t,i}Y_{t-h,i}Y_{t-k,j}Y_{t-k-h,j}] = \begin{cases} \delta_{i,j}p_i^2 & \text{if } k=0,\\ \delta_{i,j}p_i^3 & \text{if } k=h,\\ p_i^2p_j^2 & \text{else.}\end{cases}$$

$$\mathrm{E}[Y_{t+k,i}Y_{t,j}Y_{t-h,j}] = \begin{cases} \delta_{i,j}p_i^2 & \text{if } k=0,\\ p_ip_j^2 & \text{else.}\end{cases}$$

$$\mathrm{E}[Y_{t,i}Y_{t+k,j}Y_{t+k-h,j}] = \begin{cases} \delta_{i,j}p_i^2 & \text{if } k=0 \text{ or } h=k,\\ p_ip_j^2 & \text{else.}\end{cases}$$

We then once again apply these simplifications to the elements ***(ix)*** and ***(x)*** of $\boldsymbol{\Sigma}_{\boldsymbol{Y}^{(h)*}}$

(ix) $\quad i=0,\dots,m, j=m+1,\dots,2m+2:$

$$\begin{aligned}
&\mathrm{Cov}(O_tY_{t,i}, O_tO_{t-h}Y_{t,j-m-1}Y_{t-h,j-m-1})\\
&\overset{H_0}{=} \pi(h)\partial_{i,j-m-1}p_i^2 - \pi\pi(h)p_ip_{j-m-1}^2\\
&\quad + \underbrace{\pi(h)\partial_{i,j-m-1}p_i^2 + \mathrm{E}[O_{t+h}O_tO_{t-h}]p_ip_{j-m-1}^2 - 2\pi\pi(h)p_ip_{j-m-1}^2}_{h=k}\\
&\quad + p_ip_{j-m-1}^2\sum_{\substack{k=1\\k\neq h}}^{\infty}\left[\mathrm{E}[O_tO_{t+k}O_{t+k-h}] + \mathrm{E}[O_tO_{t+k}O_{t+k-h}] - 2\pi\pi(h)\right]\\
&= 2\pi(h)\partial_{i,j-m-1}p_i^2 - \pi(h)p_ip_{j-m-1}^2 - \pi\pi(h)p_ip_{j-m-1}^2\\
&\quad + p_ip_{j-m-1}^2\sum_{k=1}^{\infty}\left[\mathrm{E}[O_tO_{t+k}O_{t+k-h}] + \mathrm{E}[O_tO_{t+k}O_{t-h}] - 2\pi\pi(h)\right]\\
&= 2\pi(h)(\partial_{i;j-m-1}p_i^2 - p_ip_{j-m-1}^2) + p_ip_{j-m-1}^2\sigma_{-1;-2}
\end{aligned}$$

(x) $\quad i=m+1,\dots,2m+2, j=m+1,\dots,2m+2:$

$$\begin{aligned}
&\mathrm{Cov}(O_tO_{t-h}Y_{t,i-m-1}Y_{t-h,i-m-1}, O_tO_{t-h}Y_{t,j-m-1}Y_{t-h,j-m-1})\\
&\overset{H_0}{=} \pi(h)\partial_{i-m-1,j-m-1}p_{i-m-1}^2 - \pi(h)^2p_{i-m-1}^2p_{j-m-1}^2\\
&\quad + \underbrace{\mathrm{E}[O_tO_{t-h}O_{t+h}]\cdot\left(2\partial_{i-m-1,j-m-1}p_{i-m-1}^3\right) - 2\pi(h)^22p_{j-m-1}^2p_{i-m-1}^2}_{h=k}\\
&\quad + p_{i-m-1}^2p_{j-m-1}^2\sum_{\substack{k=1\\k\neq h}}^{\infty}2\left[\mathrm{E}[O_tO_{t-h}O_{t+k}O_{t+k-h}] - \pi(h)^2\right]\\
&= \pi(h)p_{i-m-1}^2(\partial_{i-m-1,j-m-1} - \pi(h)p_{k-m-1}^2) + 2\mathrm{E}[O_tO_{t-h}O_{t+h}]\cdot(\partial_{i-m-1,j-m-1}p_{i-m-1}^3\\
&\quad - p_{i-m-1}^2p_{j-m-1}^2) + p_{i-m-1}^2p_{j-m-1}^2\sum_{k=1}^{\infty}2\left[\mathrm{E}[O_tO_{t-h}O_{t+k}O_{t+k-h}] - \pi(h)^2\right]\\
&= \pi(h)p_{i-m-1}^2(\partial_{i-m-1,j-m-1} - p_{j-m-1}^2)\\
&\quad + 2\mathrm{E}[O_tO_{t-h}O_{t+h}]\cdot(\partial_{i-m-1,j-m-1}p_{i-m-1}^3 - p_{i-m-1}^2p_{j-m-1}^2) + p_{i-m-1}^2p_{j-m-1}^2\sigma_{-2;-2}^{(h)}
\end{aligned}$$

To now get explicit expressions for the elements $\sigma^{(h)*}$ of $\mathbf{\Sigma}_{\boldsymbol{p}^{(h)*}}$ we once again distinguish into the three cases

(i) $i, j = 0, \dots, m:$

$$\begin{aligned}
\sigma_{i,j}^{(h)*} &= \frac{p_i p_j}{\pi^2}\sigma_{-1,-1}^{(h)} - \frac{p_j}{\pi^2}\sigma_{i,-1}^{(h)} - \frac{p_i}{\pi^2}\sigma_{-1,j}^{(h)} + \frac{1}{\pi^2}\sigma_{i,j}^{(h)} \\
&= \frac{1}{\pi^2}(-p_i p_j \sigma_{-1,-1}^{(h)} + p_i p_j \sigma_{-1,-1}^{(h)} + \pi p_j(\partial_{i,j} - p_i) + \sum_{k=1}^{\infty}\pi(k)(f_{ij}(k) + f_{ji}(k) - 2f_i f_j)) \\
&\overset{H_0}{=} \frac{1}{\pi}p_j(\delta_{i,j} - p_i).
\end{aligned}$$

(ii) $i = 0, \dots, m \ j = m+1, \dots, 2m+2 (j' = j - m):$

$$\begin{aligned}
\sigma_{i,j}^{(h)*} &= -\frac{p_{j'j'}(h)}{\pi(h)}\left(\frac{-p_i}{\pi}\sigma_{-1,-2}^{(h)} + \frac{1}{\pi}\sigma_{i,-2}^{(h)}\right) + \frac{1}{\pi(h)}(-\frac{p_i}{\pi}\sigma_{-1,j} + \frac{1}{\pi}\sigma_{i,j}) \\
&\overset{H_0}{=} \frac{1}{\pi(h)\pi}(p_i p_{j'}^2 \sigma_{-1,-2}^{(h)} - p_{j'}^2 p_i \sigma_{-1,-2}^{(h)*} - p_i p_{j'}^2 \sigma_{-1,-2}^{(h)} + \sigma_{ij}) \\
&= \frac{1}{\pi(h)\pi}(\sigma_{i,j} - p_{j'}^2 p_i \sigma_{-1,-2}^{(h)*})
\end{aligned}$$

(iii) $i, j = m, \dots, 2m-1 \ (i' = i - m,\ j' = j - m):$

$$\begin{aligned}
\sigma_{i,j}^{(h)*} &= -\frac{p_{j',j'}(h)}{\pi(h)}\left(\frac{-p_{i'i'}(h)}{\pi(h)}\sigma_{-2,-2}^{(h)} + \frac{1}{\pi(h)}\sigma_{i,-2}^{(h)}\right) + \frac{1}{\pi(h)}\left(\frac{-p_{i',i'}(h)}{\pi(h)}\sigma_{-2,j}^{(h)} + \frac{1}{\pi(h)}\sigma_{i,j}^{(h)}\right) \\
&\overset{H_0}{=} \frac{1}{\pi(h)^2}\left(p_{i'}^2 p_{j'}^2 \sigma_{-2,-2}^{(h)} - p_{i'}^2 p_{j'}^2 \sigma_{-2,-2}^{(h)} - p_{i'}^2 p_{j'}^2 \sigma_{-2,-2}^{(h)} + \sigma_{i,j}^{(h)}\right) \\
&= \frac{1}{\pi(h)^2}\left(\sigma_{ij} - p_{i'}^2 p_{j'}^2 \sigma_{-2,-2}^{(h)}\right).
\end{aligned}$$

From here, we once again use the delta method to obtain closed-form expressions for the asymptotic variance and bias of $\hat{\kappa}_{\text{nom}}(h)) = \frac{A}{B}$ under H_0, where $A = \sum_{j=0}^{m}(p_{jj}(h) - p_j^2)$ and $B = 1 - \sum_{i=0}^{m} p_i^2$. The corresponding Jacobian is given by

$$\boldsymbol{D}_\kappa := (d_{1,0}, \dots, d_{1,m}, d_{2,0}, \dots, d_{2,m})$$

where for $i = 0, \dots, m$:

$$\begin{aligned}
d_{1,i} &= \frac{\partial \kappa_{\text{nom}}}{\partial p_i} = \frac{-2p_i}{B} + \frac{2p_i A}{B^2}, \\
d_{2,i} &= \frac{\partial \kappa_{\text{nom}}}{\partial p_{ii}(h)} = \frac{1}{B}.
\end{aligned}$$

Hence, the asymptotic variance of $\hat{\kappa}_{\text{nom}}(h)$ under H_0 is given by

$$\boldsymbol{D}_\kappa \mathbf{\Sigma}_{\boldsymbol{p}^{(h)*}} \boldsymbol{D}_\kappa^\top = \sum_{i,j=0}^{m} d_{1,i}\, d_{1,j}\, \sigma_{i,j}^{(h)*} + 2\sum_{i,j=0}^{m} d_{1,i}\, d_{2,j}\, \sigma_{i,\,j+m+1}^{(h)*} + \sum_{i,j=0}^{m} d_{2,i}\, d_{2,j}\, \sigma_{i+m+1,\,j+m+1}^{(h)*}$$

$$
\begin{aligned}
&= \sum_{i,j=0}^{m} \left(\frac{-2p_i}{B} + \frac{2p_i A}{B^2}\right)\left(\frac{-2p_j}{B} + \frac{2p_j A}{B^2}\right)\, \sigma_{i,j}^{(h)*} \\
&\quad + 2\sum_{i,j=0}^{m} \left(\frac{-2p_i}{B} + \frac{2p_i A}{B^2}\right)\frac{1}{B}\,\sigma_{i,\,j+m}^{(h)*} + \sum_{i,j=0}^{m} \frac{1}{B^2}\,\sigma_{i+m+1,\,j+m+1}^{(h)*} \\
&\overset{H_0}{=} \frac{1}{B^2}\left(4\sum_{i,j=0}^{m} p_i p_j\, \sigma_{i,j}^{(h)*} - 4\sum_{i,j=0}^{m} p_i\, \sigma_{i,\,j+m+1}^{(h)*} + \sum_{i,j=0}^{m} \sigma_{i+m+1,\,j+m+1}^{(h)*}\right) \\
&= \frac{1}{B^2}\Bigg(4\sum_{i,j=0}^{m} p_i p_j \frac{1}{\pi} p_j(\partial_{i,j} - p_i) + 4\sum_{i,j=0}^{m} p_i\,\frac{1}{\pi\,\pi(h)}\Big(\sigma_{i,j+m+1}^{(h)} - p_j^2 p_i\,\sigma_{-1,-2}^{(h)}\Big) \\
&\quad + \sum_{i,j=0}^{m} \frac{1}{\pi(h)^2}\,(\sigma_{i+m+1,j+m+1}^{(h)} - p_i^2 p_j^2 \sigma_{-2,-2}^{(h)})\Bigg) \\
&= \frac{1}{B^2}\Bigg(\frac{4}{\pi}\sum_{i,j=0}^{m} p_i p_j^2(\partial_{i,j} - p_i) + \frac{4}{\pi\,\pi(h)}\sum_{i,j=0}^{m} 2\pi(h) p_i(\partial_{i,j} p_i^2 - p_i p_j^2) \\
&\quad + \frac{1}{\pi(h)^2}\sum_{i,j=0}^{m} (\pi(h) p_i^2(\partial_{i,j} - 2p_j^2) + 2\pi(h,2h)\partial_{i,j} p_i^3 - p_i^2 p_j^2)\Bigg) \\
&=
\end{aligned}
$$

A. Appendix

A.1. Derivation of the Asymptotic Covariance Matrix $\boldsymbol{\Sigma}_{\boldsymbol{Z}^*}$

In this section we provide the detailed computations of the elements **(ii)** and **(iii)** of the asymptotic covariance matrix $\boldsymbol{\Sigma}_{\boldsymbol{Z}^*}$.

(ii) $j > i = -1$:

$$
\begin{aligned}
&\mathrm{Cov}(O_t, O_t Z_{t,j}) + 2\sum_{h=1}^{\infty} \mathrm{Cov}(O_t, O_{t+h} Z_{t+h,j}) \\
&\overset{\perp}{=} \Big(\mathrm{E}[O_t^2]\mathrm{E}[Z_{t,j}] - \mathrm{E}[O_t]^2\mathrm{E}[Z_{t,j}]\Big) \\
&\quad + 2\sum_{h=1}^{\infty} \Big(\mathrm{E}[O_t O_{t+h}]\mathrm{E}[Z_{t,j}] - \mathrm{E}[O_t]^2\mathrm{E}[Z_{t,j}]\Big) \\
&= (\pi - \pi^2) f_j + 2\sum_{h=1}^{\infty} (\gamma_O(h) + \pi^2 - \pi^2) f_j \\
&= \pi(1-\pi) f_j + 2 f_j \sum_{h=1}^{\infty} \gamma_O(h) = f_j \Bigg(\pi(1-\pi) + 2\sum_{h=1}^{\infty} \gamma_O(h)\Bigg) \\
&= f_j\, \sigma_{\boldsymbol{Z}^*;\,-1,-1}
\end{aligned}
$$

(iii) $i, j \geq 0$:

$$
\begin{aligned}
&\mathrm{Cov}(O_t Z_{t,i}, O_t Z_{t,j}) + \sum_{h=1}^{\infty} \Big[\mathrm{Cov}(O_t Z_{t,i}, O_{t+h} Z_{t+h,j}) + \mathrm{Cov}(O_{t+h} Z_{t+h,i}, O_t Z_{t,j})\Big] \\
&\overset{\perp}{=} \Big(\mathrm{E}[O_t^2]\mathrm{E}[Z_{t,i} Z_{t,j}] - \mathrm{E}[O_t]^2\mathrm{E}[Z_{t,i}]\mathrm{E}[Z_{t,j}]\Big) \\
&\quad + \sum_{h=1}^{\infty} \Big[\mathrm{E}[O_t O_{t+h}]\mathrm{E}[Z_{t,i} Z_{t+h,j}] - \mathrm{E}[O_t]\mathrm{E}[O_{t+h}]\mathrm{E}[Z_{t,i}]\mathrm{E}[Z_{t+h,j}] \\
&\quad + \mathrm{E}[O_t O_{t+h}]\mathrm{E}[Z_{t,j} Z_{t+h,i}] - \mathrm{E}[O_t]\mathrm{E}[O_{t+h}]\mathrm{E}[Z_{t,j}]\mathrm{E}[Z_{t+h,i}]\Big] \\
&= \Big(\pi f_{\min\{i,j\}} - \pi^2 f_i f_j\Big) + \sum_{h=1}^{\infty} \pi(h)(f_{ij}(h) + f_{ji}(h)) - 2\pi^2 f_i f_j \\
&\quad + 2 f_i f_j \sum_{h=1}^{\infty} \gamma_O(h) - 2 f_i f_j \sum_{h=1}^{\infty} \gamma_O(h) + f_i f_j \pi - f_i f_j \pi \\
&= f_i f_j \pi(1-\pi) + f_i f_j \sum_{h=1}^{\infty} \gamma_O(h) + \pi(f_{\min\{i,j\}} - f_i f_j)
\end{aligned}
$$

$$
\begin{aligned}
&+\sum_{h=1}^{\infty}\Big(\pi(h)(f_{ij}(h)+f_{ji}(h))-2f_if_j(\gamma_O(h)+\pi^2)\Big)\\
&= f_if_j\,\sigma_{\boldsymbol{Z}^*;\,-1,-1}+\pi(f_{\min\{i,j\}}-f_if_j)+\sum_{h=1}^{\infty}\pi(h)(f_{ij}(h)+f_{ji}(h)-2f_if_j)
\end{aligned}
$$

A.2. Derivation of the Asymptotic Covariance Matrix $\boldsymbol{\Sigma}_{\boldsymbol{f}^{(h)*}}$

In this section, we provide the detailed derivation of the elements of the asymptotic covariance matrix $\boldsymbol{\Sigma}_{\boldsymbol{f}^{(h)*}}$.

By the multivariate delta method, the asymptotic covariance matrix of $\sqrt{n}(\hat{\boldsymbol{f}}^{(h)*}-\mu)$ is given by

$$\boldsymbol{\Sigma}_{\boldsymbol{f}^{(h)*}}=\boldsymbol{G}\,\boldsymbol{\Sigma}_{\boldsymbol{Z}^{(h)*}}\,\boldsymbol{G}^\top.$$

Carrying out the matrix multiplication row by row yields

$$
\boldsymbol{G}\boldsymbol{\Sigma}_{\boldsymbol{Z}^{(h)*}}\boldsymbol{G}^\top=\begin{pmatrix}
\frac{-f_0}{\pi}\sigma^{(h)}_{-1,-2}+\frac{1}{\pi}\sigma^{(h)}_{0,-2} & \cdots & \frac{-f_0}{\pi}\sigma^{(h)}_{-1,2m-1}+\frac{1}{\pi}\sigma^{(h)}_{0,2m-1}\\
\vdots & \ddots & \vdots\\
\frac{-f_{m-1}}{\pi}\sigma^{(h)}_{-1,-2}+\frac{1}{\pi}\sigma^{(h)}_{m-1,-2} & \cdots & \frac{-f_{m-1}}{\pi}\sigma^{(h)}_{-1,2m-1}+\frac{1}{\pi}\sigma^{(h)}_{m-1,2m-1}\\
\frac{-f_{00}(h)}{\pi(h)}\sigma^{(h)}_{-2,-2}+\frac{1}{\pi(h)}\sigma^{(h)}_{m,-2} & \cdots & \frac{-f_{00}(h)}{\pi(h)}\sigma^{(h)}_{-2,2m-1}+\frac{1}{\pi(h)}\sigma^{(h)}_{m,2m-1}\\
\vdots & \ddots & \vdots\\
\frac{f_{m-1,m-1}(h)}{\pi(h)}\sigma^{(h)}_{-2,-2}+\frac{1}{\pi(h)}\sigma^{(h)}_{2m-1,-2} & \cdots & \frac{-f_{m-1,m-1}(h)}{\pi(h)}\sigma^{(h)}_{-2,2m-1}+\frac{1}{\pi(h)}\sigma^{(h)}_{2m-1,2m-1}
\end{pmatrix}\boldsymbol{G}^\top.
$$

To obtain explicit expressions for the entries $\sigma^{(h)*}_{i,j}$, we distinguish three cases depending on whether the indices correspond to marginal or bivariate components:

(i) $i,j=0,\dots,m-1:$

$$
\begin{aligned}
\sigma^{(h)*}_{i,j}&=\frac{f_if_j}{\pi^2}\sigma^{(h)}_{-1,-1}-\frac{f_j}{\pi^2}\sigma^{(h)}_{i,-1}-\frac{f_i}{\pi^2}\sigma^{(h)}_{-1,j}+\frac{1}{\pi^2}\sigma^{(h)}_{i,j}\\
&=\frac{1}{\pi}\Big(f_{\min\{i,j\}}-f_if_j\Big)+\sum_{h=1}^{\infty}\pi(h)\left(f_{ij}(h)+f_{ji}(h)-2f_if_j\right)\\
&\overset{H_0}{=}\frac{1}{\pi}\Big(f_{\min\{i,j\}}-f_if_j\Big)
\end{aligned}
$$

(ii) $i=0,\dots,m-1,\ j=m,\dots,2m-1\ (j'=j-m):$

$$
\begin{aligned}
\sigma^{(h)*}_{i,j}&=\frac{1}{\pi(h)\,\pi}\Big(2\pi(h)\,f_{j'}(f_{\min\{i,j'\}}-f_if_{j'})+f_if_{j'}^2\,\sigma^{(h)}_{-1,-2}-f_{j'j'}(h)f_i\sigma^{(h)}_{-1,-2}\Big)\\
&\overset{H_0}{=}\frac{2}{\pi}\,f_{j'}\Big(f_{\min\{i,j'\}}-f_if_{j'}\Big)
\end{aligned}
$$

(iii) $i,j=m,\dots,2m-1\ (i'=i-m,\ j'=j-m):$

$$
\begin{aligned}
\sigma_{i,j}^{(h)*} = {} & \frac{1}{\pi(h)^2}(f_{\min\{i',j'\}} - f_{i'}f_{j'})\Big(\pi(h)(f_{\min\{i',j'\}} - f_{i'}f_{j'}) \\
& + 2\pi(h,2h)f_{i'}f_{j'}\Big).
\end{aligned}
$$

A.3. Asymptotic Variance of $\hat{\kappa}_{\mathrm{ord}}(h)$

We derive the asymptotic variance $\sigma_\kappa^2 = \boldsymbol{D}_\kappa \boldsymbol{\Sigma}_{\boldsymbol{f}^{(h)*}} \boldsymbol{D}_\kappa^\top$ under H_0. Plugging the derivatives $d_{1;i}$ and $d_{2;i}$ and the elements of $\boldsymbol{\Sigma}_{\boldsymbol{f}^{(h)*}}$ into the delta method formula yields

$$
\begin{aligned}
& \boldsymbol{D}_\kappa \boldsymbol{\Sigma}_{\boldsymbol{f}^{(h)*}} \boldsymbol{D}_\kappa^\top \\
& = \sum_{i,j=0}^{m-1} d_{1,i}\, d_{1,j}\, \sigma_{i,j}^{(h)*} + 2\sum_{i,j=0}^{m-1} d_{1,i}\, d_{2,j}\, \sigma_{i,\,j+m}^{(h)*} + \sum_{i,j=0}^{m-1} d_{2,i}\, d_{2,j}\, \sigma_{i+m,\,j+m}^{(h)*} \\
& = \sum_{i,j=0}^{m-1} \frac{4f_i f_j}{\left(\sum_{k=0}^{m-1} f_k(1-f_k)\right)^2}\sigma_{i,j}^{(h)*} + 2\sum_{i,j=0}^{m-1} \frac{-2f_i}{\left(\sum_{k=0}^{m-1} f_k(1-f_k)\right)^2}\sigma_{i,\,j+m}^{(h)*} \\
& \quad + \sum_{i,j=0}^{m-1} \frac{1}{\left(\sum_{k=0}^{m-1} f_k(1-f_k)\right)^2}\sigma_{i+m,\,j+m}^{(h)*} \\
& = \frac{1}{\left(\sum_{k=0}^{m-1} f_k(1-f_k)\right)^2} \sum_{i,j=0}^{m-1} \left(4f_i f_j\, \sigma_{i,j}^{(h)*} - 4f_i\, \sigma_{i,j+m}^{(h)*} + \sigma_{i+m,j+m}^{(h)*}\right) \\
& = \frac{1}{\left(\sum_{k=0}^{m-1} f_k(1-f_k)\right)^2} \sum_{i,j=0}^{m-1} \Bigg(4f_i f_j\, \frac{1}{\pi}(f_{\min\{i,j\}} - f_i f_j) - 4f_i\, \frac{2}{\pi} f_j (f_{\min\{i,j\}} - f_i f_j) \\
& \quad + \frac{1}{\pi(h)^2}(f_{\min\{i,j\}} - f_i f_j)\Big[\pi(h)(f_{\min\{i,j\}} - f_i f_j) + 2\pi(h,2h) f_i f_j\Big]\Bigg) \\
& = \frac{1}{\left(\sum_{k=0}^{m-1} f_k(1-f_k)\right)^2} \frac{1}{\pi(h)} \sum_{i,j=0}^{m-1} \Bigg(-4f_i f_j \frac{\pi(h)}{\pi}(f_{\min\{i,j\}} - f_i f_j) \\
& \quad + (f_{\min\{i,j\}} - f_i f_j)^2 + (f_{\min\{i,j\}} - f_i f_j)\frac{2\pi(h,2h)}{\pi(h)} f_i f_j\Bigg) \\
& = \frac{\frac{1}{\pi(h)} \sum_{i,j=0}^{m-1} \Big(f_{\min\{i,j\}} - f_i f_j\Big)\Big(f_{\min\{i,j\}} + \Big(1 + \frac{2\pi(h,2h)}{\pi(h)} - \frac{4\pi(h)}{\pi}\Big) f_i f_j\Big)}{\left(\sum_{k=0}^{m-1} f_k(1-f_k)\right)^2}.
\end{aligned}
$$

A.4. Hessian $\kappa(h)$

In this section, we derive the partial derivatives of $d_{1,i}$ with respect to f_j.

Case $i = j$

$$
\begin{aligned}
\frac{\partial d_{1,i}}{\partial f_i} &= \frac{\partial}{\partial f_i}\left(\frac{-2f_i}{\sum_{k=0}^{m-1} f_k(1-f_k)} - \frac{\left(\sum_{k=0}^{m-1}(f_{kk}(h)-f_k^2)\right)}{\left(\sum_{k=0}^{m-1} f_k(1-f_k)\right)^2} + \frac{2f_i\left(\sum_{k=0}^{m-1}(f_{kk}(h)-f_k^2)\right)}{\left(\sum_{k=0}^{m-1} f_k(1-f_k)\right)^2}\right) \\
&= \frac{-2\left(\sum_{k=0}^{m-1} f_k(1-f_k)\right)+2f_i(1-2f_i)}{(\sum_{k=0}^{m-1} f_k(1-f_k))^2} - \frac{-2f_i\left(\sum_{k=0}^{m-1} f_k(1-f_k)\right)^2 - \left(\sum_{k=0}^{m-1} f_{kk}(h)-f_k^2\right)2(\sum_{k=0}^{m-1} f_k(1-f_k))(1-2f_i)}{(\sum_{k=0}^{m-1} f_k(1-f_k))^4} \\
&+ \frac{\left(2\left(\sum_{k=0}^{m-1}(f_{kk}(h)-f_k^2)\right)-4f_if_i\right)(\sum_{k=0}^{m-1} f_k(1-f_k))^2 - 2f_i\left(\sum_{k=0}^{m-1} f_{kk}(h)-f_k^2\right)2(\sum_{k=0}^{m-1} f_k(1-f_k))(1-2f_i)}{(\sum_{k=0}^{m-1} f_k(1-f_k))^4} \\
&= \frac{-2}{\sum_{k=0}^{m-1} f_k(1-f_k)} + \frac{2f_i(1-2f_i)}{(\sum_{k=0}^{m-1} f_k(1-f_k))^2} + \frac{2f_i}{(\sum_{k=0}^{m-1} f_k(1-f_k))^2} + \frac{2f_i\left(\sum_{k=0}^{m-1} f_{kk}(h)-f_k^2\right)2(1-2f_i)}{(\sum_{k=0}^{m-1} f_k(1-f_k))^3} \\
&+ \frac{2\left(\sum_{k=0}^{m-1} f_{kk}(h)-f_k^2\right)-4f_if_i}{(\sum_{k=0}^{m-1} f_k(1-f_k))^2} - \frac{\left(\sum_{k=0}^{m-1} f_{kk}(h)-f_k^2\right)2(1-2f_i)}{(\sum_{k=0}^{m-1} f_k(1-f_k))^3} \\
&= \frac{2f_i-4f_i^2+2f_i+2f_i\kappa(h)2(1-2f_i)-4f_i^2-\kappa(h)2(1-2f_i)}{(\sum_{k=0}^{m-1} f_k(1-f_k))^2} - \frac{2(1-\kappa(h))}{\sum_{k=0}^{m-1} f_k(1-f_k)} \\
&= \frac{4f_i-8f_i^2+\kappa(h)(2(3f_i-1-2f_i^2))}{(\sum_{k=0}^{m-1} f_k(1-f_k))^2} - \frac{2(1-\kappa(h))}{\sum_{k=0}^{m-1} f_k(1-f_k)} \\
&\overset{H_0}{=} 2\left(\frac{2f_i-4f_i^2}{(\sum_{k=0}^{m-1} f_k(1-f_k))^2} - \frac{1}{\sum_{k=0}^{m-1} f_k(1-f_k)}\right)
\end{aligned}
$$

Case $i \neq j$

$$
\begin{aligned}
\frac{\partial d_{1,i}}{\partial f_j} &= \frac{\partial}{\partial f_j}\left(\frac{-2f_i}{\sum_{k=0}^{m-1} f_k(1-f_k)} - \frac{\left(\sum_{k=0}^{m-1}(f_{kk}(h)-f_k^2)\right)}{\left(\sum_{k=0}^{m-1} f_k(1-f_k)\right)^2} + \frac{2f_i\left(\sum_{k=0}^{m-1}(f_{kk}(h)-f_k^2)\right)}{\left(\sum_{k=0}^{m-1} f_k(1-f_k)\right)^2}\right) \\
&= \frac{2f_i(1-2f_j)}{(\sum_{k=0}^{m-1} f_k(1-f_k))^2} - \frac{-2f_j\left(\sum_{k=0}^{m-1} f_k(1-f_k)\right)^2 - \left(\sum_{k=0}^{m-1} f_{kk}(h)-f_k^2\right)2(\sum_{k=0}^{m-1} f_k(1-f_k))(1-2f_j)}{(\sum_{k=0}^{m-1} f_k(1-f_k))^4} \\
&+ 2f_i\frac{-2f_j(\sum_{k=0}^{m-1} f_k(1-f_k))^2 - \left(\sum_{k=0}^{m-1}(f_{kk}(h)-f_k^2)\right)2(\sum_{k=0}^{m-1} f_k(1-f_k))(1-2f_j)}{(\sum_{k=0}^{m-1} f_k(1-f_k))^4} \\
&= \frac{2f_i(1-2f_j)}{(\sum_{k=0}^{m-1} f_k(1-f_k))^2} + \frac{2f_j}{(\sum_{k=0}^{m-1} f_k(1-f_k))^2} + \frac{(1-2f_j)\kappa(h)}{(\sum_{k=0}^{m-1} f_k(1-f_k))^2} \\
&- \frac{4f_if_j}{(\sum_{k=0}^{m-1} f_k(1-f_k))^2} - \frac{4f_i(1-2f_j)\kappa(h)}{(\sum_{k=0}^{m-1} f_k(1-f_k))^2} \\
&= \frac{2f_i-4f_if_j+2f_j+\kappa(h)-4f_i-2f_j\kappa(h)-4f_if_j+8f_if_j\kappa(h)}{(\sum_{k=0}^{m-1} f_k(1-f_k))^2} \\
&\overset{H_0}{=} 2\frac{f_i+f_j-4f_if_j}{(\sum_{k=0}^{m-1} f_k(1-f_k))^2}
\end{aligned}
$$

A.5. Bias $\kappa(h)$

Using the Hessian elements derived above and the elements of $\boldsymbol{\Sigma}_{\boldsymbol{f}^{(h)*}}$ from Section **??**, the diagonal elements of $\boldsymbol{H}_\kappa\boldsymbol{\Sigma}_{\boldsymbol{f}^{(h)*}}$ are given by

(i) $i = 0, \ldots, m-1$:

$$
\left[\boldsymbol{H}_\kappa\boldsymbol{\Sigma}_{\boldsymbol{f}^{(h)*}}\right]_{ii} = 2\left(-\frac{1}{B}\sigma_{i,i}^{(h)*} + \sum_{l=0}^{m-1}\frac{f_i+f_l-4f_if_l}{B^2}\sigma_{l,i}^{(h)*}\right) + \sum_{l=m}^{2m-1}\frac{-(1-2f_i)}{B^2}\sigma_{l,i}^{(h)*}
$$

(ii) $i = m, \dots, 2m-1$ $(i' = i - m)$:

$$\left[\boldsymbol{H}_\kappa \boldsymbol{\Sigma}_{\boldsymbol{f}^{(h)*}}\right]_{ii} = \sum_{l=0}^{m-1} \frac{-(1-2f_l)}{B^2} \sigma_{l,i}^{(h)*}$$

Summing over all diagonal elements and substituting $\sigma_{i,m+l}^{(h)*} = 2f_l \sigma_{i,l}^{(h)*}$ from case **(ii)** of $\boldsymbol{\Sigma}_{\boldsymbol{f}^{(h)*}}$ and using symmetry $\sigma_{i,l}^{(h)*} = \sigma_{l,i}^{(h)*}$:

$$\begin{aligned}
\mathrm{Bias}(\hat{\kappa}_{\mathrm{ord}}) &= \frac{1}{2}\mathrm{tr}(\boldsymbol{H}_\kappa \boldsymbol{\Sigma}_{\boldsymbol{f}^{(h)*}}) = \frac{1}{2}\sum_{i=0}^{2m-1} \left[\boldsymbol{H}_\kappa \boldsymbol{\Sigma}_{\boldsymbol{f}^{(h)*}}\right]_{ii} \\
&= \frac{1}{2}\sum_{i=0}^{m-1}\Bigg[2\Bigg(-\frac{1}{\sum_{k=0}^{m-1} f_k(1-f_k)}\sigma_{i,i}^{(h)*} + \sum_{l=0}^{m-1} \frac{f_i + f_l - 4f_i f_l}{\left(\sum_{k=0}^{m-1} f_k(1-f_k)\right)^2}\sigma_{l,i}^{(h)*}\Bigg) \\
&\quad + \sum_{l=m}^{2m-1} \frac{-(1-2f_i)}{\left(\sum_{k=0}^{m-1} f_k(1-f_k)\right)^2}\sigma_{l,i}^{(h)*}\Bigg] + \frac{1}{2}\sum_{i=m}^{2m-1}\Bigg[\sum_{l=0}^{m-1} \frac{-(1-2f_i)}{\left(\sum_{k=0}^{m-1} f_k(1-f_k)\right)^2}\sigma_{i,l}^{(h)*}\Bigg] \\
&= -\frac{\sum_{i=0}^{m-1}\sigma_{ii}^{(h*)}}{\sum_{k=0}^{m-1} f_k(1-f_k)} + \frac{\sum_{i,l=0}^{m-1}(f_i + f_l - 4f_i f_l)\sigma_{l,i}^{(h)*}}{\left(\sum_{k=0}^{m-1} f_k(1-f_k)\right)^2} - \frac{\sum_{i,l=0}^{m-1}(1-2f_i)\sigma_{i,m+l}^{(h)*}}{\left(\sum_{k=0}^{m-1} f_k(1-f_k)\right)^2} \\
&= -\frac{\frac{1}{\pi}\sum_{i=0}^{m-1}(f_i - f_i^2)}{\sum_{k=0}^{m-1} f_k(1-f_k)} + \frac{\sum_{i,l=0}^{m-1}(f_i + f_l - 4f_i f_l)\sigma_{l,i}^{(h)*} - \sum_{i,l=0}^{m-1}(1-2f_i)2f_l\sigma_{l,i}^{(h)*}}{\left(\sum_{k=0}^{m-1} f_k(1-f_k)\right)^2} \\
&= -\frac{1}{\pi} + \frac{\sum_{i,l=0}^{m-1}(f_i + f_l - 4f_i f_l - 2f_l + 4f_j f_l)\,\sigma_{l,i}^{(h)*}}{\left(\sum_{k=0}^{m-1} f_k(1-f_k)\right)^2} = -\frac{1}{\pi} + \frac{\overbrace{\sum_{i,l=0}^{m-1}(f_i - f_l)\,\sigma_{l,i}^{(h)*}}^{=0}}{\left(\sum_{k=0}^{m-1} f_k(1-f_k)\right)^2} \\
&= -\frac{1}{\pi}
\end{aligned}$$

A.6. Derivation of $\hat{\kappa}_{\mathrm{ord}}$ under i.i.d. missingness

In this section we provide the detailed computations of the simplified version of $\hat{\kappa}_{\mathrm{ord}}$ under i.i.d. missingness, which is given by

$$\begin{aligned}
\sigma_\kappa^2 &= \frac{\sum_{i,j=0}^{m-1}\left(f_{\min\{i,j\}} - f_i f_j\right)\left(f_{\min\{i,j\}} + \left(1 + \frac{2\pi(h,2h)}{\pi(h)} - \frac{4\pi(h)}{\pi}\right)f_i f_j\right)}{\pi(h)\left(\sum_{k=0}^{m-1} f_k(1-f_k)\right)^2} \\
&\overset{\text{i.i.d.}}{=} \frac{\sum_{i,j=0}^{m-1}\left(f_{\min\{i,j\}} - f_i f_j\right)\left(f_{\min\{i,j\}} + \left(1 + \frac{2\pi^3}{\pi^2} - \frac{4\pi^2}{\pi}\right)f_i f_j\right)}{\pi^2\left(\sum_{k=0}^{m-1} f_k(1-f_k)\right)^2}
\end{aligned}$$

$$= \frac{1}{\pi^2} \frac{\sum_{i,j=0}^{m-1} \Big(f_{\min\{i,j\}} - f_i f_j\Big)\Big(f_{\min\{i,j\}} + \Big(1 - 2\pi\Big) f_i f_j\Big)}{\left(\sum_{k=0}^{m-1} f_k(1 - f_k)\right)^2}$$

A.7. Asymptotic Variance and Bias of $\hat{\boldsymbol{p}}^*$

The asymptotic properties of $\hat{\boldsymbol{p}}^*$ follow from the delta method by exactly the same argument as in Section 3.3. Defining the vector-valued function $\boldsymbol{g}\colon [0,1]^{m+2} \to [0,1]^{m+1}$ by

$$g_j(x_{-1}, x_0, \ldots, x_m) = x_j/x_{-1}, \qquad j = 0, \ldots, m,$$

and evaluating the Jacobian at $\boldsymbol{\mu}_{\boldsymbol{Y}^*} := \pi(1, \boldsymbol{p}^\top)^\top$ yields

$$\boldsymbol{G} = \begin{pmatrix} -\frac{p_0}{\pi} & \frac{1}{\pi} & 0 & \ldots & 0 \\ -\frac{p_1}{\pi} & 0 & \frac{1}{\pi} & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & 0 \\ -\frac{p_m}{\pi} & 0 & \ldots & 0 & \frac{1}{\pi} \end{pmatrix},$$

which has the same structure as the Jacobian in Section 3.3, with f_i replaced by p_i throughout. Since the elements of $\boldsymbol{\Sigma}_{\boldsymbol{Y}^*}$ are obtained from those of $\boldsymbol{\Sigma}_{\boldsymbol{Z}^*}$ by replacing f_i, $f_{ij}(h)$, and $(f_{\min\{i,j\}} - f_i f_j)$ with p_i, $p_{ij}(h)$, and $p_j(\delta_{i,j} - p_i)$ respectively, it follows by the same matrix multiplication as in Section 3.3 that the elements of $\boldsymbol{\Sigma}_{\hat{\boldsymbol{p}}^*}$ are obtained from those of $\boldsymbol{\Sigma}_{\hat{\boldsymbol{f}}^*}$ by the same substitution.

For the bias, the Hessian of g_j evaluated at $\boldsymbol{\mu}_{\boldsymbol{Y}^*}$ is sparse, with non-zero entries

$$\boldsymbol{H}_{g_j} = \begin{cases} \dfrac{2p_j}{\pi^2} & \text{if } i = k = -1, \\ -\dfrac{1}{\pi^2} & \text{if } i = -1,\ k = j, \\ 0 & \text{else,} \end{cases}$$

which is identical in structure to the ordinal case. Applying Lemma 2.4 and using $\sigma_{\boldsymbol{Y}^*;-1,j} = p_j\, \sigma_{\boldsymbol{Y}^*;-1,-1}$ from the nominal analogue of case **(ii)** of Remark 3.5, the bias reduces to

$$\begin{aligned} \operatorname{Bias}(\hat{p}_j) &= \frac{1}{2} \operatorname{tr}\Big(\boldsymbol{H}_{g_j}\, \boldsymbol{\Sigma}_{\boldsymbol{Y}^*}\Big) \\ &= \frac{p_j}{\pi^2}\, \sigma_{\boldsymbol{Y}^*;-1,-1} - \frac{1}{\pi^2}\, p_j\, \sigma_{\boldsymbol{Y}^*;-1,-1} = 0. \end{aligned}$$

Hence, $\hat{\boldsymbol{p}}^*$ is unbiased to order n^{-1} under MCAR, in direct analogy to the ordinal result of Section 3.3.

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